周衍柏《理论力学教程》课后题答案

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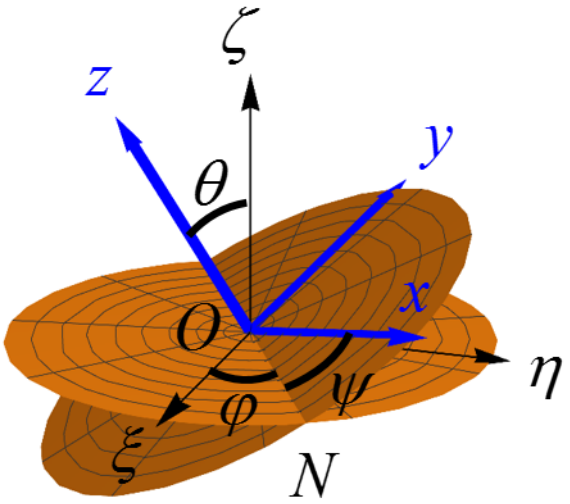

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2020年8月1日

习题一 质点运动学 习题 P.76

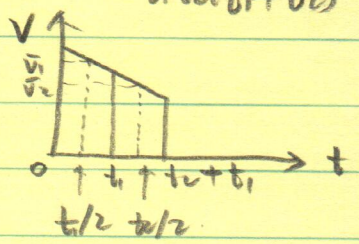
1.1 方法①: $\begin{cases} S = v_0 t_1 - \frac{1}{2} a t_1^2 \\ 2S = v_0 (t_1 + t_2) - \frac{1}{2} a (t_1 + t_2)^2 \end{cases}$

$\Rightarrow S(t_1+t_2) - 2St_1 = -\frac{1}{2} a t_1^2 (t_1+t_2) + \frac{1}{2} a (t_1+t_2)^2 t_1$

$S(t_2 - t_1) = \frac{1}{2} a t_1 (t_1+t_2) t_2$

$a = \frac{2S(t_2-t_1)}{t_1 t_2 (t_1+t_2)}$

方法②:



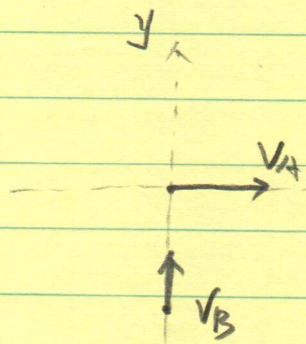
$\bar{v}_1 = \frac{S}{t_1} \qquad \bar{v}_2 = \frac{S}{t_2}$

$\bar{v}_2 - \bar{v}_1 = a\left(\frac{t_1}{2} + \frac{t_2}{2}\right)$

$\frac{S}{t_2} - \frac{S}{t_1} = a\left(\frac{t_2}{2} + \frac{t_1}{2}\right)$

$\therefore a = \frac{2S(t_1 - t_2)}{(t_1+t_2) t_1 t_2}$

1.2 ① 取 $t=0$ 为正午.



$v \equiv v_A = v_B = 15\ km/h$

$t_0 = 1.5h$

$\vec{S}_A = \vec{v}_A t = (vt, 0)$

$\vec{S}_B = S_{B0} + \vec{v}_B t = (0, v(t-t_0), 0)$

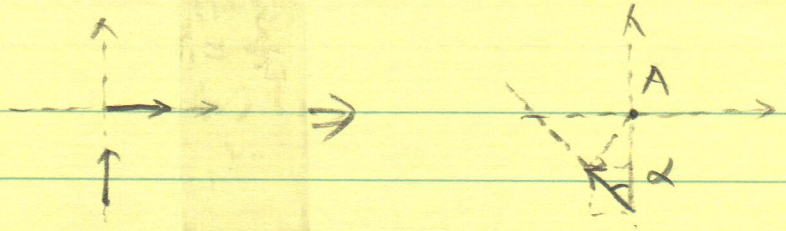
$|\vec{S}_A - \vec{S}_B| = \sqrt{v^2 t^2 + v^2 (t-t_0)^2} = v\sqrt{2t^2 - 2tt_0 + t_0^2}$

$= v\sqrt{2}\sqrt{(t - \frac{t_0}{2})^2 + \frac{t_0^2}{2}}$

$\therefore$ 当 $t = \frac{t_0}{2} = 45$ 分

$S_{min} = v\sqrt{2}\cdot\frac{t_0}{2} = \frac{v t_0}{\sqrt{2}} = \frac{15 \times 1.5}{\sqrt{2}} \approx 15.9$

② 方法二：相对法



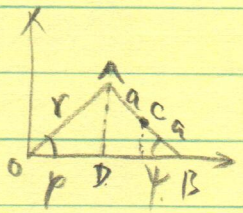
$$\begin{cases}\vec{V}_A=\vec{V}_0+\vec{V}_A'\\ \vec{V}_B=\vec{V}_0+\vec{V}_B'\end{cases}\qquad \vec{V}_A'=0,\ \vec{V}_B'=\vec{V}_B-\vec{V}_0=(-V,V,0)$$

$$\Rightarrow \alpha=\frac{\pi}{4}$$

最短距离对应垂直，即 $S_m=S_0\sin\alpha=Vt_0\cdot\frac{\pi}{2}\simeq 15.7$

1.3

$\dot\varphi=\omega$



$$\sin\psi=\frac{y}{a}$$

由几何，$r\cdot\sin\varphi=2a\sin\psi\quad\therefore\psi=\arcsin\left(\frac{r}{2a}\sin\varphi\right)\qquad\dot\psi=\frac{\omega r\cos\varphi}{2a\cos\psi}$

$\vec{r}_C=(r\cos\varphi+a\cos\psi,\ a\sin\psi)=(x,y)$

$\therefore(x-r\cos\varphi)^2+y^2=a^2$ $\qquad=\frac{2y}{r}$

用 x,y 表示 φ：$\varphi=\arcsin\left(\frac{2a}{r}\sin\psi\right)=\arcsin\left(\frac{2y}{r}\right)\qquad\sin\varphi=\frac{2a}{r}\sin\psi$

$$\therefore\cos\varphi=\frac{\sqrt{r^2-4y^2}}{r}$$

$$\therefore\left(x-\sqrt{r^2-4y^2}\right)^2+y^2=a^2$$

$$\Rightarrow 2x\sqrt{r^2-4y^2}=r^2-a^2+x^2-3y^2$$

$$4x^2r^2-16x^2y^2=(r^2-a^2)^2+x^4+9y^4+2(r^2-a^2)x^2-6(r^2-a^2)y^2-6x^2y^2$$

$$0=(r^2-a^2)+(x^2+3y^2)^2+4x^2y^2+2(a^2-r^2)x^2-4a^2x^2+6(a^2+r^2)y^2$$

即 $4x^2(a^2-y^2)=(a^2-r^2+x^2+3y^2)^2$ 此即运动轨迹

速率：$\vec{V}_0=\dot{\vec{r}}_C=(-r\dot\varphi\sin\varphi-a\dot\psi\sin\psi,\ a\dot\psi\cos\varphi)$

$$=\left(-\omega r\cdot\frac{2y}{r}-a\cdot\frac{\omega r\cos\varphi}{2a\cos\psi}\sin\psi,\ a\cdot\frac{\omega r\cos\varphi}{2a\cos\psi}\cos\psi\right)$$

$$= \left(-\omega r\cdot\sin\varphi - \frac{\omega r}{2}\frac{\cos\varphi\sin\psi}{\cos\psi},\ \frac{\omega r}{2}\cdot\cos\varphi\right)$$

$$|\vec{v}| = \omega r\sqrt{\frac{1}{4}\cos^2\varphi + \frac{1}{4}\cos^2\varphi\cdot\frac{\sin^2\psi}{\cos^2\psi} + \sin^2\varphi + \frac{\sin\varphi\cos\varphi\sin\psi}{\cos\psi}}$$

$$= \frac{\omega r}{2\cos\psi}\sqrt{\cos^2\varphi + \sin\varphi\cos\psi\sin(\varphi+\psi)}$$

另外 $\vec{r}_C = (r\cos\varphi + a\cos\psi,\ \frac{r}{2}\sin\varphi) = (x, y)$

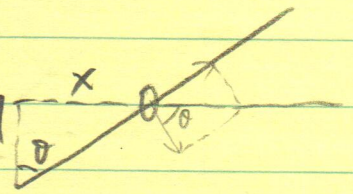
$$\therefore \left(\frac{x}{2} - \frac{a}{2}\cos\psi\right)^2 + y^2 = \frac{r^2}{4} \qquad \cos\psi = \frac{\sqrt{a^2-y^2}}{a},\ \frac{a}{2}\cos\psi = \frac{\sqrt{a^2-y^2}}{2}$$

$$x^2 + a^2 - y^2 - 2x\sqrt{a^2-y^2} + 4y^2 = r^2$$

$$\therefore 2x\sqrt{a^2-y^2} = a^2 - r^2 + x^2 + 3y^2$$

$$\Rightarrow 4x^2(a^2-y^2) = (a^2-r^2+x^2+3y^2)^2$$

1.4 正确的解是(C)：杆在平面内匀角速转动



$$x = d\tan\theta \qquad (\tan\theta)' = \frac{1}{\cos^2\theta} = \sec^2\theta$$

$$\begin{cases} V_\theta = \omega r = \dfrac{\omega d}{\cos\theta} \\ V_\theta = v\cdot\cos\theta \end{cases}$$

另解之一 $v = \dot{x} = \dfrac{d}{\cos^2\theta}\dot{\theta} = \dfrac{\omega d}{\cos^2\theta} = \dfrac{\omega(x^2+d^2)}{d}$

$$a = \ddot{x} = \omega d(-2)\frac{-\sin\theta\,\dot{\theta}}{\cos^3\theta}$$

$$= 2d\omega^2\cdot\frac{x}{d}\cdot\frac{d^2+x^2}{d^2} = \frac{2\omega^2 x(d^2+x^2)}{d^2}$$

1.5：

$$v = v_0 + \int_0^t a(\tau)\,d\tau = \int_0^t c\left(1-\sin\frac{\pi\tau}{2T}\right)d\tau = ct + \frac{2cT}{\pi}\cdot\cos\frac{\pi\tau}{2T}\Big|_0^t$$

$$= c\cdot\left[t + \frac{2T}{\pi}\left(\cos\frac{\pi t}{2T} - 1\right)\right]$$

$$s = \int_0^t v(\tau)\,d\tau = c\int_0^t\left[t' + \frac{2T}{\pi}\left(\cos\frac{\pi t'}{2T} - 1\right)\right]dt'$$

$$= c\cdot\left[\frac{t^2}{2} + \left(\frac{2T}{\pi}\right)^2\sin\frac{\pi t}{2T} - \frac{2T}{\pi}t\right] = c\left[\frac{t^2}{2} + \frac{2T}{\pi}\left(\frac{2T}{\pi}\sin\frac{\pi t}{2T} - t\right)\right]$$

1.6

$\begin{cases} V_r = \lambda r \\ V_\theta = \mu\theta \end{cases}$ 与 $\begin{cases} V_r = \dot{r} \\ V_\theta = r\dot{\theta} \end{cases}$

$\dot{r} = \lambda r \Rightarrow r = C_1 e^{\lambda t}$

$\mu\theta = C_1 e^{\lambda t}\cdot\dot{\theta}$

$\Rightarrow \int \frac{\mu}{C_1} e^{-\lambda t}\, dt = \int \frac{d\theta}{\theta} = \ln\theta$

$\theta = C_2 \exp\left[\frac{\mu}{C_1}\exp(-\lambda t)\right]$

$\therefore \vec{a}_r = \ddot{r} - r\dot{\theta}^2 = \lambda^2 r - r\left(\frac{\mu\theta}{r}\right)^2 = \boxed{\lambda^2 r - \frac{\mu^2\theta^2}{r}}$

$\vec{a}_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta}$

$\mu\theta = r\dot{\theta} \Rightarrow \mu\dot{\theta} = \dot{r}\dot{\theta} + r\ddot{\theta} \Rightarrow \ddot{\theta} = \frac{1}{r}(\mu\dot{\theta} - \dot{r}\dot{\theta})$

$= \frac{1}{r}\left(\mu\cdot\frac{\mu\theta}{r} - \lambda r\cdot\frac{\mu\theta}{r}\right)$

$= \frac{\mu^2\theta}{r^2} - \frac{\lambda\mu\theta}{r}$

$\vec{a}_\theta = \frac{\mu^2\theta}{r} - \lambda\mu\theta + 2\lambda r\cdot\frac{\mu\theta}{r} = \boxed{\frac{\mu^2\theta}{r} + \lambda\mu\theta} = \mu\theta\left(\lambda + \frac{\mu}{r}\right)$

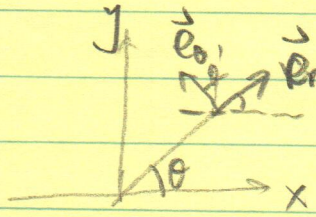
1.7

$x = r\cos\theta$

$\dot{x} = \dot{r}\cos\theta - r\dot{\theta}\sin\theta$

$\ddot{x} = \ddot{r}\cos\theta - \dot{r}\dot{\theta}\sin\theta - \dot{r}\dot{\theta}\sin\theta - r\ddot{\theta}\sin\theta - r\dot{\theta}\dot{\theta}\cos\theta$

$= \ddot{r}\cos\theta - r\dot{\theta}^2\cos\theta - (r\ddot{\theta} + 2\dot{r}\dot{\theta})\sin\theta$



$y = r\sin\theta$

$\dot{y} = \dot{r}\sin\theta + r\dot{\theta}\cos\theta$

$\ddot{y} = \ddot{r}\sin\theta + \dot{r}\dot{\theta}\cos\theta + \dot{r}\dot{\theta}\cos\theta + r\ddot{\theta}\cos\theta - r\dot{\theta}\dot{\theta}\sin\theta$

$= \ddot{r}\sin\theta - r\dot{\theta}^2\sin\theta + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\cos\theta$

$\vec{e}_r = \cos\theta\,\vec{i} + \sin\theta\,\vec{j} \qquad \vec{e}_\theta = -\sin\theta\,\vec{i} + \cos\theta\,\vec{j}$

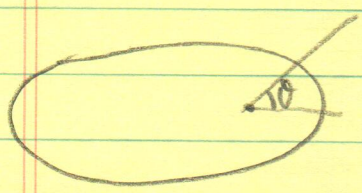
$$\therefore \begin{pmatrix}\vec{e}_r\\ \vec{e}_\theta\end{pmatrix} = \begin{pmatrix}\cos\theta & \sin\theta\\ -\sin\theta & \cos\theta\end{pmatrix}\begin{pmatrix}\vec{i}\\ \vec{j}\end{pmatrix} \Rightarrow$$

$$\vec{a} = (a_x, a_y)\begin{pmatrix}\vec{i}\\ \vec{j}\end{pmatrix} = (a_x\ a_y)\begin{pmatrix}\cos\theta & -\sin\theta\\ +\sin\theta & \cos\theta\end{pmatrix}\begin{pmatrix}\vec{e}_r\\ \vec{e}_\theta\end{pmatrix} = (\vec{a}_r\ \ \vec{a}_\theta)\begin{pmatrix}\vec{e}_r\\ \vec{e}_\theta\end{pmatrix}$$

$$\therefore \begin{pmatrix}a_r\\ a_\theta\end{pmatrix} = \begin{pmatrix}\cos\theta & \sin\theta\\ -\sin\theta & \cos\theta\end{pmatrix}\begin{pmatrix}a_x\\ a_y\end{pmatrix} = \begin{pmatrix}\cos\theta & \sin\theta\\ -\sin\theta & \cos\theta\end{pmatrix}\begin{pmatrix}(\ddot{r}-r\dot{\theta}^2)\cos\theta - 2\dot{r}\dot{\theta}\sin\theta\\ (\ddot{r}-r\dot{\theta}^2)\sin\theta + 2\dot{r}\dot{\theta}\cos\theta\end{pmatrix}$$

$$= \begin{pmatrix}\ddot{r}-r\dot{\theta}^2\\ r\ddot{\theta}+2\dot{r}\dot{\theta}\end{pmatrix}$$

1.8



$$r = \frac{a(1-e^2)}{1+e\cos\theta}, \quad \dot{r} = \frac{a(1-e^2)}{(1+e\cos\theta)^2}\dot{\theta}e\sin\theta$$

$$v_r = \dot{r} = \omega r\cdot\frac{re\sin\theta}{a(1-e^2)} = \omega r\cdot\frac{e\sin\theta}{1+e\cos\theta}$$

$$v_\theta = r\dot{\theta} = \omega r$$

$$\therefore v = \sqrt{v_r^2+v_\theta^2} = \omega r\left[1+\frac{e^2\sin^2\theta}{(1+e\cos\theta)^2}\right]^{\frac{1}{2}} = \frac{\omega r}{1+e\cos\theta}\left(1+e^2+2e\cos\theta\right)^{\frac{1}{2}}$$

$$a^2(1-e^2) = b^2 \quad \therefore e = \sqrt{1-\frac{b^2}{a^2}} = \frac{\sqrt{a^2-b^2}}{a}, \quad e\cos\theta = \frac{b^2}{ar}-1$$

$$\therefore v = \frac{\omega r^2}{a(1-e^2)}\cdot\left(1+\frac{a^2-b^2}{a^2}+2\frac{b^2}{ar}-2\right)^{\frac{1}{2}}$$

$$-\frac{b^2}{a^2}+\frac{2b^2}{ar}$$

$$= \frac{\omega r}{b}\cdot\sqrt{2ar-r^2} = \frac{\omega r}{b}\sqrt{(2a-r)r}$$

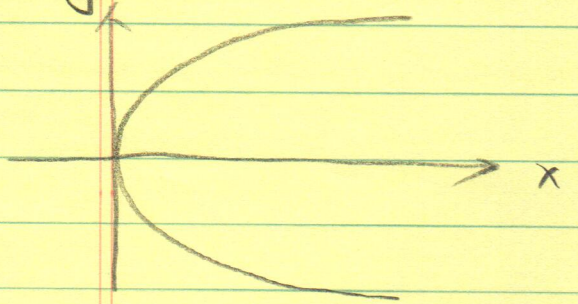
1.9. 取図省略

$\vec{v} = \dot{s}\vec{i}$

$\vec{a} = \frac{dv}{dt}\vec{i} + \frac{v^2}{\rho}\vec{j} = \frac{v^2}{\rho}\vec{j}$

$\therefore \vec{v}\cdot\vec{a} = 0$. 証明.

1.10



$y\,dy = \rho\,dx$

$\frac{dv}{dt} = -2k\frac{v^2}{\rho}$

$\therefore -\frac{dv}{v^2} = \frac{2k}{\rho}dt$

$\therefore \frac{1}{v} = \frac{2kt}{\rho} + C$, ~~$v = \frac{1}{C + 2kt/\rho}$~~

$\rho = -\frac{ds}{d\theta} = -\frac{ds}{dv}\cdot\frac{dv}{d\theta} =$

$a_r = \frac{dv}{dt}$

$a_\theta = \frac{v^2}{\rho}$

$\frac{dv}{dt} = \frac{dv}{ds}\cdot\frac{ds}{dt} = v\cdot\frac{dv}{ds}$.

$a_t = \frac{dv}{dt} = v\cdot\frac{dv}{ds} = -2k\,a_n = -2k\frac{v^2}{\rho} = 2kv^2\frac{d\theta}{ds}$

$\boxed{\frac{1}{\rho} = -\frac{d\theta}{ds}}$ $\therefore \frac{dv}{v} = 2k\,d\theta$

$\frac{dy}{dx} = \frac{\rho}{y}$

x が $x = \frac{\rho}{2}$, $y = \rho$. $\frac{dy}{dx} = 1$, $\theta = \frac{\pi}{4}$.

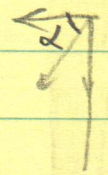
$\therefore (\ln v - \ln u) = 2k\left[\frac{\pi}{4} - \left(-\frac{\pi}{4}\right)\right]$

$\therefore v = u\,e^{k\pi}$.

1.11

$a_t = \frac{dv}{dt}$

$a_n = \frac{v^2}{r}$



$\therefore \tan\alpha = \frac{a_n}{a_t}$ 又 $\frac{dv}{dt} = \frac{v^2}{r}\cot\alpha$

$\therefore -\frac{dv}{v^2} = \frac{dt}{r}\cot\alpha$

$\therefore -\frac{1}{v} = C + \frac{t}{r}\cot\alpha$

$t=0,\ C = -\frac{1}{v_0}$

$\therefore \frac{1}{v} = \frac{1}{v_0} - \frac{t}{r}\cot\alpha$

1.12

$v = r\dot{\theta} \quad \therefore \quad d\theta = \frac{v}{r}dt$

$\therefore d\theta = \frac{dt}{\frac{r}{v_0} - t\cdot\cot\alpha}$ $\quad \theta - \theta_0 = -\frac{1}{\cot\alpha}\ln\left(\frac{r}{v_0} - t\cdot\cot\alpha\right) + C$

$\frac{r}{v} = \frac{r}{v_0} - t\cot\alpha = C\cdot e^{-\cot\alpha\,(\theta-\theta_0)}$

$\therefore v = \frac{r}{C}e^{\cot\alpha\cdot\theta}$

$\because \theta = 0,\ v = v_0 \quad \therefore \frac{r}{C} = v_0 \quad \therefore \boxed{v = v_0 e^{(\theta-\theta_0)\cot\alpha}}$

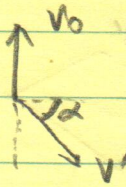
1.13. $V = V_0 + V'$.

A ⇌ B

① $V_0 = 0$, $V = V'$, $t_0 = \frac{2l}{V'}$

② 往返后有 $t = \frac{l}{V_0+V'} + \frac{l}{-V_0+V'} = \frac{2V'l}{V'^2-V_0^2} = \frac{2l}{V'} \cdot \frac{V'^2}{V'^2-V_0^2} = \frac{t_0}{1-V_0^2/V'^2}$

③



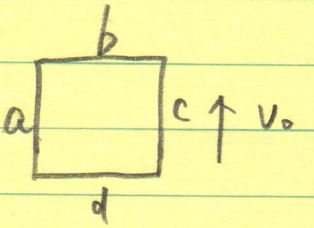
$V' \sin\alpha = V_0 \Rightarrow \alpha = \arcsin\frac{V_0}{V'}$

$\therefore \vec{V} = \vec{V_0} + \vec{V'} = V' \cos\alpha \cdot \vec{i}$

$= V' \sqrt{1-\sin^2\alpha}\,\vec{i} = V' \sqrt{1-V_0^2/V_1^2}\,\vec{i}$

$\therefore t = \frac{2l}{V} = \frac{2l}{V'} \cdot \frac{1}{\sqrt{1-V_0^2/V'^2}} = \frac{t_0}{\sqrt{1-V_0^2/V_1^2}}$

1.14.



$V_a = V_0 + V'$

$V_b = V' \sqrt{1-V_0^2/V_1^2}$

$V_c = V_0 - V'$

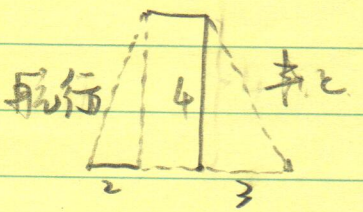
$V_d = V' \sqrt{1-V_0^2/V_1^2}$.

$\therefore t = \frac{t_0}{1-V_0^2/V_1^2} + \frac{t_0}{\sqrt{1-V_0^2/V_1^2}} = \frac{2l}{V'} \left(\frac{1}{1-V_1^2/V_1^2} + \frac{1}{\sqrt{1-V_0^2/V_1^2}} \right) \simeq 0.25$

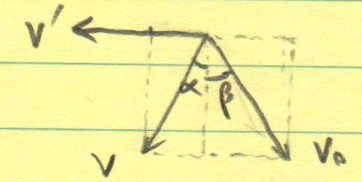
$\simeq 15.3$分

$\frac{1}{1-V_0^2/V_1^2} = \frac{1}{1-28^2/100^2} =$

1.15.



雨点行为相对静止观察后



$\tan\alpha = \frac{2}{4}$

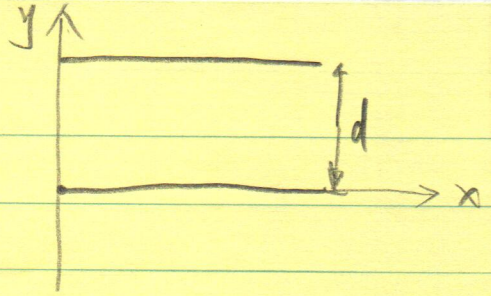
$\tan\beta = \frac{3}{4}$

静后：$V_0 \cos\beta = V \cos\alpha$

风后：$V' - V_0 \sin\beta = V \sin\alpha$

$\Rightarrow V' = V_0 \sin\beta + \frac{V_0 \cos\beta}{\cos\alpha} \sin\alpha = V_0 (\sin\beta + \cos\beta \cdot \tan\alpha) = V_0 \left(\frac{3}{5} + \frac{4}{5}\cdot\frac{2}{4}\right) = V_0 = 8\,m/s$

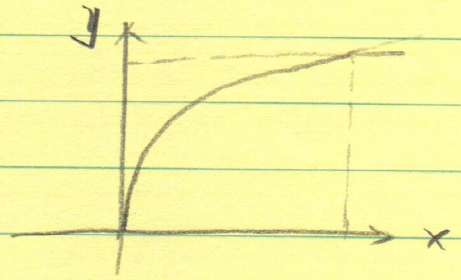
1.16



① $\vec{v} = \frac{2c}{d} \cdot y\,\vec{i} + v\,\vec{j}$ $\quad x \le \frac{d}{2}$

② $\begin{cases} \frac{dx}{dt} = \frac{2c}{d} \cdot y \\ \frac{dy}{dt} = v \end{cases} \Rightarrow \begin{cases} y = vt \\ x = \int_0^t \frac{2c}{d} \cdot vt'\,dt' = \frac{cv}{d}t^2 \end{cases}$

$\therefore x = \frac{c}{dv}y^2$ 抛物线.



② $\vec{v} = \frac{2c}{d}(d-y)\,\vec{i} + v\,\vec{j}$

$\frac{d}{2}/v = \frac{d}{2v}$

$\begin{cases} \frac{dx}{dt} = \frac{2c}{d}(d-y) \\ \frac{dy}{dt} = v \end{cases} \Rightarrow$

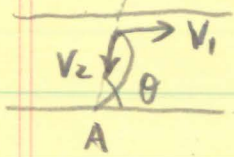
$y = vt$,

$x = \int \frac{2c}{d}(d-vt)\,dt = 2ct - \frac{cv}{d}t^2 + C_2$

$y = \frac{d}{2}$, $x = \frac{cd}{4v} \Rightarrow C_2 = \frac{-cd}{2v}$

$\therefore y = 2ct - \frac{cv}{d}t^2 - \frac{cd}{2v}$

1.17



地面系: $\begin{cases} V_r = \dot{r} = V_1\cos\theta - V_2 & ① \\ V_\theta = r\dot{\theta} = -V_1\sin\theta & ② \end{cases}$

<9A>

$\vec{V} = \vec{V}_1 + \vec{V}_2$.

①/②, 得

$$\frac{dr}{r d\theta} = -\cot\theta + \frac{V_2}{V_1}\frac{1}{\sin\theta}$$

$$\therefore \frac{dr}{r} = \underbrace{-\frac{\cos\theta}{\sin\theta}d\theta}_{I} + \underbrace{\frac{k}{\sin\theta}d\theta}_{II}$$

$$\int I = -\int\frac{d\sin\theta}{\sin\theta} = -\ln\sin\theta + C$$

$$\int II = \int\frac{k\sin\theta\,d\theta}{\sin^2\theta} = -k\int\frac{d\cos\theta}{1-\cos^2\theta} = -\frac{k}{2}\int\left(\frac{1}{1-\cos\theta} + \frac{1}{1+\cos\theta}\right)d\cos\theta$$

$$= -\frac{k}{2}\left[-\ln(1-\cos\theta) + \ln(1+\cos\theta)\right] + C'$$

$$= -\frac{k}{2}\ln\frac{1+\cos\theta}{1-\cos\theta} + C'$$

$$\therefore \ln r = -\ln\sin\theta\left(\frac{1+\cos\theta}{1-\cos\theta}\right)^{\frac{k}{2}} + C''$$

$$= -\ln 2\frac{\cos^{k+1}\frac{\theta}{2}}{\sin^{k+1}\frac{\theta}{2}} + C''$$

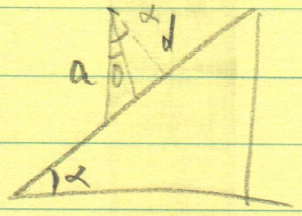
$$\therefore r = D\cdot\frac{\sin^{k+1}\frac{\theta}{2}}{\cos^{k+1}\frac{\theta}{2}} = D\frac{\sin^{k+1}\alpha}{\cos^{k+1}\alpha}$$

$$\frac{1+\cos\theta}{1-\cos\theta} = \left(\frac{\cos\frac{\theta}{2}}{\sin\frac{\theta}{2}}\right)^2$$
$$\sin\theta = 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}$$

代入初条件, $r_0 = D\cdot\frac{\sin^{k+1}\alpha_0}{\cos^{k+1}\alpha_0} \Rightarrow D = r_0\frac{\cos^{k+1}\alpha_0}{\sin^{k+1}\alpha_0}$

$$\therefore r = r_0\left(\frac{\sin\alpha}{\sin\alpha_0}\right)^{k+1}\cdot\left(\frac{\cos\alpha_0}{\cos\alpha}\right)^{k+1}$$

1.18.



$S=\frac{1}{2}at^2 \Rightarrow$ 여기서 $t=\sqrt{\frac{2S}{a}}$

여기서, $S=\frac{d}{\cos(\alpha-\theta)}$, $a=g\cos\theta$

$\therefore t^2=\frac{2S}{a}=\frac{2d}{g}\frac{1}{\cos(\alpha-\theta)\cos\theta}$

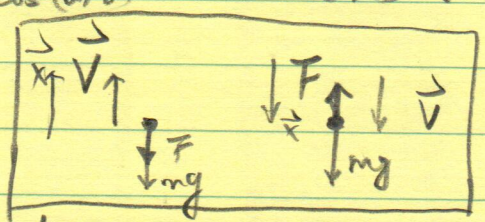
여기서, $\frac{dt^2}{d\theta}=0$ 을 구하면 $\frac{-\sin\theta\cos(\alpha-\theta)+\sin(\alpha-\theta)\cos\theta}{[\cos(\alpha-\theta)\cos\theta]^2}=\frac{\sin(\alpha-2\theta)}{[\quad]^2}$

$\therefore \alpha-2\theta=0 \Rightarrow \alpha=2\theta \quad \therefore \theta=\frac{\alpha}{2}$

$\therefore t=\sqrt{\frac{2d}{g}\cdot\frac{1}{\cos^2(\alpha/2)}}=\frac{1}{\cos(\alpha/2)}\sqrt{\frac{2d}{g}}$

1.19

$\vec{F}=m\vec{a}$



① 올라갈 때

$\frac{dV}{dt}=-g-k^2gV^2$

$\Rightarrow \frac{-dV}{1+k^2V^2}=-g\,dt$

양변 적분, $\frac{1}{k}\arctan(kV)=-gt+C'$

$\therefore kV=\tan(C-kgt)$

$\int\tan x\,dx=-\ln\cos x$

$t=0,\ V=0 \Rightarrow C=0 \Rightarrow kV=-\tan(kgt)$

$S(t)-S(t_0)=\int_t^0 V\,dt \Rightarrow S(t)=S(t_0)+\int_0^t V\,dt=\int_0^t\frac{1}{k}\tan(kgt)\,dt=\frac{1}{k^2g}\ln\cos(kgt)$

최고점에서, $kV_0=-\tan(kgt_0) \Rightarrow \cos kgt_0=\frac{1}{\sqrt{1+\tan^2}}=\frac{1}{\sqrt{1+k^2V_0^2}}$

$\therefore S(t_0) = \frac{1}{k^2 g} \ln \sqrt{1+k^2 v_0^2}$

② 下降 $\frac{dv}{dt} = g - k^2 g v^2 \Rightarrow \frac{dv}{1-k^2v^2} = +g\,dt$

$\Rightarrow \frac{1}{k}\operatorname{arctanh}(kv) = +gt + C'$

$\Rightarrow kv = +\tanh(kgt + c)$

$t=0, v=v_0 \Rightarrow kv = \tanh(kgt)$ $\quad \int \tanh t\,dt = \log(\cosh t)$

$\therefore S'(t) = \int_0^t v\,dt = \frac{1}{k}\int_0^t \tanh(kgt)\,dt = \frac{1}{k^2 g}\ln\cosh(kgt)$

$1 - \tanh^2 x = \frac{1}{\cosh^2 x}$

$S'(t) = \frac{1}{k^2 g}\ln\frac{1}{\sqrt{1-k^2v^2}}$

③ 上抛与下落至抛回的过程，有 $S'(v) = S(v_0)$

$\Rightarrow \sqrt{1+k^2v_0^2} = \frac{1}{\sqrt{1-k^2v^2}}$

$\Rightarrow 1 + k^2 v_0^2 - (1+k^2v_0^2)\,k^2v^2 = 1$

$\Rightarrow v^2 = \frac{v_0^2}{1+k^2v_0^2} \Rightarrow v = \frac{v_0}{\sqrt{1+k^2v_0^2}}$

解法二：直接求 $S(v)$

上升：$v = \frac{ds}{dt} = \frac{ds}{dv}\frac{dv}{dt} = \frac{ds}{dv}(-g - k^2 g v^2)$

$\Rightarrow ds = -\frac{1}{g}\frac{v\,dv}{1+k^2v^2} \Rightarrow S = -\frac{1}{2gk^2}\ln(1+k^2v^2)$，$v=0$ $S=0$

>0

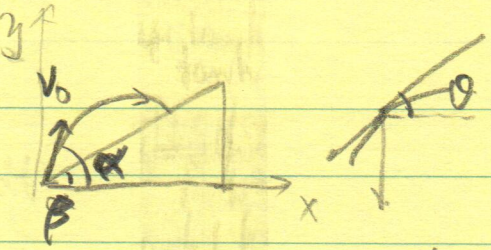
下降：$v = \frac{ds}{dt} = \frac{ds}{dv}\frac{dv}{dt} = \frac{ds}{dv}(g - k^2 g v^2)$

$\Rightarrow ds = \frac{1}{g}\frac{v\,dv}{1+k^2v^2} \Rightarrow S = \frac{-1}{2gk^2}\ln(1-k^2v^2)$

<0

两种都有 $1+k^2v^2 = \frac{1}{1-k^2v^2}$

1.20 ① 在直角...



$$\begin{cases} m\dot{V}_x = 0 \\ m\dot{V}_y = mg \end{cases} \Rightarrow \begin{cases} V_x = V_0\cos\alpha \\ V_y = V_0\sin\alpha - gt \end{cases} \Rightarrow \begin{cases} x = V_0 t\cos\alpha \\ y = V_0 t\sin\alpha - \frac{1}{2}gt^2 \end{cases}$$

落回斜面, $\theta = \beta$, 则 $\tan\beta = \frac{y}{x} = \frac{V_0\sin\alpha - \frac{1}{2}gt}{V_0\cos\alpha}$

$$\Rightarrow t = \frac{2V_0}{g}(\sin\alpha - \cos\alpha\tan\beta)$$

$$l^2 = x^2 + y^2 = V_0^2t^2 + \frac{1}{4}g^2t^4 - V_0\sin\alpha\, gt^3$$

$$= t^2\cdot\left[V_0^2 + \frac{1}{4}\cdot 4V_0^2(\sin\alpha - \cos\alpha\tan\beta)^2 - V_0\frac{2V_0}{g}(\sin\alpha - \cos\alpha\tan\beta)\sin\alpha\right]$$

$$= V_0^2t^2\left[1 + (\sin\alpha - \cos\alpha\tan\beta)^2 - 2(\sin\alpha - \cos\alpha\tan\beta)\sin\alpha\right]$$

$$= V_0^2t^2\left[1 + \sin^2\alpha + \cos^2\alpha\tan^2\beta - 2\sin\alpha\cos\alpha\tan\beta - 2\sin^2\alpha + 2\sin\alpha\cos\alpha\tan\beta\right]$$

$$= V_0^2t^2\left[1 + \tan^2\beta\right]\cdot\cos^2\alpha$$

$$= \frac{4V_0^4}{g^2}(\sin\alpha - \cos\alpha\tan\beta)^2\frac{\cos^2\alpha}{\cos^2\beta}$$

$$= \frac{4V_0^4}{g^2}(\sin^2\alpha + \cos^2\alpha\tan^2\beta - 2\sin\alpha\cos\alpha\tan\beta)\cdot\frac{\cos^2\alpha}{\cos^2\beta}$$

$$= \frac{4V_0^4}{g^2}\cdot\frac{\sin^2(\alpha-\beta)\cos^2\alpha}{\cos^4\beta}$$

$$\therefore l = \frac{2V_0^2}{g}\cdot\frac{\sin(\alpha-\beta)\cos\alpha}{\cos^2\beta}$$

$\beta > \alpha$

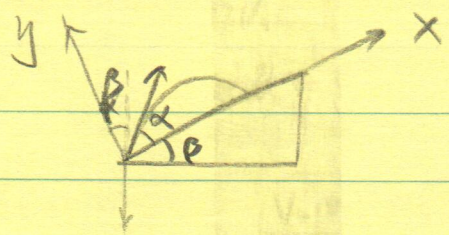
$\alpha < \alpha < \pi$

$$\frac{dl}{d\alpha} = 0 \Rightarrow \cos(\alpha-\beta)\cos\alpha - \sin(\alpha-\beta)\sin\alpha = \cos(2\alpha-\beta) = 0$$

$$\Rightarrow 2\alpha - \beta = \frac{\pi}{2} \Rightarrow \alpha = \frac{\pi}{4} + \frac{\beta}{2}$$

$$l_{max} = \frac{2V_0^2}{g}\cdot\frac{\sin\left(\frac{\pi}{4} - \frac{\beta}{2}\right)\cos\left(\frac{\pi}{4} + \frac{\beta}{2}\right)}{\cos^2\beta} = \frac{V_0^2}{g}\cdot\frac{1-\sin\beta}{\cos^2\beta} = \frac{V_0^2}{g(1+\sin\beta)}$$

② 建系，设初状：



$$\begin{cases} \dot{v}_x = -g\sin\beta \\ \dot{v}_y = -g\cos\beta \end{cases} \Rightarrow \begin{cases} v_x = v_0\cos\alpha - gt\sin\beta \\ v_y = v_0\sin\alpha - gt\cos\beta \end{cases} \begin{cases} x = v_0t\cos\alpha - \frac{1}{2}gt^2\sin\beta \\ y = v_0t\sin\alpha - \frac{1}{2}gt^2\cos\beta \end{cases}$$

落回斜面 $\Rightarrow y=0 \Rightarrow t = \dfrac{2v_0\sin\alpha}{g\cos\beta}$

$$\therefore l = x = \frac{2v_0^2\sin\alpha\cos\alpha}{g\cos\beta} - \frac{g\sin\beta}{2}\cdot\frac{4v_0^2\sin^2\alpha}{g^2\cos^2\beta}$$

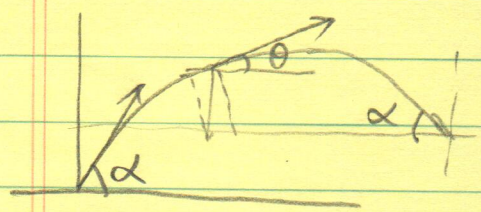
$$= \frac{2v_0^2\sin\alpha}{g\cos^2\beta}\left(\cos\alpha\cos\beta - \sin\alpha\sin\beta\right) = \frac{2v_0^2\sin\alpha\cos(\alpha+\beta)}{g\cos^2\beta}$$

l_{max} 对 α 求导 $\dfrac{dl}{d\alpha}=0 \Rightarrow \cos\alpha\cos(\alpha+\beta) - \sin\alpha\sin(\alpha+\beta) = \cos(2\alpha+\beta) = 0$

$$\Rightarrow 2\alpha+\beta = \frac{\pi}{2} \Rightarrow \alpha = \frac{\pi}{4} - \frac{\beta}{2}$$

$$\Rightarrow l_{max} = \frac{2v_0^2\sin(\frac{\pi}{4}-\frac{\beta}{2})\cos(\frac{\pi}{4}+\frac{\beta}{2})}{g\cos^2\beta} = \frac{v_0^2}{g}\cdot\frac{1-\sin\beta}{\cos^2\beta} = \frac{v_0^2}{g(1+\sin\beta)}$$

1-21



① 利用极坐标：

$$\begin{cases} m\dot{v} = -mg\sin\theta - mkv \\ m\dfrac{v^2}{\rho} = mg\cos\theta \end{cases}$$

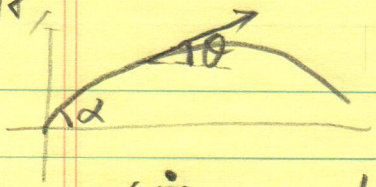
$$\dot{v} = \frac{dv}{d\theta}\cdot\frac{d\theta}{dt} = \frac{dv}{d\theta}\cdot\dot{\theta}$$

$$\rho = \frac{ds}{d\theta} = \frac{ds}{dt}\cdot\frac{dt}{d\theta} = v\cdot\frac{1}{\dot{\theta}}$$

$$\therefore mv\dot{\theta} = mg\cos\theta \Rightarrow \dot{\theta} = \frac{g\cos\theta}{v} \Rightarrow \frac{dt}{d\theta} = \frac{v}{g\cos\theta}$$

$$\therefore \frac{dv}{d\theta}\cdot\frac{g\cos\theta}{v} = -g\sin\theta - kv$$

② 直射



$$\begin{cases} m\dot{V}_x = -mkV_x \\ m\dot{V}_y = -mg - mkV_y \end{cases} \Rightarrow \begin{cases} V_x = C_1 e^{-kt} \\ V_y = C_2 e^{-kt} - \frac{g}{k} \end{cases}$$

$$\frac{dV_y}{V_y + \frac{g}{k}} = -k\,dt$$

$$V_y + \frac{g}{k} = C_2 e^{-kt}$$

$$\boxed{C_2 - \frac{g}{k} = V_0 \sin\alpha}$$

$$\Rightarrow \begin{cases} V_x = V_0 \cos\alpha\, e^{-kt} \\ V_y = V_0 \sin\alpha\, e^{-kt} + \frac{g}{k}(e^{-kt} - 1) \end{cases}$$

$$V_y = -V_x \cdot \tan\alpha$$

$$\Rightarrow V_0 \sin\alpha + \frac{g}{k}(1 - e^{kt}) = -V_0 \cos\alpha \cdot \tan\alpha$$

$$\Rightarrow \frac{2kV_0 \sin\alpha}{g} = 1 - e^{kt}$$

$$\Rightarrow t = \frac{1}{k}\ln\left(1 + \frac{2kV_0\sin\alpha}{g}\right)$$

1-22.

$$\vec{E} = E\vec{j}, \quad \vec{B} = B\vec{k}$$

$$m\vec{\dot{V}} = q(\vec{E} + \vec{V}\times\vec{B}), \quad \vec{V}\times\vec{B} = (V_x\vec{i} + V_y\vec{j} + V_z\vec{k})\times B\vec{k} = V_y B\vec{i} - V_x B\vec{j}$$

$$\begin{cases} \dot{V}_x = \frac{q}{m} V_y B \\ \dot{V}_y = \frac{q}{m}(E - V_x B) \\ \dot{V}_z = 0 \end{cases} \Rightarrow \ddot{V}_y = -\frac{q}{m} B \cdot \dot{V}_x = -\frac{q^2B^2}{m^2} V_y$$

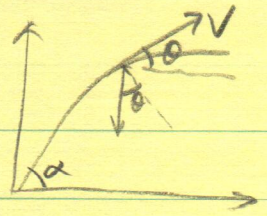
$$\Rightarrow V_y = C_1 \cos\frac{qB}{m}t + C_2 \sin\frac{qB}{m}t, \quad V_z = C = 0$$

$$V_x = \frac{1}{B}\left(E - \frac{m\dot{V}_y}{q}\right) = \frac{1}{B}\left(E - C_2 B \cdot \cos\frac{qBt}{m}\right)$$

当 $t=0$, 有 $V_x = V \Rightarrow \frac{1}{B}(E - C_2 B) = V \Rightarrow C_2 = \frac{E}{B} - V$.

$$\therefore x = \int V\,dt = \frac{E}{B}t - (E - VB)\frac{m}{qB^2}\sin\frac{qBt}{m} \qquad y = \frac{m}{qB^2}(E - VB)\left(1 - \cos\frac{qBt}{m}\right)$$

1.21. 第1/2：采用自然坐标系.



$$\begin{cases} m\dot{v} = -mkv - mg\sin\theta \\ m\frac{v^2}{\rho} = mg\cdot\cos\theta \end{cases}$$

$$\dot{v} = \frac{dv}{dt} = \frac{dv}{ds}\frac{ds}{dt} = v\cdot\frac{dv}{ds}, \quad -\frac{1}{\rho} = \frac{d\theta}{ds}$$

$$\begin{cases} v\frac{dv}{ds} = -kv - g\sin\theta \\ v^2\frac{d\theta}{ds} = -g\cos\theta \end{cases} \Rightarrow \frac{1}{v}\frac{dv}{d\theta} = \frac{kv + g\sin\theta}{g\cdot\cos\theta}$$

即有 $\frac{g}{v} = -k\sin\theta + C_1 g\cos\theta$

$\theta = \alpha$ 时, $v = v_0 \Rightarrow \frac{g}{v_0} + k\sin\alpha = C_1\cdot g\cos\alpha$

$$\Rightarrow C_1 = \frac{1}{v_0\cos\alpha} + \frac{k}{g}\tan\alpha$$

$$\therefore \frac{1}{v} = -\frac{k}{g}\sin\theta + \left(\frac{1}{v_0\cos\alpha} + \frac{k}{g}\tan\alpha\right)\cdot\cos\theta$$

$$\therefore \frac{dt}{d\theta} = \frac{dt}{ds}\frac{ds}{d\theta} = \frac{-1}{v}\rho = -\frac{v}{g\cos\theta}$$

$$= -\frac{1}{-k\sin\theta\cos\theta + \left(\frac{g}{v_0}\frac{1}{\cos\alpha} + k\tan\alpha\right)\cos^2\theta}$$

$$= \frac{1}{k\sin\theta\cos\theta - \left(\frac{g}{v_0}\frac{1}{\cos\alpha} + k\tan\alpha\right)\cos^2\theta} = \frac{1}{a_1\sin\theta\cos\theta - a_2\cos^2\theta}$$

$$t = \int\frac{dt}{d\theta}d\theta + C = \int\frac{d\theta}{a_1\sin\theta\cos\theta - a_2\cos^2\theta} + C = \frac{1}{a_1}\ln\left(a_2 - a_1\tan\theta\right) + C$$

$\theta = \alpha$, 有 $t = \infty$, $\Rightarrow 0 = \frac{1}{k}\ln\left(\frac{g}{v_0}\frac{1}{\cos\alpha} + k\tan\alpha - k\tan\alpha\right) + C$

$$\therefore C = \frac{1}{k}\ln\frac{v_0\cos\alpha}{g}$$

$\theta = -\alpha$, $t = \frac{1}{k}\ln\left(\frac{g}{v_0}\frac{1}{\cos\alpha} + k\tan\alpha + k\tan\alpha\right)_{\wedge}^{+C} = \frac{1}{k}\ln\frac{g + 2kv_0\sin\alpha}{v_0\cos\alpha} + C$

$$= \frac{1}{k}\ln\left(1 + \frac{2kv_0\sin\alpha}{g}\right)$$

1-23. ① B=0, 有

$$\begin{cases} \dot{v}_x = 0 \\ \dot{v}_y = \frac{qE}{m} \\ \dot{v}_z = 0 \end{cases} \Rightarrow \begin{cases} x = vt \\ y = \frac{1}{2}\frac{qE}{m}t^2 \end{cases}$$

$$\Rightarrow \boxed{y = \frac{qE}{2mv^2}x^2}$$

抛物线.

② $\begin{cases} \dot{v}_x = \frac{q}{m}v_y B \\ \dot{v}_y = -\frac{q}{m}v_x B \end{cases} \Rightarrow \begin{cases} x = \frac{mv}{qB}\sin\frac{qBt}{m} \\ y = -\frac{mv}{qB}(1-\cos\frac{qBt}{m}) \end{cases}$

$$\Rightarrow x^2 + \left(y + \frac{mv}{qB}\right)^2 = \left(\frac{mv}{qB}\right)^2.$$ 圆.

1-24. 对2m:

$$\begin{cases} 2m(\ddot{x}_1 - \ddot{x}) = 2mg - T \\ m\ddot{x} = mg + \frac{mg}{a}(x_1 - x - a) - T \\ m\ddot{x}_1 = mg - \frac{mg}{a}(x_1 - x - a) = mg\left(2 - \frac{x_1 - x}{a}\right) \end{cases}$$

2m ⋮ m ⋮ m

$$\begin{cases} m\ddot{x} = \frac{T}{2} - mg \\ m\ddot{x}_1 = mg\left(\frac{x_1 - x}{a}\right) - T \\ m\ddot{x}_1 = mg\left(2 - \frac{x_1 - x}{a}\right) \end{cases}$$

$$T = \frac{2}{3}mg\left(1 + \frac{x_1 - x}{a}\right)$$

$$\begin{cases} \ddot{x} = g\left(-\frac{2}{3} + \frac{x_1 - x}{3a}\right) \\ \ddot{x}_1 = g\left(2 - \frac{x_1 - x}{a}\right) \end{cases}$$

$t=0$, 有 $x_1 - x = a$.

$$\therefore \ddot{x}_1 - \ddot{x} = \frac{4}{3}g\left(2 - \frac{x_1 - x}{a}\right) \qquad \xi = x_1 - x$$

$$\ddot{\xi} + \frac{4g}{3a}\xi - \frac{8g}{3} = 0$$

令 $\xi = \tilde{\xi} + 2a$, 则 $\ddot{\tilde{\xi}} + \frac{4g}{3a}\tilde{\xi} = 0$

$$\therefore \xi = 2a + C_1\cos 2\sqrt{\frac{g}{3a}}t + C_2\sin 2\sqrt{\frac{g}{3a}}t$$

$t=0$, $\xi = a$, 有 $C_1 = -a$; $t=0$, $\dot{\xi} = 0$, 有 $C_2 = 0$. $\therefore \xi = a\left(2 - \cos 2\sqrt{\frac{g}{3a}}t\right)$

$$\therefore \ddot{x} = -\frac{g}{3}\cos 2\sqrt{\frac{g}{3a}}t, \quad \dot{x} = -\frac{1}{2}\sqrt{\frac{ag}{3}}\sin 2\sqrt{\frac{g}{3a}}t, \quad x = \frac{a}{4}\left(\cos 2\sqrt{\frac{g}{3a}}t - 1\right)$$

$$\therefore x_1 = \xi + x = \frac{7a}{4} - \frac{3a}{4}\cos 2\sqrt{\frac{g}{3a}}t, \text{ 周期 } \tau = \frac{2\pi}{\omega} = \sqrt{\frac{3a}{g}}$$

$$T=\frac{2}{3}mg\left(1+\frac{\xi}{a}\right)=\frac{2}{3}mg\left(3-\cos 2\sqrt{\frac{g}{3a}}t\right)=2mg\left(1-\frac{1}{3}\cos 2\sqrt{\frac{g}{3a}}t\right)$$

$$T'=mg\left(\frac{x_1-x}{a}+1\right)=mg\left(2-\cos 2\sqrt{\frac{g}{3a}}t+1\right)=mg\left(1-\cos 2\sqrt{\frac{g}{3a}}t\right)$$

1.25. 机械能守恒. $\frac{1}{2}k(x^2-\lambda_0^2)=mg(x-\lambda_0)+\frac{1}{2}mv_0^2$

$k\lambda_0=mg$

$\Rightarrow k=\frac{mg}{\lambda_0}$

$$\frac{1}{2}\frac{mg}{\lambda_0}(x^2-\lambda_0^2)=mg(x-\lambda_0)+\frac{1}{2}mv_0^2$$

$$x^2-\lambda_0^2=2\lambda_0 x-2\lambda_0^2+\frac{\lambda_0}{g}v_0^2$$

$$\therefore x^2-2\lambda_0 x+\lambda_0^2-\frac{\lambda_0}{g}v_0^2=0$$

$$\Delta=4\lambda_0^2-4\lambda_0^2+\frac{4\lambda_0}{g}v_0^2=\frac{4\lambda_0}{g}v_0^2$$

$$x=\frac{1}{2}\left(2\lambda_0\pm\sqrt{\frac{\lambda_0}{g}}\cdot 2v_0\right)$$

$$=\lambda_0+v_0\sqrt{\frac{\lambda_0}{g}}$$

$$T=kx=mg\left(1+\frac{v_0}{\sqrt{g\lambda_0}}\right)$$

1.26. $\begin{cases} kb=mg \\ kc=m'g \end{cases}$

$m\ddot{x}=mg-k(x-a)$

$x=\xi+\frac{mg}{k}+a$

$b+a$

$$\therefore x=C_1\cos\sqrt{\frac{k}{m}}t+C_2\sin\sqrt{\frac{k}{m}}t+\frac{mg}{k}+a$$

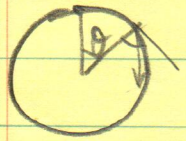
$\dot{x}|_{t=0}=0\Rightarrow C_2=0$

$x|_{t=0}=a+b+C\Rightarrow C_1+b+a=a+b+C$

$\Rightarrow C_1=C-b-a$

$$\therefore x=a+b+c\cos\sqrt{\frac{g}{k}}t$$

1.27.



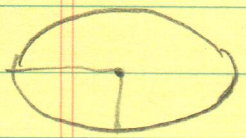
$\frac{1}{2}mv^2 = mgr(1-\cos\theta)$

$mg\cos\theta = \frac{mv^2}{r} = 2mg(1-\cos\theta)$

$\Rightarrow \cos\theta = 2-2\cos\theta \Rightarrow \cos\theta = \frac{2}{3}$

$\therefore \theta = \arccos\frac{2}{3}$

1.28.



$\begin{cases} x = a\cos t \\ y = b\sin t \end{cases}$

$\theta = \arctan\frac{y}{x} = \arctan\frac{b\sin t}{a\cos t}$

$\therefore \frac{d\theta}{dt} = \frac{1}{1+\frac{b^2\sin^2 t}{a^2\cos^2 t}} \cdot \frac{b(\cos^2 t+\sin^2 t)}{a\cos^2 t} = \frac{b}{a}\cdot\frac{1}{\cos^2 t+\frac{b^2}{a^2}\sin^2 t}$

$t=\frac{\pi}{2}$ 时 $\dot\theta = \frac{a}{b}$

$\frac{ds}{dt} = \sqrt{a^2\sin^2 t + b^2\cos^2 t} \to a$

$t=0$ 时 $\dot\theta = \frac{b}{a}$, $ds = b$, $\frac{ds}{d\theta} = a$

$\therefore \frac{ds}{d\theta} = \left.\frac{ds/dt}{d\theta/dt}\right|_{t=\frac{\pi}{2}} = \frac{a}{a/b} = b$ ✗ 因为 θ 到 $\frac{\pi}{2}$ 时速度不为 0

$\therefore \frac{1}{2}mv^2 = mgb$ $\therefore F = mg + \frac{mv^2}{\rho} = mg + \frac{2mgb}{b} = 3mg$ ✗

$\frac{x^2}{a^2}+\frac{y^2}{b^2} = 1 \Rightarrow y = \frac{b}{a}\sqrt{a^2-x^2}$

$\therefore y' = \frac{b}{a}\frac{x}{\sqrt{a^2-x^2}}$, $y'' = \frac{b}{a}\frac{\sqrt{a^2-x^2}+\frac{x^2}{\sqrt{a^2-x^2}}}{a^2-x^2} = \frac{ab}{(a^2-x^2)^{3/2}}$

$\therefore 1+y'^2 = \frac{b^2x^2+(a^2-x^2)a^2}{(a^2-x^2)a^2}$ $\left.y''\right|_0 = \frac{b}{a^2}$, $y'=0$

$\therefore \rho = \frac{a^2}{b}$ $\therefore F = mg + \frac{mv^2}{\rho} = mg + \frac{2mgb}{a^2/b} = mg\left(1+\frac{2b^2}{a^2}\right)$

1.29 $y = y_{max} \Rightarrow y_{20}$

$$\therefore \quad mv^2 = 2mga(1+\cos 2\theta) = 4mga\cdot\cos^2\theta$$

$$y' = \frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{2a\cdot\sin 2\theta}{2a(1+\cos 2\theta)} = \tan\theta$$

$$y'' = \frac{d}{dx}(y') = \left(\frac{d}{d\theta}y'\right)\frac{d\theta}{dx} = \frac{1}{\cos^2\theta}\cdot\frac{1}{2a(1+\cos 2\theta)} = \frac{1}{4a\cdot\cos^4\theta}$$

$$1+y'^2 = \frac{1}{\cos^2\theta}$$

$$\therefore \quad \frac{1}{\rho} = \frac{y''}{(1+y'^2)^{3/2}} = \frac{1}{4a\cdot\cos^4\theta}\cdot\cos^3\theta = \frac{1}{4a\cos\theta}$$

$$mg\cos\theta - N = \frac{mv^2}{\rho}$$

$$\therefore \quad N = mg\cdot\cos\theta + mg\cos\theta = 2mg\cos\theta$$

1.30. $\angle\theta = 0 \Rightarrow \theta = \frac{\pi}{2}$.

$$ds = 4a\cdot\cos\theta\, d\theta$$

$$\text{접선}: \quad \dot{v} = g\sin\theta - \mu N$$

$$\text{법선}: \quad mg\cos\theta - N = \frac{mv^2}{\rho}$$

$$ds = v\cdot dt$$

$$\therefore \quad dt = \frac{ds}{v} = \frac{4a\cos\theta\, d\theta}{v}$$

$$\dot{v} = g\sin\theta - \mu\, g\cos\theta + \mu\cdot\frac{v^2}{\rho} \qquad \frac{\mu mv^2}{4a\cdot\cos\theta}$$

$$\therefore \quad \frac{dv}{dt} = \frac{v\,dv}{4a\cos\theta\, d\theta}$$

$$\therefore \quad v\,dv = \left(4ag\sin\theta\cos\theta - 4\mu ag\cos^2\theta + \mu mv^2\right)d\theta$$

$$\frac{1}{2}dv^2 - \mu\, v^2 d\theta = \underline{4ag\cos\theta(\sin\theta - \mu\cos\theta)\,d\theta} \qquad f(\theta)$$

$v^2 = \xi$,

$$\xi' - 2\mu\xi + 2f(\theta) = 0$$

运用常数变易法：$f(\theta)=0$，有 $\xi_0=Ce^{-2\mu\theta} \Rightarrow \xi=C(\theta)e^{-2\mu\theta}$

$\therefore C'e^{-2\mu\theta}+Ce^{-2\mu\theta}\cdot(-2\mu)-2\mu\xi+2f(\theta)=0$

$\therefore C'=-2f(\theta)e^{2\mu\theta}$

$\therefore C=\int -2f(\theta)e^{2\mu\theta}\,d\theta$

$=8ga\int\cos\theta(\sin\theta-\mu\cos\theta)e^{2\mu\theta}\,d\theta$

$=4ga\int[\sin2\theta-\mu(1+\cos2\theta)]e^{2\mu\theta}\,d\theta$

$=4ga\int\left[-\frac{i}{2}(e^{i2\theta}-e^{-i2\theta})-\mu-\frac{\mu}{2}(e^{i2\theta}+e^{-i2\theta})\right]e^{2\mu\theta}\,d\theta$

$=2ga\int\left[(-i-\mu)e^{2(i-\mu)\theta}+(i-\mu)e^{2(-i-\mu)\theta}-2\mu e^{-2\mu\theta}\right]d\theta$

$=2ga\left[\frac{-i-\mu}{2(i-\mu)}e^{2(i-\mu)\theta}+\frac{i-\mu}{2(-i-\mu)}e^{2(-i-\mu)\theta}-\frac{2\mu}{-2\mu}e^{-2\mu\theta}\right]$

$=ga\left[\frac{\mu+i}{\mu-i}e^{-2(\mu-i)\theta}+\frac{\mu-i}{\mu+i}e^{-2(\mu+i)\theta}+2e^{-2\mu\theta}\right]$

$=ga\,\frac{e^{-2\mu\theta}}{\mu^2+1}\Big((\mu^2-1+2\mu i)(\cos2\theta+i\sin2\theta)+(\mu^2-1-2\mu i)(\cos2\theta-i\sin2\theta)+2\mu^2+2\Big)$

$=\frac{ga}{1+\mu^2}e^{-2\mu\theta}\left(2(\mu^2-1)\cos2\theta-4\mu\sin2\theta+2\mu^2+2\right)+C'$

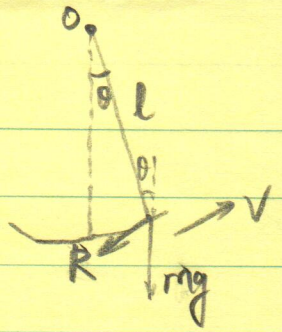
$v^2=C\cdot e^{2\mu\theta}$

$\theta=0,\ v=0 \Rightarrow \frac{ga}{1+\mu^2}(2\mu^2-2+2\mu^2+2)+C'=0 \Rightarrow C'=-\frac{4\mu^2}{1+\mu^2}ga$

$\theta=\frac{\pi}{2},\ v=0 \Rightarrow \frac{ga}{1+\mu^2}(-2\mu^2+2+2\mu^2+2)-\frac{4\mu^2}{1+\mu^2}ga\cdot e^{\pi\mu}=0$

$\Rightarrow \mu^2e^{\pi\mu}=1$

1.31



牛顿运动方程：

$$ml\ddot{\theta} = -mg\sin\theta - 2mkl\dot{\theta}$$

即 $\ddot{\theta} + 2k\dot{\theta} + \frac{g}{l}\sin\theta = 0$

θ 很小，$\sin\theta \approx \theta$

$$\therefore \ddot{\theta} + 2k\dot{\theta} + \frac{g}{l}\theta = 0$$

$$r^2 + 2kr + \frac{g}{l} = 0$$

$$r_{1,2} = \frac{1}{2}\left(-2k \pm \sqrt{4k^2 - 4\frac{g}{l}}\right) = -k\left(1 \mp \sqrt{1 - \frac{g}{k^2 l}}\right)$$

$k^2 < \frac{g}{l}$，即 $\frac{g}{k^2 l} > 1$，令 $\sqrt{1 - \frac{g}{k^2 l}} = i\sqrt{\frac{g}{k^2 l} - 1} = i\beta$

则 $r_{1,2} = -k(1 \mp i\beta)$

$$\therefore \theta = C_1 e^{-kt - ik\beta t} + C_2 e^{-kt + ik\beta t}$$

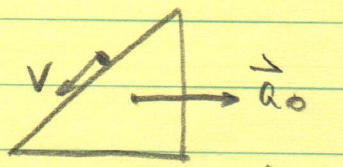
$t = 0,\ \theta = 0 \Rightarrow C_1 = -C_2$

$$\therefore \theta = C_1 e^{-kt}(-2i)\sin k\beta t$$

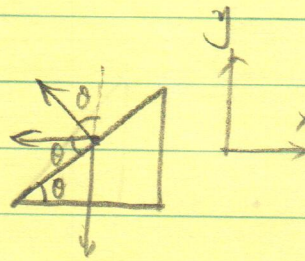
$$= \tilde{C}_1 e^{-kt}\sin k\beta t$$

$$\therefore T = \frac{2\pi}{k\beta} = 2\pi\frac{1}{\sqrt{\frac{g}{l} - k^2}} = 2\pi\sqrt{\frac{l}{g - k^2 l}}$$ 为伪周期

1.32.



$$\vec{a} = \vec{a}' + \vec{a}_0$$



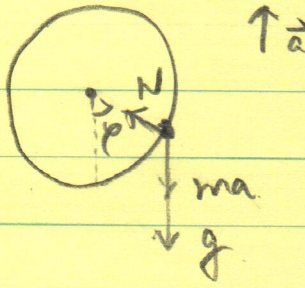
取楔子为参考系，质点受惯性力 $\vec{F} = -m\vec{a}_0 = \mp ma_0\vec{i}$

先讨论相对运动，加上惯性力的作用。

$$\therefore ma' = mg\sin\theta \pm ma_0\cos\theta$$ （"+"对应 $\vec{a}_0$ 向右）

垂直斜面方向，有 $F\sin\theta + N = mg\cos\theta$ $\therefore N = m(g\cos\theta \mp a_0\sin\theta)$ （"-"对应 $\vec{a}_0$ 向右）

1.33.



↑ $\vec{a}$

由动能定理:

$$m(g\pm a)r(\cos\varphi-\cos\varphi_0)=\frac{1}{2}mv_r^2-\frac{1}{2}mv_{r0}^2$$

$$\therefore v_r=\sqrt{v_{r0}^2+2(g\pm a)(\cos\varphi-\cos\varphi_0)r}.$$

径向列方程: $N-m(g\pm a)\cos\varphi=m\frac{v_r^2}{r}$

$$\therefore N=m(g\pm a)\cos\varphi+\frac{mv_{r0}^2}{r}+2m(g\pm a)(\cos\varphi-\cos\varphi_0)$$

$$=\frac{mv_0^2}{r}+m(g\pm a)(3\cos\varphi-2\cos\varphi_0)$$

1.34

$k=F\cdot v \quad F=\frac{k}{v}$

① $m\dot{v}=\frac{k}{v}-f$

$$\therefore \frac{mv\,dv}{k-fv}=dt$$

$$\therefore t=\frac{m}{f}\int_{v_0}^{v}\frac{v\,dv}{\frac{k}{f}-v}=-\frac{m}{f}\left[v+\frac{k}{f}\ln\left(\frac{k}{f}-v\right)\right]_{v_0}^{v}$$

$$=\frac{mk}{f^2}\ln\frac{k-v_0f}{k-vf}-\frac{m(v-v_0)}{f}$$

$\int\frac{x}{a-x}dx=-x-a\ln(a-x)$

$\int_0^x\frac{\xi}{a-\xi}d\xi=\int_0^{x-a}\frac{\xi}{a-\xi}d(\xi-a)$

$=\frac{1}{2}\int_{a^2}^{(x-a)^2}\frac{d(\xi-a)^2}{a-\xi}=\frac{1}{2}\int_{a^2}^{(x-a)^2}\frac{d\xi}{\xi-a}$

$=\xi\frac{1}{2}\Big|_{a^2}^{(x-a)^2}$

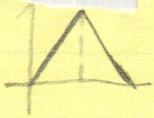
$=a-x-a=-x$

② $m\dot{v}=\frac{k}{v}-\alpha v$

$$\Rightarrow\frac{mv\,dv}{k-\alpha v^2}=dt$$

$$t=\frac{m}{\alpha}\int\frac{v\,dv}{\frac{k}{\alpha}-v^2}=\frac{m}{2\alpha}\int\frac{dv^2}{\frac{k}{\alpha}-v^2}=-\frac{m}{\alpha}\ln\left(\frac{k}{\alpha}-v^2\right)\Big|_{v_0}^{v}=\frac{m}{2\alpha}\ln\frac{k-\alpha v_0^2}{k-\alpha v^2}$$

$$=\frac{mv}{2f}\ln\frac{kv-fv_0^2}{kv-fv^2}$$

1.35. $F = \begin{cases} \frac{2t}{\tau} P \\ (2 - \frac{2t}{\tau}) P \end{cases}$  $\frac{\tau - t}{\tau/2}$

$$v_1 = v_0 + \int_0^t \frac{F}{m} dt = \frac{t^2}{m\tau} P$$

$$v_2 = \frac{\tau P}{4m} + \int_{\frac{\tau}{2}}^t (2 - \frac{2t}{\tau}) \frac{P}{m} dt$$

$$= \frac{\tau P}{4m} + \frac{P}{m} (2t - \frac{t^2}{\tau}) \Big|_{\frac{\tau}{2}}^t$$

$$= \frac{\tau P}{4m} + \frac{P}{m} (2t - \frac{t^2}{\tau} - \tau + \frac{\tau}{4})$$

$$= -\frac{\tau P}{2m} + \frac{P}{m} (2t - \frac{t^2}{\tau}) = \frac{P}{2m\tau} (-\tau^2 + 4t\tau - 2t^2)$$

$$W = \Delta T = \frac{1}{2} m v_f^2 = \frac{1}{2} m \cdot (\frac{P\tau}{2m})^2 = \frac{P^2 \tau^2}{8m}$$

1.36. ① $\partial_x F_y - \partial_y F_x = 6abz^3 - 40bx^3 y - (6abz^3 - 40bx^3 y) = 0$

$\partial_x F_z - \partial_z F_x = 18aby z^2 - 18abz^2 y = 0$

$\partial_y F_z - \partial_z F_y = 18abxz^2 - 18abxz^2 = 0$

$\therefore \nabla \times \vec{F} = 0$. 保守力.

$(0,0,0) \to (x,0,0) \to (x,y,0) \to (x,y,z)$

$$A(\vec{r}) = \int F_x(x,0,0)\, dx + \int F_y(x,y,0)\, dy + \int F_z(x,y,z)\, dz$$

$$= 0 - \int_0^y 10bx^4 y\, dy + \int_0^z 18abxyz^2\, dz$$

$$= -5bx^4 y^2 + 6abxyz^3$$

$\therefore V(\vec{r}) = -A(\vec{r}) = 5bx^4 y^2 - 6abxyz^3$

② $\nabla \times \vec{F} = 0$, 保守力; $\nabla \times \vec{F} \neq 0$, 非保守力

保守力时, $V = \int_l \vec{F} \cdot d\vec{r} = -\int_0^x F_x(x,0,0)\, dx - \int_0^y F_y(x,y,0)\, dy - \int_0^z F_z(x,y,z)\, dz$

1.37. ① $\nabla = \partial_r \hat{e}_r + \frac{1}{r}\partial_\theta \hat{e}_\theta$

$\therefore \vec{F} = -\nabla V(r) = -\frac{d}{dr}\left(\frac{ke^{-\alpha r}}{r}\right)\hat{e}_r$

$= -k\left(\frac{-\alpha e^{-\alpha r}}{r} - \frac{e^{-\alpha r}}{r^2}\right)\hat{e}_r$

$= k(\alpha r + 1)\cdot\frac{e^{-\alpha r}}{r^2}\hat{e}_r$.

A 多了 $e^{-\alpha r}$ 项之，除 r 作大近似下降

B 多了 $\frac{1}{r}$ 项

② $\frac{mv^2}{r} = -k(\alpha r+1)\frac{e^{-\alpha r}}{r^2}$

$\therefore J = r\,mv = \sqrt{\frac{mv^2}{r}\cdot(r^3 m)} = \sqrt{-mk(\alpha r^2 + r)e^{-\alpha r}}$

$E = \frac{1}{2}mv^2 + V(r) = \frac{e^{-\alpha r}}{r}\left(-\frac{k\alpha r}{2} - \frac{k}{2}\right) + \frac{ke^{-\alpha r}}{r}$

$= -k\frac{e^{-\alpha r}}{2r}(\alpha r - 1)$

1.38. 之用 $\nabla \times \vec{F} = 0$，有：

$\partial_x F_y - \partial_y F_x = a_{21} - a_{12} = 0$

$\partial_x F_z - \partial_z F_x = a_{31} - a_{13} = 0$

$\partial_y F_z - \partial_z F_y = a_{32} - a_{23} = 0$

$V = -\int_0^x a_{11}x\,dx - \int_0^y (a_{21}x + a_{22}y)\,dy - \int_0^z (a_{31}x + a_{32}y + a_{33}z)\,dz$

$= -\frac{1}{2}a_{11}x^2 - a_{21}xy - \frac{1}{2}a_{22}y^2 - a_{31}xz - a_{32}yz - \frac{1}{2}a_{33}z^2$

$= -\frac{1}{2}\left(a_{11}x^2 + a_{22}y^2 + a_{33}z^2 + 2a_{12}xy + 2a_{13}xz + 2a_{23}yz\right)$

$\vec{F} = \hat{a} \cdot \vec{r}$.

例 $V = -\frac{1}{2} \vec{r} \cdot \hat{a}\vec{r}$

1.39 $F(r) = -\frac{c}{r^{\frac{1}{2}}} \hat{e}_r$ $\qquad (r^{-\frac{1}{2}})' = -\frac{1}{2} r^{-\frac{3}{2}}$.

$\therefore V(r) = V_0 - \int F(r)\, dr = V_0 - \frac{2c}{\sqrt{r}}$

$r \to \infty$, 取 $V=0$, 有 $V_0 = 0$

机械能守恒: $\Rightarrow E_1 = 0 - \left(-\frac{2c}{\sqrt{a}}\right) = \frac{2c}{\sqrt{a}}$

$E_2 = -\frac{2c}{\sqrt{a}} - \left(-\frac{2c}{\sqrt{a}/4}\right) = \frac{2c}{\sqrt{a}}$

$\therefore E_1 = E_2 \Rightarrow V_1 = V_2$

1.40 $F(r) = -\frac{k}{r} \hat{e}_r$

$\therefore m\frac{dv}{dt} = +\frac{k}{r} \quad \Rightarrow \quad mr\,dv = +k\,dt$

即 $m\frac{dv}{dr} v = -\frac{k}{r}$

$\therefore \frac{1}{2} m\, dv^2 = -\frac{k\,dr}{r}$

$\therefore \frac{1}{2} mv^2 = -k \ln r + C$ $\qquad mv^2 = 2k \ln \frac{a}{r} \Rightarrow v = \sqrt{\frac{2k}{m} \ln \frac{a}{r}}$

$r = a$, $v = 0$ $\therefore C = k \ln a$.

$\therefore dv = \sqrt{\frac{k}{m}} \cdot \frac{\frac{1}{2} \cdot \frac{r}{a} \cdot \frac{-a}{r^2}}{\sqrt{\ln \frac{a}{r}}}\, dr$

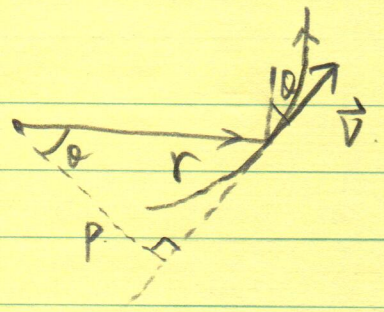
$= -\sqrt{\frac{k}{2m}} \frac{\frac{1}{r}}{\sqrt{\ln \frac{a}{r}}}\, dr$

$\therefore t = \int dt = \frac{+m}{k} \int r\, dv = -\sqrt{\frac{m}{2k}} \int_a^0 \frac{dr}{\sqrt{\ln \frac{a}{r}}}$

令 $u = \sqrt{\ln \frac{a}{r}} \Rightarrow r = a \cdot e^{-u^2}$ $\therefore dr = -2au\, e^{-u^2} du$

$\therefore t = \sqrt{\frac{m}{2k}} \int_0^\infty \frac{a \cdot 2au \times e^{-u^2} du}{u} = \sqrt{\frac{m}{2k}} (2a) \int_0^\infty e^{-u^2} du = a\sqrt{\frac{\pi m}{2k}}$

1.44



有心力: $F = m(\ddot{r} - r\dot{\theta}^2)$

$=$

$mh = J = |\vec{r} \times m\vec{v}| = mrv\cos\theta = mpv \Rightarrow p = \frac{h}{v}$

$\therefore p^{-2} = \frac{v^2}{h^2}$

$E = \frac{1}{2}mv^2 + V(r)$

$\Rightarrow v^2 = \frac{2(E-V)}{m}$

$= \frac{2(E-V)}{mh^2}$

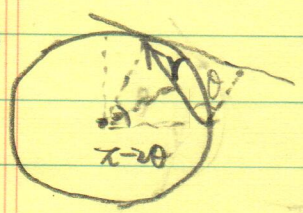
$J = mh$

$\therefore \frac{dp^{-2}}{dr} = \frac{2}{mh^2} \cdot \left(-\frac{dV}{dr}\right) = \frac{2}{mh^2} F$

$\therefore F = \frac{mh^2}{2} \frac{dp^{-2}}{dr}$

1.42

①



$p = r\cos\theta = \frac{r^2}{2a}$

$r = 2a \cdot \sin\left(\frac{\pi}{2} - \theta\right) = 2a\cos\theta$

$\therefore p^{-2} = \frac{4a^2}{r^4}$

$\therefore \frac{dp^{-2}}{dr} = -\frac{16a^2}{r^5}$

$\therefore F \propto \frac{1}{r^5}$

② $r = e^{a\theta}$, $h = pv$ $\quad \frac{dr}{d\theta} = ae^{a\theta} = ar$ $\quad \dot{\theta} = \frac{h}{r^2}$

$h = pv = r^2\dot{\theta} \Rightarrow p = \frac{h}{v}$

$\frac{dp^{-2}}{dr} = \frac{d}{dr}\left(\frac{v^2}{h^2}\right) = \frac{d}{dr}\left(\frac{\dot{r}^2 + r^2\dot{\theta}^2}{h^2}\right) = \frac{1}{h^2}\frac{d}{dr}\left[\left(\left(\frac{dr}{d\theta}\right)^2 + r^2\right)\frac{h^2}{r^4}\right]$

$= \frac{a^2+1}{h^2}\frac{d}{dr}\left(\frac{h^2}{r^2}\right) = \frac{-2(a^2+1)}{r^3}$

$\therefore F = \frac{mh^2}{2} = mh^2(a^2+1)\frac{1}{r^3}$

1.43 $r^2 = a^2\cos 2\theta$ 求 F.

$r^2\dot\theta = h \Rightarrow \dot\theta = \frac{h}{r^2}$ <4>

$F = m\cdot a_r = m(\ddot r - r\dot\theta^2) = m\left(\ddot r - \frac{h^2}{r^3}\right)$

$2r\dot r = 2a^2\sin 2\theta\,\dot\theta$

$\therefore \frac{dr}{d\theta} = -\frac{a^2}{r}\sin 2\theta$

$\dot r = \frac{dr}{dt} = \frac{dr}{d\theta}\dot\theta = -\frac{a^2h}{r^3}\sin 2\theta$

$\ddot r = \frac{d\dot r}{dt} = \frac{d\dot r}{d\theta}\cdot\dot\theta = -2\frac{a^2h}{r^3}\cos 2\theta\,\frac{h}{r^2} + \frac{3a^2h}{r^4}\sin 2\theta\left(\frac{-a^2}{r}\sin 2\theta\right)\frac{h}{r^2}$

$= -\frac{2h^2}{r^5} + \frac{3a^4h^2}{r^8}\sin^2 2\theta$

$= -\frac{2h^2}{r^5} - \frac{3a^4h^2}{r^8}\times\left(1 - \frac{r^4}{a^4}\right) = -\frac{3a^4h^2}{r^7} + \frac{h^2}{r^3}$

$\therefore F = m(\ddot r - \frac{h^2}{r^3}) = -\frac{3ma^4h^2}{r^7}$

1.44

$U = -m\left(\frac{\mu^2}{r} + \frac{V}{2r^2}\right)$

$V = -m\frac{\mu^2}{r} - \frac{mV}{2r^2} + \frac{J^2}{2mr^2}$

$= -m\frac{\mu^2}{r} + \frac{(J^2 - m^2V)}{2mr^2} = -m\frac{\mu^2}{r} + m\frac{h^2 - V}{2r^2}$

相当于 $k^2 \to r^2$, $J^2 \to J^2 - m^2V$, 引力场.

$\dot\theta = \frac{d\theta}{dr}\cdot v = \frac{h}{r^2} \Rightarrow \int d\theta = \int \frac{h/r^2}{v}\,dr$

$\therefore \theta = \int_{r_0}^{r} \frac{J/r^2\,dr}{\sqrt{2mE + \frac{2m^2\mu^2}{r} - \frac{J^2}{r^2}}} = \frac{J}{J'}\arccos\frac{\frac{J'}{r} - mk^2/J'}{\sqrt{2mE + \frac{m^2k^4}{J'^2}}}$

$\left(\frac{J'}{J}\right)^2 = \frac{h^2 - V}{h^2} = k^2$ $\quad\therefore \cos k\theta = \frac{J'/r - mk^2/J'}{\frac{mk^2}{J'}\sqrt{\frac{2EJ'^2}{mk^4} + 1}} = \frac{1}{e}\left(\frac{J'^2}{mk^2 r} - 1\right)$

$= \frac{1}{e}\left(\frac{p}{r} - 1\right)$

$\therefore r = \frac{p}{1 + e\cos k\theta}$

$\therefore e = \sqrt{1 + \frac{2mE(h^2-V)}{k^4}}$, $p = \frac{m(h^2-V)}{\mu^2}$, $k = \sqrt{\frac{h^2-V}{h^2}}$ <27>

1.45. $r^2\dot{\theta} = h \Rightarrow \dot{\theta} = \frac{h}{r^2}$ $\therefore r\dot{\theta} = \frac{h}{r}$

$r = \frac{p}{1+e\cos\theta} \Rightarrow$ 近日点 $\frac{p}{1+e}$, 远 $\frac{p}{1-e}$

$\therefore \frac{V_{近}}{V_{远}} = \frac{1+e}{1-e}$

1.46 $v^2 = \frac{a^2}{r^2} = \dot{r}^2 + (r\dot{\theta})^2 = \dot{r}^2 + \frac{h^2}{r^2}$ $\therefore \dot{r} = \pm\frac{\sqrt{a^2-h^2}}{r}$

$\frac{dr}{dt} = \frac{dr}{d\theta}\dot{\theta} = \frac{dr}{d\theta}\cdot\frac{h}{r^2} = \pm\frac{\sqrt{a^2-h^2}}{r}$

$\frac{h\,dr}{r} = \pm\sqrt{a^2-h^2}\,d\theta$

$\therefore h\ln r = \pm\sqrt{a^2-h^2}\,\theta + c$.

$r = r_0, \theta = 0 \Rightarrow c = h\ln r_0$

$\therefore h\ln\frac{r}{r_0} = \pm\sqrt{a^2-h^2}\,\theta$

$\Rightarrow \ln\frac{r}{r_0} = \pm\frac{\sqrt{a^2-h^2}}{h}\theta$.

$\int \frac{dx}{(1+\cos x)^2} = \frac{(2+\cos x)\sin x}{3(1+\cos x)^2}$

1.47 $r = \frac{p}{1+\cos\theta}$ 近日点 $\Rightarrow \theta = 0$

$\Rightarrow r = \frac{p}{2} = \frac{a}{n}$ $\therefore p = \frac{2a}{n}$ $\therefore r = \frac{2a}{n(1+\cos\theta)}$.

$\frac{mv^2}{a} = \frac{k^2 m}{a^2} \Rightarrow v^2 = \frac{k^2}{a}$ $\therefore T = \frac{2\pi a}{v} = \frac{2\pi a\sqrt{a}}{k} = \frac{2\pi}{k}a^{\frac{3}{2}}$

$r = a \Rightarrow \frac{2}{n(1+\cos\theta_0)} = 1 \Rightarrow \cos\theta_0 = \frac{2-n}{n} = \frac{2}{n} - 1$

$\frac{h}{r^2} = \dot{\theta} = \frac{d\theta}{dt} \Rightarrow t = 2\int\frac{r^2 d\theta}{h} = 2\int_0^{\theta_0}\frac{4a^2\,d\theta}{n^2 h(1+\cos\theta)^2} = \frac{8a^2}{n^2 h}\cdot\frac{\sqrt{2-\frac{2}{n}}\left(\frac{2}{n}+1\right)}{3\cdot\left(\frac{2}{n}\right)^{\frac{3}{2}}}$

$= \frac{4a^2}{3h}\sqrt{n-1}\cdot(n+2)$

$p = \frac{h^2}{k^2} = \frac{2a}{n}$ $\therefore h = k\sqrt{\frac{2a}{n}}$

$$= \frac{4a^{3/2}}{3k\sqrt{2}\eta}\sqrt{\frac{n-1}{n}}(n+2) = \frac{4a^{3/2}}{3\sqrt{2}k}\sqrt{\frac{n-1}{n}}\frac{n+2}{n}$$

$$\therefore \frac{t}{T} = \frac{\phi}{3\sqrt{2}k}\sqrt{\frac{n-1}{n}}\frac{n+2}{n} \times \frac{k}{2\pi} = \frac{\sqrt{2}}{3\pi}\sqrt{\frac{n-1}{n}}\frac{n+2}{n} = \frac{2}{3\pi}\sqrt{\frac{n-1}{2n}}\frac{n+2}{n}$$

② $$\frac{d\left(\frac{t}{T}\right)}{dn} = 0 = \frac{\sqrt{2}}{3\pi}\left(\frac{\sqrt{n-1}\,(n+2)}{n^{3/2}}\right)' = \frac{\sqrt{2}}{3\pi}\cdot\frac{\left(\frac{n+2}{2\sqrt{n-1}} + \sqrt{n-1}\right)n^{3/2} - \sqrt{n-1}\,(n+2)\frac{3}{2}\sqrt{n}}{n^3}$$

$$= \frac{\sqrt{2}}{3\pi n^3}\frac{\sqrt{n}}{\sqrt{n-1}}\left(\frac{n^2}{2} + n + n^2 - n - \frac{3}{2}(n-1)(n+2)\right)$$

$$= \frac{\sqrt{2}}{3\pi n^3}\sqrt{\frac{n}{n-1}}\cdot\left(\frac{3n^2}{2} - \frac{3}{2}n^2 - \frac{3}{2}n + 3\right)$$

$\propto$ ——— $\left(-\frac{3}{2}n+3\right) = 0 \Rightarrow n = 2$

$$\therefore \frac{t}{T} = \frac{2}{3\pi}\cdot\frac{1}{2}\cdot\frac{4}{2} = \frac{2}{3\pi}$$

1.48 $$r = \frac{p}{1+e\cos\theta}$$ 3p

$$\begin{cases} r_m = \frac{p}{1+e} \\ r_{ms} = \frac{p}{1-e} \end{cases} \qquad \frac{p}{r_m} + \frac{p}{r_{ms}} = 2 \qquad \therefore p = \frac{2}{\frac{1}{r_m}+\frac{1}{r_{ms}}}$$

$$\therefore p = \frac{2r_m \cdot r_{ms}}{r_m + r_{ms}} \simeq 7690$$

$r_{ms} = 6839$
$r_m = 8784$

$$e = \frac{p}{r_{ms}} - 1 \simeq 0.124 \qquad e = \sqrt{1 + \frac{2E}{m}\left(\frac{h}{k^2}\right)^2} = \sqrt{1 + \frac{2Ep}{mk^2}}$$

$$\therefore E = (e^2-1)\frac{mk^2}{2p}$$

$$E = \frac{1}{2}mv^2 - \frac{k^2 m}{r} \qquad \therefore \frac{1}{2}v^2 = \frac{k^2}{r} - (1-e^2)\frac{k^2}{2p}$$

$$\therefore v = k\sqrt{\frac{2}{r} - (1-e^2)\frac{1}{p}}$$

$\frac{k^2}{R^2} = g \quad \therefore k = R\sqrt{g}$.
$\simeq 633.57$

$\therefore v_{远} \simeq 8.12\,km/s$.

$v_{近} \simeq 6.32\,km/s$

面积速度 $\sigma = \frac{1}{2} r \cdot r\dot{\theta} = \frac{1}{2} r^2 \dot{\theta} = \frac{1}{2} r \cdot v_\theta$

$\therefore \tau = \frac{S}{v_s} = \frac{2\pi ab}{a v_a} = \frac{2\pi b}{v_a} = 113\ min$

1.49 椭圆轨道，机械能为 $-\frac{mk^2}{2a}$

$\therefore \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2) - \frac{mk^2}{r} = -\frac{mk^2}{2a}$ ①

$a - r = ae\cos E$, $\int \frac{2\pi}{\tau} dt = T$

$\Rightarrow \dot{r} = ae\sin E \cdot \dot{E}$ ②

$(r^2\dot{\theta} = h)$

由①式，$\dot{r}^2 = -\frac{k^2}{a} + \frac{2k^2}{r} - r^2\dot{\theta}^2$

$= -\frac{k^2}{a} + \frac{2k^2}{r} - \frac{h^2}{r^2}$

$= \frac{-k^2}{a} + \frac{2k^2}{r} - \frac{k^2 a(1-e^2)}{r^2}$

$= \frac{k^2}{r^2 a}(-r^2 + 2ra - a^2 + a^2e^2) = \frac{k^2}{r^2 a}[a^2e^2 - (r-a)^2]$

$r = \frac{p}{1+e\cos\theta}$, $p = \frac{h^2}{k^2}$

$\theta = 0$, 有 $r = a(1-e)$

$\therefore a(1-e) = \frac{p}{1+e}$ $\therefore p = a(1-e^2)$

$\therefore h^2 = k^2 a(1-e^2)$

$= \frac{k^2}{r^2 a} \cdot a^2e^2\sin^2 E = \frac{k^2 a}{r^2} e^2 \sin^2 E = \frac{k^2 a e^2 \sin^2 E}{a^2(1-e\cos E)^2}$

由②式，$\dot{r}^2 = a^2e^2\sin^2 E \cdot \dot{E}^2$

$\therefore a^2e^2\sin^2 E \cdot \dot{E}^2 = \frac{k^2 a e^2 \sin^2 E}{a^2(1-e\cos E)^2}$ $\therefore \dot{E}^2 = \frac{k^2}{a^3(1-e\cos E)^2}$

$\therefore (1-e\cos E)\frac{dE}{dt} = \frac{k}{a^{\frac{3}{2}}} = \frac{2\pi}{\tau}$ $\tau = \frac{2\pi \cdot a^{\frac{3}{2}}}{k}$

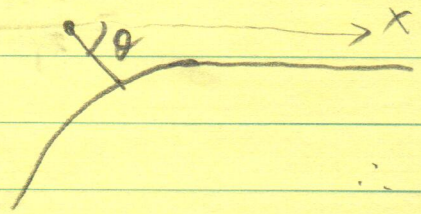
$\therefore (1-e\cos E)\, dE = \frac{2\pi}{\tau} dt$

两边积分，有

$E - e\sin E = \int \frac{2\pi}{\tau} dt$

$= T$

1.60



$$m(\ddot{r} - r\dot{\theta}^2) = \frac{mc}{r^3}$$

$$\therefore \ddot{r} = \frac{mc}{r^3} + r\frac{h^2}{r^4} = \frac{mc+h^2}{r^3}$$

$$\therefore \frac{d\dot{r}}{dt} = \frac{d\dot{r}}{d\theta}\frac{h}{r^2}$$

$$\frac{mh^2}{2r^2} + \frac{1}{2}m\dot{r}^2 + \frac{mc}{2r^2} = \frac{1}{2}mV_\infty^2$$

$$\therefore \dot{r} = \sqrt{V_\infty^2 - \frac{c+h^2}{r^2}}$$

$$\therefore \frac{d\dot{r}}{d\theta} = \frac{\frac{1}{2}\frac{2(c+h^2)dr}{r^3\,d\theta}}{\sqrt{V_\infty^2 - \frac{c+h^2}{r^2}}} = \frac{\frac{c+h^2}{r^3}}{\sqrt{V_\infty^2 - \frac{c+h^2}{r^2}}}\frac{dr}{d\theta}$$

$$\therefore \frac{c+h^2}{r^3} = \frac{h}{r^2}\frac{c+h^2}{r^3}\frac{1}{\sqrt{V_\infty^2 - \frac{c+h^2}{r^2}}}\frac{dr}{d\theta}$$

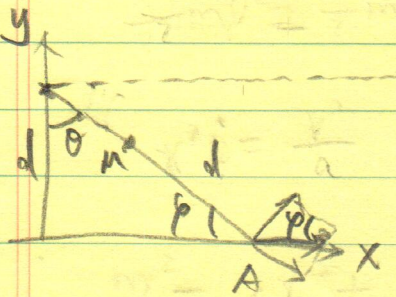
$$1 = \frac{h}{r}\frac{1}{\sqrt{V_\infty^2 r^2 - (c+h^2)}}\frac{dr}{d\theta}$$

$$\therefore \frac{dr}{d\theta} = \frac{r}{h}\sqrt{V_\infty^2 r^2 - (c+h^2)} \qquad h = rV_\infty = \rho V_\infty$$

$$\frac{dr}{d\theta} = 0 \Rightarrow r = \sqrt{\frac{c+h^2}{V_\infty^2}} = \sqrt{\frac{c+\rho^2 V_\infty^2}{V_\infty^2}} = \sqrt{\rho^2 + \frac{c}{V_\infty^2}}$$

补充例题：

1.1



$$x = \frac{d\cos\varphi}{\sin\varphi} - d\cos\varphi = \left(\frac{d}{y} - 1\right) d\cos\varphi$$

$$y = d\sin\varphi$$

$$\frac{x}{\left(\frac{d}{y}-1\right)^2} + y^2 = d^2$$

$$\vec{V}_A = c\sin\varphi\,\hat{e}_\theta + c\cos\varphi\,\hat{e}_r \qquad \theta = \frac{\pi}{2} - \varphi$$

$$= r\dot{\theta}\,\hat{e}_\theta + \dot{r}\,\hat{e}_r$$

$$\therefore -\frac{d}{\sin\varphi}\cdot\dot{\varphi} = c\sin\varphi \Rightarrow \dot{\varphi} = -\frac{c}{d}\sin^2\varphi$$

$$\frac{-d\cdot\cos\varphi}{\sin^2\varphi}\cdot\dot{\varphi} = c\cos\varphi \qquad \text{又有} \qquad r = \frac{d}{\sin\varphi} - d$$

$$\therefore V_r = \left(\frac{d}{\sin\varphi} - d\right)' = c\cos\varphi$$

$$V_\theta = r\dot{\theta} = -r\dot{\varphi} = \left(\frac{1}{\sin\varphi} - 1\right) c\sin^2\varphi = c\sin\varphi(1-\sin\varphi)$$

$$\therefore V = \sqrt{V_r^2 + V_\theta^2} = c\sqrt{1 + \sin^4\varphi - 2\sin^3\varphi}$$

$$\ddot{r} = \frac{d\dot{r}}{dt} = -c\sin\varphi\cdot\dot{\varphi} = \frac{c^2}{d}\sin^3\varphi$$

$$\ddot{\theta} = -\ddot{\varphi} = \frac{c}{d}\cdot 2\sin\varphi\cos\varphi\cdot\dot{\varphi} = -\frac{c^2}{d^2}\sin 2\varphi\sin^2\varphi$$

$$\therefore a_r = \ddot{r} - r\dot{\theta}^2 = \frac{c^2}{d}\sin^3\varphi - \left(\frac{d}{\sin\varphi} - d\right)\frac{c^2}{d^2}\sin^4\varphi$$

$$= \frac{c^2}{d}\sin^4\varphi$$

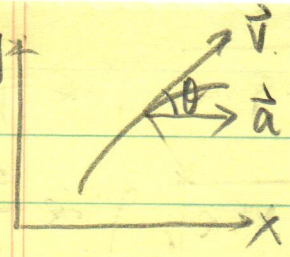
$$a_\theta = 2\dot{r}\dot{\theta} + r\ddot{\theta} = 2c\cos\varphi\cdot\frac{c}{d}\sin^2\varphi + \left(\frac{d}{\sin\varphi} - d\right)\left(-\frac{c^2}{d^2}\sin 2\varphi\sin^2\varphi\right)$$

$$= \frac{c^2}{d}\sin^2\varphi(2\cos\varphi - 2\cos\varphi + \sin 2\varphi)$$

$$= \frac{2c^2}{d}\sin^3\varphi\cos\varphi \Rightarrow a = \sqrt{a_r^2 + a_\theta^2} = \frac{c^2}{d}\sin^3\varphi\sqrt{\sin^2\varphi + 4\cos^2\varphi}$$

$$= \frac{c^2}{d}\sin^3\varphi\sqrt{1 + 3\cos^2\varphi}$$

1.2



$$\begin{cases} v\sin\theta = c \\ \frac{v^2}{\rho} = a\sin\theta \end{cases}$$

$$\Rightarrow a = \frac{1}{\sin\theta}\frac{v^2}{\rho} = \frac{v^3}{\rho c}$$

1.3 取C点为 $t=0$, $x=0$ 点, 则 $mg = kb$

BD间位移为 x 时: $F = mg - k(b+x) = -\frac{mg}{b}x$ 建立图示坐标

$\therefore m\ddot{x} = -mgx/b$ $\therefore x = A\cos\sqrt{\frac{g}{b}}t + B\sin\sqrt{\frac{g}{b}}t$

$t=0$, $x=0$ $\Rightarrow A=0$

机械能守恒 $\Rightarrow \frac{1}{2}kb^2 + mga = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{g(2a+b)}$

$\therefore \dot{x}\big|_{t=0} = B\sqrt{\frac{g}{b}} = v_{max} = \sqrt{g(2a+b)} \Rightarrow B = \sqrt{2ab+b^2}$

$\therefore x = \sqrt{2ab+b^2}\sin\sqrt{\frac{g}{b}}t$ 振动角频率为 $\sqrt{\frac{g}{b}}$

时间分为三段. ① $\frac{1}{2}gt_1^2 = a \Rightarrow t_1 = \sqrt{\frac{2a}{g}}$, $v = gt_1 = \sqrt{2ga}$

② $\dot{x} = \sqrt{g(2a+b)}\cos\sqrt{\frac{g}{b}}t_2 = v = \sqrt{2ga}$

$\Rightarrow \cos\sqrt{\frac{g}{b}}t_2 = \sqrt{\frac{2a}{2a+b}}$ $\therefore \tan(\quad) = \sqrt{\frac{b}{2a}}$

$\therefore t_2 = \sqrt{\frac{b}{g}}\arctan\sqrt{\frac{b}{2a}}$

③ $t_3 = \frac{T}{4} = \frac{\pi}{2}\sqrt{\frac{b}{g}}$

$\therefore \Delta t = t_1 + t_2 + t_3 = \sqrt{\frac{2a}{g}} + \sqrt{\frac{b}{g}}\left(\arctan\sqrt{\frac{b}{2a}} + \frac{\pi}{2}\right)$.

1.4 解：$\begin{cases} m\dot{v} = mg\sin\theta \\ m\frac{v^2}{\rho} = mg\cdot\cos\theta - N \end{cases}$

$y^2 = 2px \quad \therefore y' = \frac{p}{y}, \quad y'' = \frac{p}{y^2}(-y') = -\frac{p^2}{y^3}$

$\therefore \frac{1}{\rho} = \frac{|y''|}{(\sqrt{1+y'^2})^3} = \frac{\frac{p^2}{y^3}}{(1+\frac{p^2}{y^2})^{\frac{3}{2}}} = \frac{p^2}{(p^2+y^2)^{\frac{3}{2}}} = \frac{1}{(1+4x^2)^{\frac{3}{2}}}$

$m\dot{v} = mv\frac{dv}{ds} = \frac{dT}{ds} = mg\sin\theta$

$\Rightarrow dT = mg\,dy \quad \therefore T = mg(h-y) \quad \therefore v^2 = 2g(h-y)$

$\cos\theta = \frac{1}{\sqrt{1+\tan^2\theta}} = \frac{y}{\sqrt{y^2+p^2}}$

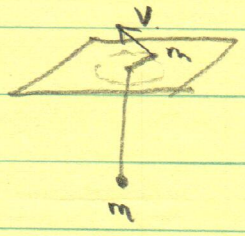
$\therefore m\,2g(h-y)\frac{p^2}{(p^2+y^2)^{\frac{3}{2}}} = mg\cdot\frac{y}{\sqrt{y^2+p^2}}$

$\therefore \frac{2(h-y)p^2}{p^2+y^2} = y \quad \Rightarrow y^3 + 3p^2y - 2p^2h = 0$ 为轨迹的上升方程

1.5. 有心力 ⇒ 角动量守恒. ⇒ $r^2\dot\theta = h = r\cdot V_\theta = aV$ <34>

$V=\sqrt{\frac{9ga}{2}}$

$\therefore h^2 = a^2V^2 = \frac{9}{2}ga^3$



$$\begin{cases} m(\ddot r - r\dot\theta^2) = -T & \text{对质点A} \\ m\ddot r = T - mg & \text{对质点B} \end{cases}$$

$$r\dot\theta^2 = \frac{h^2}{r^3}$$

$$\therefore m\ddot r = -T + \frac{mh^2}{r^3}$$

$$\Rightarrow T - mg = -T + \frac{mh^2}{r^3} \Rightarrow T = \frac{1}{2}mg + \frac{mh^2}{2r^3}$$

$$\therefore m\ddot r = -\frac{1}{2}mg + \frac{mh^2}{2r^3} = -\nabla V(r)$$

$$V(r) = \frac{1}{2}mgr + \frac{mh^2}{4r^2} + V_0$$ 引力势能

$E = \frac{1}{2}m\dot r^2 + V(r)$. 初始, $\dot r = 0$, 设 $E = 0$. 有

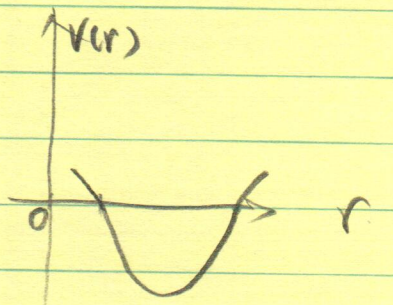
$$\frac{1}{2}mga + \frac{mh^2}{4a^2} + V_0 = \frac{1}{2}mga + \frac{1}{4}mV^2 + V_0 = \frac{13}{8}mga + V_0$$

$$\therefore V(r) = \frac{1}{2}mgr + \frac{m}{4r^2}\frac{9}{2}ga^3 - \frac{13}{8}mga$$

$$= \frac{mg}{8r^2}(4r^3 - 13r^2a + 9a^3)$$ $\to 4(r-a)r^2 - 9a(r+a)(r-a)$

$$= \frac{mg}{8r^2}(r-a)(r-3a)(4r+3a)$$ $\to (r-a)(4r^2 - 9ar - 9a^2)$ $\to (r-a)(r-3a)(4r+3a)$

$\therefore V(r) = 0 \Rightarrow \begin{cases} r = a \\ r = 3a \\ r = -\frac{3}{4}a. \end{cases}$



最后一个不物理舍去, 余去.

初始 $r=a$, $a<r<3a$ 此 $V(r)<0$ 势阱 ∴ 粒子在势阱内

$\therefore a < r < 3a$

对大质点受力有：$\begin{cases} m\ddot{r} = -T + \frac{J^2}{mr^3} \\ m\ddot{r} = \frac{T}{\lambda} - g \end{cases}$ m, λm

$\therefore \frac{1+\lambda}{\lambda} T = g + \frac{J^2}{mr^3} \Rightarrow T = \frac{\lambda}{1+\lambda} g + \frac{\lambda}{1+\lambda} \frac{J^2}{mr^3}$

$\therefore m\ddot{r} = -\frac{\lambda mg}{1+\lambda} g + \frac{1}{1+\lambda} \frac{J^2}{mr^3}$

$= -\nabla V(r)$

$$\boxed{V(r) = \frac{\lambda}{1+\lambda} mgr + \frac{1}{1+\lambda} \frac{J^2}{2mr^2} + V_0}$$

初始时刻，$\dot{r}=0$, $E=0$, 有： $J = mva$

$$\frac{\lambda}{1+\lambda} mga + \frac{1}{1+\lambda} \frac{J^2}{2ma^2} + V_0 = \frac{\lambda}{1+\lambda} mga + \frac{1}{1+\lambda} \frac{mv^2}{2} + V_0 = 0$$

$$\therefore V(r) = \frac{\lambda}{1+\lambda} mg(r-a) + \frac{m}{1+\lambda} \left(\frac{v^2 a^2}{2r^2} - \frac{v^2}{2} \right) = 0$$

$$\therefore 2\lambda g(r-a) + \frac{v^2 a^2}{r^2} - v^2 = 0$$

$$\therefore 2\lambda g r^3 - (2\lambda g a + v^2) r^2 - v^2 a^2 = 0 \qquad 令 x = \frac{r}{a}$$

$$\therefore x^3 - \left(1 + \frac{v^2}{2\lambda g a}\right) x^2 - \frac{v^2}{2\lambda g a} = 0$$

$x^3 - ax^2 - b = 0 = (x+c_1)(x^2 + c_2 x + c_3)$

$= x^3 + (c_1 + c_2) x^2 + (c_1 c_2 + c_3) x + c_1 c_3 = 0$

$\Rightarrow c_1 + c_2 = -a$, $c_1 c_2 + c_3 = 0$, $c_1 c_3 = -b$

$c_1^2 c_2 = b$

积分公式：$\int \frac{dx}{1+x^2} = \arctan x$　　$\int \frac{dx}{\sqrt{1-x^2}} = \arcsin x = -\arccos x$

$\int \frac{dx}{1-x^2} = \operatorname{arctanh} x$　　$\int \frac{dx}{\sqrt{1+x^2}} = \operatorname{arcsinh} x$.

一阶微分方程：① 可分离变量：$y' = g(x)h(y) \Rightarrow \int \frac{dy}{h(y)} = \int g(x)dx + C$

② 齐次微分方程：$y' = f(x,y)$，$f(\lambda x, \lambda y) = t^n f(x,y)$，$n=0$ 为齐次微分方程.

取 $f(x,y) = P(\frac{y}{x})$，$v = \frac{y}{x}$，则

$\int \frac{dv}{P(v)-v} = \ln x + C_1 \Rightarrow y = xv = \cdots$

③ 线性微分方程：$y' + P(x)y + Q(x) = 0$，$y = -e^{\int P(x)dx}\left[\int Q(x) e^{-\int P(x)dx} dx\right] + C$

④ 伯努利方程：$y' + P(x)y + Q(x)y^n = 0$，同除 y^{-n}，取 $v = y^{1-n}$，

$\Rightarrow \frac{1}{1-n} v' + P(x)v + Q(x) = 0$　利用②中公式，~~再代回~~ $y = v^{\frac{1}{1-n}}$

二阶微分方程：降阶法：$y'' = F(x, y', x)$

① $F = F(x)$，直接积分：$y = \int [\int f(x)dx] dx + C_1 x + C_2$.

② $F = F(x, y')$，取 $v = y'$，则 $v' = F(x, v)$. 一阶

③ $F = F(y, y')$，取 $y'' = \frac{dv}{dx} = \frac{dv}{dy} v$，则 $v\frac{dv}{dy} = F(y, v)$，一阶.

二阶线性常微分方程：$y'' + a_1(x)y' + a_2(x)y = 0$

已知特解 y_1，另一特解为 $y_2 = y_1 \int e^{-\int P(x)dx} dx$，$P(x) = \frac{1}{y}(2y_1' + a_1 y_1)$

（取 $y = uy_1$，代回，变成 u' 的一阶，可用降阶法）

二阶线性常系数常微分方程：齐次：$y'' + a_1 y' + a_2 y = 0$

特征方程：$r^2 + a_1 r + a_2 = 0$

$$y = \begin{cases} C_1 e^{r_1 x} + C_2 e^{r_2 x}, & \text{(两不同实根)} \\ (C_1 + C_2 x)e^{rx} & \text{(二重实根)} \\ e^{\alpha x}(C_1 e^{i\beta x} + C_2 e^{-i\beta x}) & \text{(两不同复根)} \end{cases}$$

可推广到 n 阶齐次

二阶线性常系数非齐次 ① 待定系数法 ② 常数变易法

$y''+a_1y'+a_2y=f(x)$

$f(x)=P_m e^{\lambda x}$

一般解法

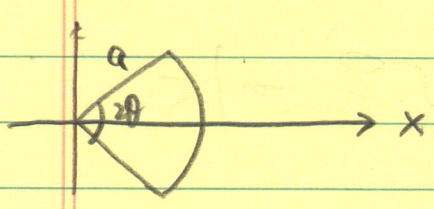
设 $C_1C_2 \to C_1(x)\ C_2(x)$

二阶线性变系数非齐次：欧拉方程 $x^n y^{(n)}+\cdots+a_{n-1}xy'+a_ny=f(x)$

令 $x=e^t$，$dx=x\,dt$，可变为常系数

$y^{(n)}+\cdots+b_ny=f(e^t)$

2.1.



由对称性，质心必在x轴上.

任质心公式 $\vec{r}_c = \frac{1}{m}\int \rho(\vec{r})\,\vec{r}\,dV$，有

$$x_c = \frac{1}{m}\int \rho x \cdot \underbrace{dxdy}_{rdrd\varphi} = \frac{\rho}{m}\int r\cos\varphi \cdot r\,dr\,d\varphi = \frac{\rho}{m}\int_0^a r^3 dr \int_{-\theta}^{\theta}\cos\varphi\,d\varphi$$

$$= \frac{2}{3}\cdot\frac{\rho}{m}\cdot a^3\cdot\sin\theta$$

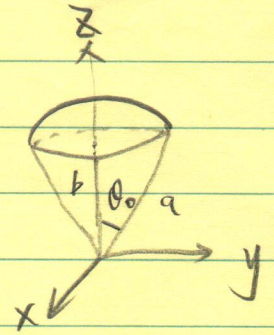
扇形质量 $m = \rho S = \rho\cdot\frac{1}{2}a^2\cdot 2\theta = \rho\theta a^2$

代入 x_c 中，有

$$x_c = \frac{2}{3}a\cdot\frac{\sin\theta}{\theta}$$

半圆弧，对应 $\theta = \frac{\pi}{2}$，$x_c = \frac{4}{3\pi}\cdot a$

2.2. 根据对称性，质心必在对称轴z轴上.



$$z_c = \frac{1}{m}\int \rho z\,dV$$

柱坐标 $dV = r\,dr\,d\varphi\,dz$

$$\therefore z_c = \frac{\rho}{m}\int z\cdot r\,dr\,d\varphi\,dz = \frac{\rho}{m}\int_b^a z\,dz\int_0^{\sqrt{a^2-z^2}} r\,dr\cdot\int_0^{2\pi} d\varphi$$

$$= \frac{\rho}{m}\int_b^a z\cdot\frac{1}{2}(a^2-z^2)\cdot 2\pi\,dz$$

$$= \frac{\rho}{m}\cdot\left(\frac{a^2z^2}{2}-\frac{z^4}{4}\right)\Big|_b^a\cdot\pi$$

$$= \frac{\rho}{m}\cdot\frac{\pi}{4}(a^2-b^2)^2 = \frac{\pi\rho}{4m}(a+b)^2(a-b)^2$$

$$m_z=\int\rho\,dV_z=\rho\cdot\int r\,dr\,d\varphi\,dz=\rho\int_b^a dz\int_0^{\sqrt{a^2-z^2}} r\,dr\cdot\int_0^{2\pi}d\varphi$$

<2>

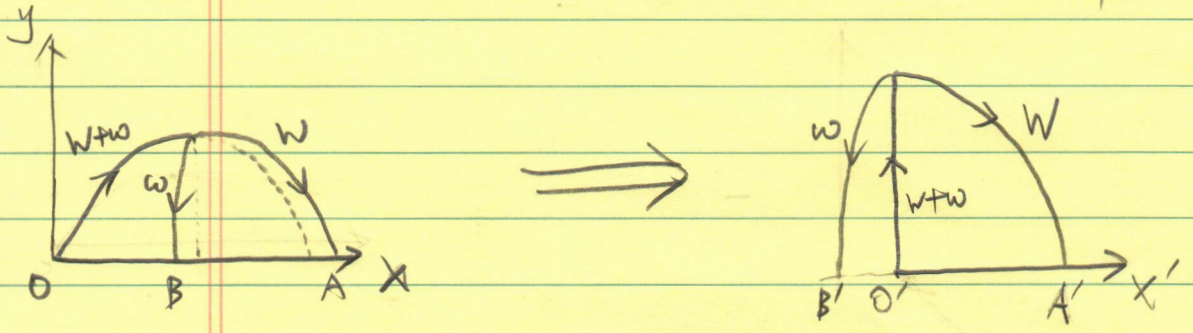
$$=\rho\int_b^a(a^2-z^2)\cdot\pi\cdot dz=\pi\rho\left(a^2z-\frac{z^3}{3}\right)\Big|_b^a=\frac{\pi\rho}{3}(2a+b)(a-b)^2$$

$$\therefore z_c=\frac{\pi\rho}{4}(a+b)^2(a-b)^2:\frac{3}{\pi\rho}\frac{1}{(2a+b)(a-b)^2}=\frac{3(a+b)^2}{4(2a+b)}$$

2.3. 设物和人组成整体，质心在X方向受力，作匀速直线运动，$x_c=$

$$\therefore x_c=v_0t\cdot\cos\alpha$$

取$(x_c,0)$为参考O'-x'y'z'，在随船参考中质心竖直上抛。



根据x方向动量守恒 $\Rightarrow Wv_1-w(u-v_1)=0\Rightarrow v_1=\frac{w}{W+w}u$

下落时间 = 上升时间 $=\frac{v_0\sin\alpha}{g}$

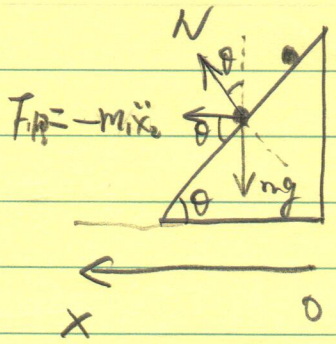
∴ W从最高点抛出的距离为

$$O'A'=v_1\cdot t=\frac{w}{(W+w)g}uv_0\sin\alpha.$$

2.4 m_1 与 m_2 组成的系统水平方向动量

$\therefore 0 = \ddot{X}_c = m_1\ddot{x}_1 + m_2\ddot{x}_2$

$\therefore \ddot{x}_2 = -\frac{m_1}{m_2}\ddot{x}_1$



在支架上，m_1 的加速度为

$$a_x' = \ddot{x}_1 - \ddot{x}_2 = \frac{m_1+m_2}{m_2}\ddot{x}_1 \quad , \quad a' = \frac{a_x}{\cos\theta}$$

同时根据受力分析 $\begin{cases} N + F_{惯}\cdot\sin\theta - mg\cos\theta = 0 & ① \\ F_{惯}\cdot\cos\theta + mg\sin\theta = m_1 a' & ② \end{cases}$

由②得 $-m_1\cdot\ddot{x}_2\cos\theta + mg\sin\theta = m_1\cdot\frac{m_1+m_2}{m_2}\ddot{x}_1\frac{1}{\cos\theta}$

即 $\frac{m_1}{m_2}\ddot{x}_1\cos\theta + g\sin\theta = \frac{m_1+m_2}{m_2}\cdot\frac{1}{\cos\theta}\ddot{x}_1$

$\therefore \frac{m_1}{m_2}\ddot{x}_1\cos^2\theta + g\sin\theta\cos\theta = \frac{m_1+m_2}{m_2}\ddot{x}_1$

$g\sin\theta\cos\theta = \left(\frac{m_1}{m_2}\sin^2\theta + 1\right)\ddot{x}_1$

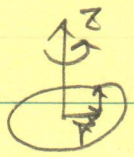
$\Rightarrow \ddot{x}_1 = \frac{m_2\sin\theta\cos\theta}{m_2+m_1\sin^2\theta}\cdot g$, $\ddot{x}_2 = -\frac{m_1}{m_2}\ddot{x}_1 = -\frac{m_1\sin\theta\cos\theta}{m_2+m_1\sin^2\theta}g$

在地面系中，$R_1\cdot\sin\theta = m_1\ddot{x}_1 \Rightarrow R_1 = \frac{m_1m_2\cos\theta}{m_2+m_1\sin^2\theta}g$

根据支架竖直方向上有 $\ddot{y}_2 = 0 \Rightarrow R_2 = m_2g + R_1\cdot\cos\theta = m_2g + \frac{m_1m_2\cos^2\theta}{m_2+m_1\sin^2\theta}g$

$= \frac{m_2(m_1+m_2)}{m_2+m_1\sin^2\theta}g$.

2.5: 圆柱绕z轴的转动惯量



$$d\vec{J} = dm \cdot \vec{r} \times \vec{v}$$

$$= \rho \cdot r\,dr\,d\varphi\, r^2 \omega\, \hat{e}_z$$

$$= \rho \omega r^3 dr\, d\varphi\, \hat{e}_z$$

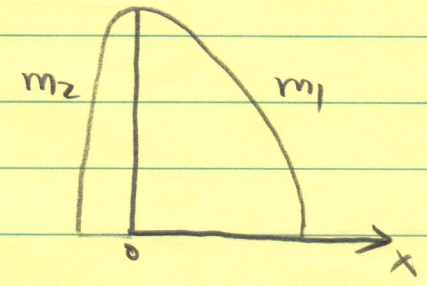
∴ 总角动量

$$\vec{J} = \int d\vec{J} = \rho\omega \cdot \int_0^a r^3\,dr \int_0^{2\pi} d\varphi\, \hat{e}_z$$

$$= \frac{\pi}{2} a^4 \rho \omega \cdot \hat{e}_z \qquad m = \pi a^2 \cdot \rho$$

$$= \frac{1}{2} m a^2 \omega\, \hat{e}_z.$$

2.6 仍利用2.3中的结果，求最高点



$$\begin{cases} m_1 v_1 + m_2 v_2 = 0 \\ \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2 = E \end{cases}$$

求动量：

$$\begin{cases} P_1 + P_2 = 0 \\ \frac{P_1^2}{2m_1} + \frac{P_2^2}{2m_2} = E \end{cases} \Rightarrow P_1 = \sqrt{\frac{2m_1 m_2 E}{m_1 + m_2}},\ P_2 = -\sqrt{\frac{2m_1 m_2 E}{m_1 + m_2}}$$

$$\therefore v_1 - v_2 = \frac{P_1}{m_1} - \frac{P_2}{m_2} = \sqrt{\frac{2E}{m_1 + m_2}} \cdot \left(\sqrt{\frac{m_2}{m_1}} + \sqrt{\frac{m_1}{m_2}} \right) = \sqrt{\frac{2E}{m_1 + m_2}} \frac{m_1 + m_2}{\sqrt{m_1 m_2}}$$

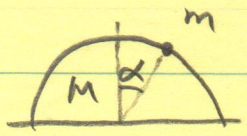
$$= \sqrt{\frac{2E(m_1 + m_2)}{m_1 m_2}} = \sqrt{2E\left(\frac{1}{m_1} + \frac{1}{m_2}\right)}$$

$$\Delta t = \frac{V}{g}$$

$$\therefore \Delta L = (v_1 - v_2)\Delta t = \frac{V}{g} \sqrt{2E\left(\frac{1}{m_1} + \frac{1}{m_2}\right)}$$

2.7 根据质心的定义：

$$(M+m)X_c = m\cdot x_1 + M\cdot x_2$$



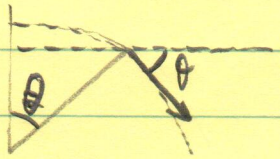
对地水平方向，$mV_{1x} + MV_2 = 0$ ①

根据机械能守恒，$mga(\cos\alpha - \cos\theta) = \frac{1}{2}m(V_{1x}^2 + V_{1y}^2) + \frac{1}{2}MV_2^2$ ②

在 M 系上看 $\vec{V}_1' = (V_{1x} - V_2)\hat{i} + V_{1y}\hat{j}$

$$= (1 + \frac{m}{M})V_{1x}\hat{i} + V_{1y}\hat{j}$$ ③

由几何关系，



$$\vec{V}_1' = a\dot{\theta}(\cos\theta\,\hat{i} - \sin\theta\,\hat{j})$$ ④

由②、④，$V_{1y} = -a\dot{\theta}\sin\theta$，$V_{1x} = \frac{M}{M+m}a\dot{\theta}\cos\theta$，$V_2 = -\frac{m}{M+m}a\dot{\theta}\cos\theta$

代入②式化简为：$mga(\cos\alpha - \cos\theta) = \frac{1}{2}\cdot a^2\dot{\theta}^2\left[m\sin^2\theta + \frac{mM}{M+m}\cos^2\theta + \frac{M\cdot m^2}{(M+m)^2}\cos^2\theta\right]$

$$= \frac{m}{2}a^2\dot{\theta}^2\left[1 - \frac{Mm}{(M+m)^2}\cos^2\theta\right]$$

$$\Rightarrow \dot{\theta} = \sqrt{\frac{2g(\cos\alpha - \cos\theta)}{a\left(1 - \frac{Mm}{(M+m)^2}\cos^2\theta\right)}}$$

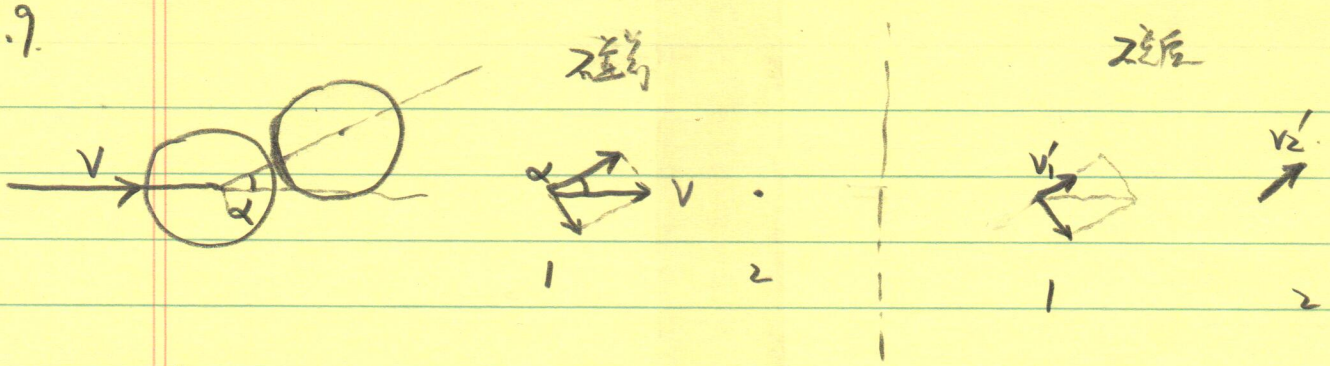
2.8 由动量守恒：$m\vec{V}_A = m\vec{V}_A' + m\vec{V}_B'$ $\Rightarrow$ $\vec{V}_A = \vec{V}_A' + \vec{V}_B'$ ①

由动能守恒：$\frac{1}{2}m\vec{V}_A^2 = \frac{1}{2}m\vec{V}_A'^2 + \frac{1}{2}m\vec{V}_B'^2$ $\Rightarrow$ $\vec{V}_A^2 = \vec{V}_A'^2 + \vec{V}_B'^2$ ②

① 平方，有：$\vec{V}_A^2 = \vec{V}_A'^2 + \vec{V}_B'^2 + 2\vec{V}_A'\cdot\vec{V}_B'$

与② 比较，必有 $2\vec{V}_A'\cdot\vec{V}_B' = 0$ $\therefore$ $\vec{V}_A' \perp \vec{V}_B'$

2.9.



相碰前后沿接触点法线方向：$e=\dfrac{V_2'-V_1'}{V\cos\alpha} \Rightarrow V_2'-V_1'=eV\cos\alpha$

动量守恒：$MV=MV_1'+MV_2' \Rightarrow V=V_1'+V_2'$

$$\therefore \begin{cases} V_2'=\dfrac{1+e}{2}V\cdot\cos\alpha \\ V_1'=\dfrac{1-e}{2}V\cos\alpha \end{cases}$$

∴ 粒子1沿碰前方向

$$\tan\beta=\frac{-V_1'\cdot\sin\alpha+V\cdot\sin\alpha\cdot\cos\alpha}{V_1'\cos\alpha+V\cdot\sin\alpha\cdot\sin\alpha}=\frac{-\frac{1-e}{2}\sin\alpha\cos\alpha+\sin\alpha\cos\alpha}{\frac{1-e}{2}\cos^2\alpha+\sin^2\alpha}$$

$$=\frac{(1+e)\sin\alpha\cos\alpha}{(1-e)\cos^2\alpha+2\sin^2\alpha}=\frac{(1+e)\tan\alpha}{1-e+2\tan^2\alpha}$$

$$f(x)=\frac{(1+e)x}{1-e+2x^2} \qquad f'(x)=\frac{(1+e)(1-e+2x^2)-(1+e)x\cdot 4x}{(1-e+2x^2)^2}=\frac{(1-e^2)-2(1+e)x^2}{(1-e+2x^2)^2}=0$$

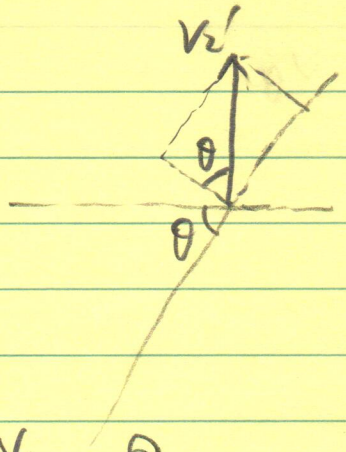
$$\Rightarrow x^2=\frac{1-e}{2}=\tan^2\alpha .$$

$$\therefore \tan\beta_2=\frac{(1+e)x}{1-e+2x^2}=\frac{(1+e)\sqrt{\frac{1-e}{2}}}{1-e+1-e}=\frac{1+e}{2\sqrt{2(1-e)}}$$

$$\therefore 1+\tan^2\beta=\frac{(1+e)^2+8(1-e)}{8(1-e)}=\frac{9-6e+e^2}{8(1-e)}=\frac{(3-e)^2}{8(1-e)}$$

$$\therefore \sin\beta=\frac{\tan\beta}{\sqrt{1+\tan^2\beta}}=\frac{1+e}{3-e} \qquad \therefore \beta=\arcsin\left(\frac{1+e}{3-e}\right)$$

2.10 垂直平面内动量守恒



$$\Rightarrow \quad m_1 V_1 \sin\theta = m_1 V_1' \sin\theta + m_2 V_2' \qquad ①$$

由碰撞恢复系数定义

$$e = \frac{V_2' \sin\theta - V_1'}{V_1} \Rightarrow V_1' = V_2' \sin\theta - eV_1 \qquad ②$$

将②代入①中，有：

$$m_1 V_1 \sin\theta = (m_1 \sin^2\theta + m_2) V_2' - e m_1 V_1 \sin\theta.$$

$$\Rightarrow V_2' = \frac{m_1(1+e)\sin\theta}{m_2 + m_1 \sin^2\theta} V_1$$

$$V_1' = V_2' \sin\theta - eV_1 = \frac{m_1(1+e)\sin^2\theta - e(m_2 + m_1\sin^2\theta)}{m_2 + m_1\sin^2\theta} V_1 = \frac{m_1\sin^2\theta - em_2}{m_2 + m_1\sin^2\theta} V_1$$

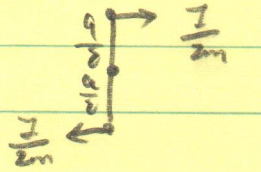
① 初始质心静止在原点，由质心动量

$$V_c = \frac{I}{2m}$$

质心不受力，$F=0$，质心作匀速直线运动 $\therefore \vec{X}_c(t) = V_c t \cdot \hat{e}_x$

② 质心为参考系：$V_1' = V_1 - V_c = \frac{I}{m} - \frac{I}{2m} = \frac{I}{2m}$

$$V_2' = 0 - V_c = -\frac{I}{4m}$$



质心圆运动 两质点绕质心作匀速圆周运动，$\omega = \frac{V}{r} = \frac{I}{ma}$

$$\therefore \vec{r}_1'(t) = \frac{a}{2}\sin\omega t\,\hat{e}_x + \frac{a}{2}\cos\omega t\,\hat{e}_y$$

$$\vec{r}_2'(t) = -\frac{a}{2}\sin\omega t\,\hat{e}_x - \frac{a}{2}\cos\omega t\,\hat{e}_y$$

③ 实际运动

$$\therefore \vec{r}_1 = \vec{r}_1'(t) + \vec{X}_c(t) = \left(\frac{It}{2m} + \frac{a}{2}\sin\frac{It}{ma}\right)\hat{e}_x + \frac{a}{2}\cos\frac{It}{ma}\,\hat{e}_y$$

$$\vec{r}_2 = \vec{r}_2'(t) + \vec{X}_c(t) = \left(\frac{It}{2m} - \frac{a}{2}\sin\frac{It}{ma}\right)\hat{e}_x - \frac{a}{2}\cos\frac{It}{ma}\,\hat{e}_y$$

的摆线轨迹.

④ $\vec{V}_1 = \frac{I}{2m}\left(1+\cos\frac{It}{ma}\right)\hat{e}_x - \frac{I}{2m}\sin\frac{It}{ma}\,\hat{e}_y \quad \therefore V_1^2 = \frac{I^2}{4m^2}\left(2+2\cos\frac{It}{ma}\right)$

$\vec{V}_2 = \frac{I}{2m}\left(1-\cos\frac{It}{ma}\right)\hat{e}_x + \frac{I}{2m}\sin\frac{It}{ma}\,\hat{e}_y \quad \therefore V_2^2 = \frac{I^2}{4m^2}\left(2-2\cos\frac{It}{ma}\right)$

$$\therefore \frac{T_1}{T_2} = \frac{V_1^2}{V_2^2} = \frac{1+\cos\frac{It}{ma}}{1-\cos\frac{It}{ma}} = \cot^2\frac{It}{2ma}$$

$1+\cos\theta = 2\cos^2\frac{\theta}{2}$

$1-\cos\theta = 2\sin^2\frac{\theta}{2}$

2.12 质心速度 $V_c = \dfrac{m_1 v_1}{m_1+m_2}$

质心中变量用 $\tilde{v}$ 表示，则

$\tilde{v}_1 = v_1 - v_c = \dfrac{m_2 v_1}{m_1+m_2}$ $\quad$ $\tilde{v}_2 = v_2 - v_c = \dfrac{-m_1 v_1}{m_1+m_2}$

由碰撞恢复系数定义 $e = \dfrac{|\tilde{v}_1' - \tilde{v}_2'|}{\tilde{v}_1 - \tilde{v}_2}$ ①

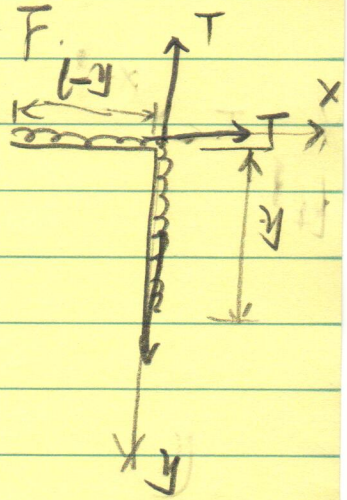
由动量守恒，$m_1\tilde{v}_1' + m_2\tilde{v}_2' = 0$ ②

由①②，得 $\tilde{v}_1' = -e\tilde{v}_1 = -\dfrac{e m_2}{m_1+m_2} v_1$

$\tilde{v}_2' = -e\tilde{v}_2 = -\dfrac{e m_1}{m_1+m_2} v_1$

$T = \frac{1}{2}m_1\tilde{v}_1^2 + \frac{1}{2}m_2\tilde{v}_2^2 = \frac{1}{2}\dfrac{m_1 m_2}{m_1+m_2} v_1^2 = \frac{1}{2}\mu v_1^2$

2.13 ① 运用变质量物体动力学方程 $\dfrac{d}{dt}(m\vec{v}) - \dfrac{dm}{dt}\vec{u} = \vec{F}$



其中 m 为下垂部分质量，$m = \sigma y$

$\vec{F}$ 为动力与保守力合力

因 u 沿 y 方向合量 $u_y = 0$，由上述方程组为

$\dfrac{d}{dt}(\sigma y \cdot \dot{y}) = \sigma y \cdot g - T$ ①

水平部分 $\dfrac{d}{dt}[\sigma(l-y)\dot{y}] = T$ ② → 变质量物体运动方程

联立①②，有：

$\sigma\dot{y}^2 + \sigma y\ddot{y} = \sigma y g - [-\sigma\dot{y}^2 + \sigma(l-y)\ddot{y}] \Rightarrow \ddot{y} = \dfrac{y}{l}\cdot g$

而 $\ddot{y} = \dfrac{d\dot{y}}{dt} = \dfrac{d\dot{y}}{dy}\dfrac{dy}{dt} = \dfrac{1}{2}\dfrac{d\dot{y}^2}{dy}$

$$\therefore \frac{1}{2}\frac{dy^2}{dy} = \frac{y}{l}\cdot g \quad \Rightarrow \frac{1}{2}dy^2 = \frac{g}{l}\cdot y\,dy$$

$$\Rightarrow \frac{v^2}{2} = \frac{g}{l}\cdot y^2\Big|_{\frac{l}{2}}^{l} = \frac{3gl}{4}$$

$$\Rightarrow v = \frac{1}{2}\sqrt{3gl}$$

② 运用机械能守恒，取不同为势能零点

$$-\frac{m}{2}g\cdot\frac{l}{4} = -mg\cdot\frac{l}{2} + \frac{1}{2}mv^2$$

$$\therefore v = \frac{1}{2}\sqrt{3gl}$$

2.14 自由下落小段绳，空中部分视为自由质点，有 $v = \sqrt{2gy}$

由题意，当空中剩余 x 长绳，则其已下落 $l+h-x$

$$\therefore v = \sqrt{2g(l+h-x)}, \quad m =$$

由变质物体运动方程，因 $\vec{u}=0$，则

$$\frac{d}{dt}(mv) = mg - N$$

$$\Rightarrow \frac{d}{dt}(\sigma x v) = \sigma x g - N$$

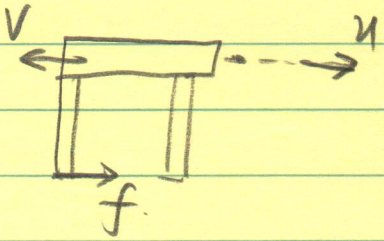
$$\therefore \sigma v^2 + \sigma x\dot{v} = \sigma x g - N$$

自由下落，$\dot{v} = g$，故 $N = \sigma v^2 = \sigma g(2h+2l-2x)$

考虑到已落到地面那段绳子 b.

$$N' = N + m'g = N + \sigma(l-x)g = \sigma g[2h+3(l-x)].$$

2.15. t时刻 机枪与剩余子弹(质量为 $M+M'-mt$，速度为 V)列



运动方程为：

$$\frac{d}{dt}\left[(M+M'-mt)V\right]-(-m)(-u)=-\mu\cdot(M+M'-mt)g$$

$$\therefore\ d\left[(M+M'-mt)V\right]=\left[mu-\mu(M+M'-mt)g\right]dt$$

$t=0,\ V=0$

末态 $t=\frac{M'}{m}$，两边积分：

$$\int_0^V d\left[(M+M'-mt)V\right]=\int_0^{\frac{M'}{m}}\left[mu-\mu(M+M'-mt)g\right]dt$$

$$\therefore\ MV=M'u-\mu g(M+M')\frac{M'}{m}+\mu mg\frac{1}{2}\left(\frac{M'}{m}\right)^2$$

$$=M'u-\mu g\frac{1}{m}\left[MM'+M'^2-\frac{M'^2}{2}\right]$$

$$=M'u-\frac{\mu g}{2m}\left[M'^2+2MM'\right]$$

$$=M'u-\frac{\mu g}{2m}\left[(M'+M)^2-M^2\right]$$

$$\therefore\ V=\frac{M'}{M}u-\frac{\mu g}{2mM}\left[(M'+M)^2-M^2\right]$$

2.16 由题意，$\frac{dm}{dt}=kS$

即 $\rho\frac{d}{dt}\left(\frac{4}{3}\pi r^3\right)=k\cdot 4\pi r^2\ \Rightarrow\ \dot{r}=\frac{k}{\rho}$ 为常数

设 $\lambda=\frac{k}{\rho}=\dot{r}$，$r=a+\lambda t$

质量物体运动方程：$\frac{d}{dt}(mV)=mg$　　$m=\frac{4}{3}\pi r^3$

$$\therefore\ \frac{d}{dt}(r^3V)=r^3g\quad\therefore\ d(r^3V)=(r^3g)dt$$

习 $d(r^3v) = (a+\lambda t)^3 g\,dt$

两边积分, $\int_0^{r^3v} d(r^3v) = \int_0^t (a+\lambda t)^3 g\,dt$

$\therefore r^3v = \frac{1}{4\lambda}[(a+\lambda t)^4 - a^4]g$

因 $r = a+\lambda t$, 故

$$v = \frac{g}{4\lambda}\left[(a+\lambda t) - \frac{a^4}{(a+\lambda t)^3}\right].$$

2.17 喷气速度为火箭与气体之相对速度, $v_0 = 2074\ m/s$

火箭速度为 v, 则 t 时喷气体速度(对地面)为 $v - v_0 = u$

$\lambda = \frac{m_0}{60}$, $m = m_0 - \lambda t$, c为装满燃料后与所燃烧燃料质量之比值

由火箭运动方程

$$\frac{d}{dt}(mv) - \dot{m}u = F$$

即 $\frac{d}{dt}(mv) + \lambda(v - v_0) = -mg$.

$\therefore \frac{d}{dt}[(m_0 - \lambda t)v] + \lambda(v - v_0) = (m_0 - \lambda t)g$

$-\lambda v + (m_0 - \lambda t)\dot{v} + \lambda(v - v_0) = (m_0 - \lambda t)g$

$$\dot{v} = \frac{\lambda v_0}{m_0 - \lambda t} - g$$

$\therefore v = -v_0 \ln(m_0 - \lambda t)\Big|_0^t - gt$

$= -v_0 \ln\left(1 - \frac{\lambda}{m_0}t\right) - gt$

设火箭燃料为质量的 $\frac{n}{n+1}$ 倍时燃料量 $\frac{n}{n+1}m_0$, $T = \frac{n}{n+1}\frac{m_0}{\lambda} = \frac{60n}{n+1}$.

太阳卫星，要求 $v \geq$ 第二宇宙速度，$v_2 = 11.2\,km/s = 11200\,m/s$

由 $v \geq v_2$，得 $11200 = 2074 \cdot \ln\left(1 - \frac{n}{n+1}\right) - \frac{60n}{n+1}$

当 n 很大时，右边 $\frac{n}{n+1} \to 1$，$\frac{60n}{n+1} \simeq 60$

$\therefore 11200 = 2074 \cdot \ln(n+1) - 600$

解得 $n \approx 295$

2.18 用 $\frac{d\tilde{m}}{dt} = -\alpha m_0$，常数，故 $\tilde{m} = (1-\alpha t) m_0$

系统的动量定理为

$$\frac{d}{dt}(\tilde{m}\tilde{v}) - \frac{d\tilde{m}}{dt} \cdot (\tilde{v} - v) = -\tilde{m} g$$

$$\therefore -\alpha m_0 \tilde{v} + \tilde{m}\dot{\tilde{v}} + \alpha m_0 (\tilde{v} - v) = -\tilde{m} g$$

$$\therefore \tilde{m}\dot{\tilde{v}} = \alpha m_0 v - \tilde{m} g$$

火箭若要上升，则有 $\dot{\tilde{v}} > 0$，即 $\alpha m_0 v - \tilde{m} g > 0$

$\therefore$ 至少 $\alpha v > g$

将 m 代入，有 $\dot{\tilde{v}} = \frac{\alpha m_0}{m} v - g = \frac{\alpha v}{1-\alpha t} - g$

$$\therefore \tilde{v} = -v \ln(1-\alpha t) - gt$$

$\tilde{v}$ 最大，对应 $\alpha m_0 t = m_0 - m$，$\therefore t = \frac{1}{\alpha}\left(1 - \frac{m}{m_0}\right)$，

$$\therefore \tilde{v} = v \ln \frac{m_0}{m} - \frac{g}{\alpha}\left(1 - \frac{m}{m_0}\right)$$

$h_1 = \int_0^t \tilde{v}\,dt = \frac{v}{\alpha}\left[(1-\alpha t)\ln(1-\alpha t) + \alpha t\right] - \frac{1}{2}gt^2$

$1-\alpha t = \frac{m}{m_0}$

$\therefore h_1 = \frac{v}{\alpha}\left[\frac{m}{m_0}\ln\left(\frac{m}{m_0}\right) + 1 - \frac{m}{m_0}\right] - \frac{g}{2}\cdot\frac{1}{\alpha^2}\left(1-\frac{m}{m_0}\right)^2$

$h_2 = \frac{\tilde{v}^2}{2g} = \frac{1}{2g}\left[v^2\left(\ln\frac{m_0}{m}\right)^2 + \frac{g^2}{\alpha^2}\left(1-\frac{m}{m_0}\right)^2 - \frac{2vg}{\alpha}\left(1-\frac{m}{m_0}\right)\ln\frac{m_0}{m}\right]$

$= \frac{v^2}{2g}\left(\ln\frac{m_0}{m}\right)^2 + \frac{g}{2\alpha^2}\left(1-\frac{m}{m_0}\right)^2 - \frac{v}{\alpha}\left(1-\frac{m}{m_0}\right)\ln\frac{m_0}{m}$

$\therefore h = h_1 + h_2 = \frac{v^2}{2g}\left(\ln\frac{m_0}{m}\right)^2 + \frac{v}{\alpha}\left[1 - \frac{m}{m_0} - \ln\frac{m_0}{m}\right]$

2.19. $\bar{T} = \frac{1}{T}\int_0^T \frac{1}{2}mv^2\,dt = \frac{1}{T}\int_0^T \frac{1}{2}m\vec{v}\cdot d\vec{r} = \frac{1}{T}\left[\frac{1}{2}m\vec{v}\cdot\vec{r}\Big|_0^T - \frac{1}{2}\int_0^T \vec{r}\,d(m\vec{v})\right]$

$\underline{\qquad\qquad}$ 0

$= -\frac{1}{2T}\int_0^T \vec{r}\cdot\vec{F}\cdot dt = -\frac{1}{2T}\int_0^T \vec{r}\cdot\left(-\frac{GM\cdot m}{r^2}\cdot\hat{e}_r\right)\cdot dt$

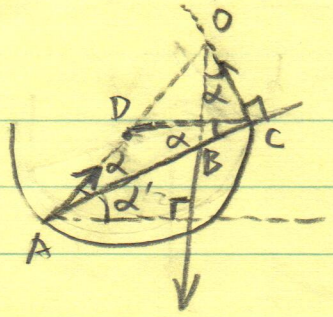
$= \frac{-1}{2T}\int_0^T \underbrace{\frac{-GM\cdot m}{r}}_{V(r)}\,dt = -\frac{1}{2}\bar{V}$

$\therefore \bar{T} = -\frac{1}{2}\bar{V}$ 此为位力定理之应用

$V = Ar^{n+1}$

$F \sim r^n$, $V \sim r^{n+1}$, 则 $\bar{T} = \frac{1}{2}(n+1)\bar{V}$

3.1



平行的，光线行三角形交于一点

CD为光线宽度

$\cos\alpha = \frac{c}{2r}$

$\sin^2\alpha = \frac{4r^2-c^2}{4r^2}$

$AB = AC - BC$

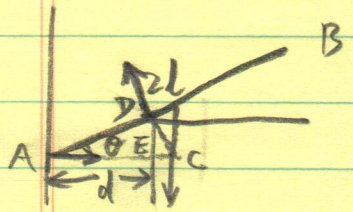
$= c - OC\cdot\tan\alpha$

$= c - c\cdot\tan\alpha\tan\alpha$

$= c(1-\tan^2\alpha) = c\times\left(1-\frac{4r^2-c^2}{c^2}\right) = \frac{2c^2-4r^2}{c}$

$\therefore l = 2AB = \frac{4}{c}(c^2-2r^2)$

3.2



$d = AE = AD\cdot\cos\theta =$

$\hookrightarrow AC\cdot\cos\theta$

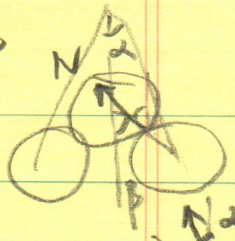
$\hookrightarrow l\cdot\cos\theta$

$= l\cdot\cos^3\theta$

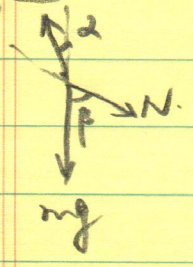
$\therefore \cos\theta = \left(\frac{d}{l}\right)^{\frac{1}{3}}$

$\therefore \theta = \arccos\left(\frac{d}{l}\right)^{\frac{1}{3}}$

3.4



$$2N\cos\beta = mg \Rightarrow N = \frac{mg}{2\cos\beta}$$



水平: $T\sin\alpha = N\sin\beta$

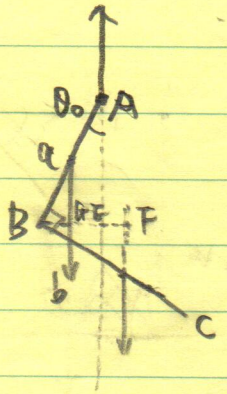
$$\therefore T = \frac{mg}{2}\cdot\frac{1}{\sin\alpha}\cdot\frac{\sin\beta}{\cos\beta}$$

$$T\cos\alpha = N\cos\beta + mg$$

由 $\frac{mg}{2}\cdot\frac{\cos\alpha}{\sin\alpha}\cdot\frac{\sin\beta}{\cos\beta} = \frac{mg}{2} + mg$

$$\tan\beta = 3\cdot\tan\alpha$$

3.3



取A為基準, 力矩平衡

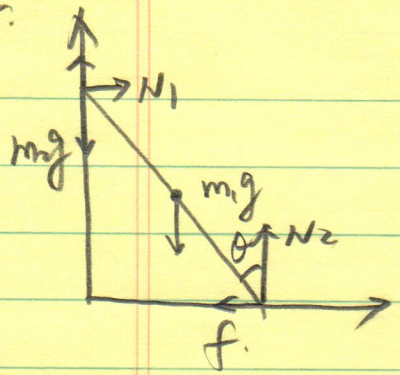
$$\Rightarrow \sigma a\cdot GE = \sigma b\, EF.$$

由 $a\times\frac{a}{2}\sin\theta_0 = b\times\left(\frac{b}{2}\cos\theta_0 - a\cdot\sin\theta_0\right)$

$$\therefore (a^2+2ab)\sin\theta_0 = b^2\cos\theta_0$$

$$\therefore \tan\theta_0 = \frac{b^2}{a^2+2ab}$$

3.5



① $\mu_1 \mu_2 N_2 + N_2 = m_1 g + m_2 g$

$\Rightarrow N_2 = \frac{(m_1+m_2)g}{\mu_1\mu_2+1} = \frac{4 m_1 g}{1+\mu_1\mu_2}$

② 토크 평형:

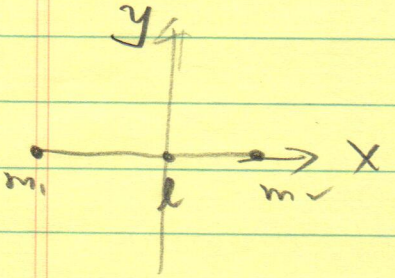
$m_1 g \cdot l \sin\theta + \mu_2 N_2 \cdot 2l \cos\theta = N_2 \cdot \sin\theta \cdot 2l$

$(2N_2 - m_1 g)\sin\theta = 2\mu_2 N_2 \cos\theta$

$\Rightarrow \frac{8 - (1+\mu_1\mu_2)}{1+\mu_1\mu_2}\sin\theta = \frac{2\mu_2}{1+\mu_1\mu_2}\cos\theta$

$\therefore \tan\theta = \frac{8\mu_2}{7-\mu_1\mu_2} = \frac{8\times\frac{1}{2}}{7-\frac{1}{8}} = \frac{24}{41}$

3.6 (1)



m_1, $\left(-\frac{m_2}{m_1+m_2}l, 0, 0\right)$

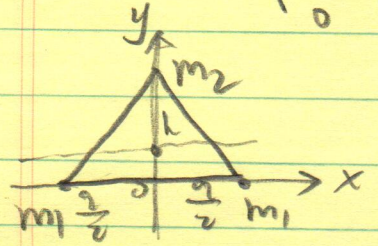
m_2, $\left(+\frac{m_1}{m_1+m_2}l, 0, 0\right)$

$\therefore I_{xx} = 0$, $I_{yy} = I_{zz} = m_1\left(\frac{m_2}{m_1+m_2}l\right)^2 + m_2\left(\frac{m_1}{m_1+m_2}l\right)^2$

$= \frac{m_1^2 + m_2^2}{m_1+m_2} l^2 = I_2$

$\therefore I = \begin{pmatrix} 0 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_2 \end{pmatrix}$

(2)



m_1 : $\left(\frac{a}{2}, -b, 0\right)$

m_2 : $\left(-\frac{a}{2}, -b, 0\right)$

m_2 : $(0, h-b, 0)$

$2m_1 b = m_2(h-b)$

$\therefore b = \frac{m_2}{2m_1+m_2}h$

$h - b = \frac{2m_1}{2m_1+m_2}h$

$I_{xx} = m_1(b^2+b^2) + m_2(h-b)^2 = \frac{2m_1 m_2^2 h^2 + m_2 4m_1^2 h^2}{(2m_1+m_2)^2} = \frac{2m_1 m_2 h^2}{2m_1+m_2}$

$I_{yy} = m_1 \cdot \frac{a^2}{2}$, $I_{zz} = \left(\frac{a^2}{2}+b^2\right)m_1 + m_2(h-b)^2 = I_{xx} + I_{yy}$

3.7: 令 $\xi = \frac{x}{a}$, $\eta = \frac{y}{b}$, $\zeta = \frac{z}{c}$

$\Rightarrow \xi^2+\eta^2+\zeta^2 = 1$ $\qquad \rho \cdot \frac{4}{3}\pi abc = m$

$$I_{xx} = \rho\int (y^2+z^2)\,dV = \rho\int (b^2\eta^2 + c^2\zeta^2)\,dV\cdot abc$$

$$= \rho abc \int (b^2 \sin^2\theta \sin^2\varphi + c^2\cos^2\theta)\, r^4 \sin\theta\, dr\, d\theta\, d\varphi$$

$$= \rho abc \cdot \frac{1}{5}\left[\int b^2 \sin^3\theta \sin^2\varphi\, d\theta d\varphi + c^2 \int \cos^2\theta \sin\theta\, d\theta d\varphi\right]$$

$$= \frac{\rho}{5} abc \left[\pi b^2 \left(\frac{\cos^3\theta}{3} - \cos\theta\right)\Big|_1^{-1} + 2\pi c^2 \cdot \frac{-\cos^3\theta}{3}\Big|_1^{-1}\right]$$

$$= \frac{\rho}{5} abc \cdot \left[\frac{4\pi}{3} b^2 + \frac{4\pi}{5} c^2\right]$$

$$= \rho \cdot \frac{4\pi}{3} abc \cdot \frac{1}{5}(b^2+c^2) = \frac{1}{5} m(b^2+c^2)$$

$I_{yy} = \frac{1}{5} m(a^2+c^2)$ $\qquad I_{zz} = \frac{1}{5} m(a^2+b^2)$

3.8. $I_{xx} = I_{yy} = I_{zz} = \int \rho\cdot(x^2+y^2)\, r^2 \sin\theta\, dr\, d\theta\, d\varphi$

$$= \rho_0 \int \left(1-\alpha\frac{r^2}{R^2}\right) r^4 \sin^3\theta\, dr\, d\theta\, d\varphi$$

$$= 2\pi\rho_0 \int \left(1-\alpha\frac{r^2}{R^2}\right)\cdot r^4\, dr \left(\cos\theta - \frac{\cos^3\theta}{3}\right)\Big|_{-1}^{1}$$

$$= \frac{8\pi\rho_0}{3} \times \left(\frac{R^5}{5} - \alpha\frac{R^7}{7R^2}\right)$$

$$= \frac{8\pi\rho_0 R^5}{3}\cdot\frac{7-5\alpha}{35}$$

$$m = \int \rho\, dV = 4\pi\rho_0 \int \left(1-\alpha\frac{r^2}{R^2}\right) r^2\, dr = 4\pi\rho_0\left(\frac{R^3}{3} - \frac{\alpha R^5}{5R^2}\right)$$

<5>

$= \frac{4\pi\rho_0 R^3}{15}$ (5-32)

$\therefore I_{xx} = mR^2$

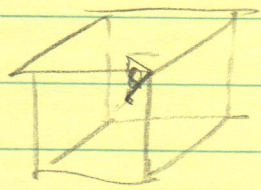
$\Rightarrow R^2 = \frac{I_{xx}}{m} = \frac{8}{3\times 35} \times \frac{15}{4} \times \frac{7-2\alpha}{5-2\alpha} R^2 = \frac{14-4\alpha}{35-21\alpha} R^2$

$\therefore k = \sqrt{\frac{14-4\alpha}{35-21\alpha}} R$

3.9 정육면체 $\rho = \frac{m}{a^3}$

$I_{xx} = \frac{m}{a^3} \int (y^2+z^2) \hat{\rho}\, dx dy dz$

$= \frac{m}{a^3} \cdot a \cdot a \cdot \frac{1}{3} \frac{a^3}{4} \times 2 = \frac{1}{6} ma^2$



$\therefore I = \begin{pmatrix} \frac{1}{6}ma^2 & 0 & 0 \\ 0 & \frac{1}{6}ma^2 & 0 \\ 0 & 0 & \frac{1}{6}ma^2 \end{pmatrix}$

$\sin\theta = \sqrt{\frac{2}{3}} \quad \cos\theta = \sqrt{\frac{1}{3}}$

$\rho = \frac{2}{4}$

$\hat{e}_n = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right) = \frac{1}{\sqrt{3}}(1,1,1)$

$\therefore I = \hat{e}_n \cdot I \cdot \hat{e}_n = \frac{1}{6}ma^2 = mk^2$

$\therefore k = \frac{a}{\sqrt{6}}$ $d = \sqrt{3}\cdot a$

$\therefore k = \frac{a}{\sqrt{6}} = \frac{d}{\sqrt{3}\sqrt{6}} = \frac{d}{3\sqrt{2}}$

3.10 $I = \frac{m}{\pi a^2} \int r^2 \cdot 2\pi r dr = \frac{2m}{a^2} \cdot \frac{a^4}{4} = \frac{1}{2}ma^2$

$f = \frac{\mu mg}{\pi a^2}$ $\therefore M = \int r \cdot f ds = \int r \cdot \frac{\mu mg}{\pi a^2} r dr d\varphi = \frac{2\mu mg}{a^2} \frac{a^3}{3} = \frac{2\mu mga}{3}$

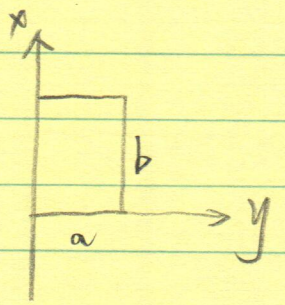
$\therefore \dot{J} = M \Rightarrow \Delta t = \frac{J}{M} = \frac{\frac{1}{2}ma^2\omega_0}{\frac{2}{3}mmga} = \frac{3a\omega_0}{4\mu g}$.

3.11 $\vec{J} = I\vec{\omega}$ $\quad J = I\omega$. $\quad M = -k\omega$.

$\therefore I\dot{\omega} = -k\omega \quad \therefore \omega = \omega_0 e^{-kt/I}$

$\omega(\Delta t) = \frac{\omega_0}{2} = \omega_0 e^{-k\Delta t/I} \quad \therefore \frac{k\Delta t}{I} = \ln 2 \quad \therefore \Delta t = \frac{I}{k}\ln 2$

$n = \frac{1}{2\pi}\int_0^{\Delta t} \omega(t)\,dt = \frac{\omega_0}{2\pi}\cdot\frac{I}{k}\left(-e^{-kt/I}\right)\Big|_0^{\Delta t} = \frac{I\omega_0}{4\pi k}$

3.12 

$I = \int \rho \cdot y^2\,ds$

$= \int_0^a\int_0^b \frac{m}{ab}\, y^2\,dx\,dy$

$= \frac{m}{ab}\cdot b\cdot\frac{a^3}{3} = \frac{1}{3}ma^2$

$f = kv^2 = k\omega^2 y^2$

$\therefore M = -\int f\,ds = -\int_0^a\int_0^b y\,k\omega^2 y^2\,dx\,dy = -\frac{k}{4}\omega^2 a^4 b$

$\therefore J = I\omega \qquad \dot{J} = M \Rightarrow I\dot{\omega} = -\frac{k}{4}\omega^2 a^4 b$

$\dot{\omega}\frac{1}{3}ma^2 = -\frac{k}{4}\omega^2 a^4 b$

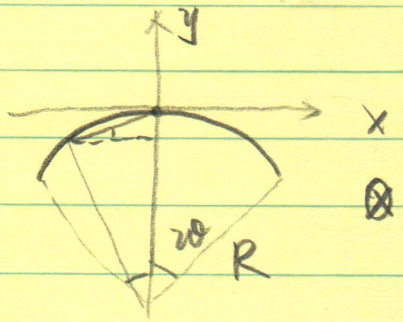
$\dot{\omega} = -\frac{3ka^3b}{4m}\omega^2$

$\therefore \frac{d\omega}{\omega^2} = -\frac{3ka^3b}{4m}dt \quad \therefore -\frac{1}{\omega} + c = \frac{3ka^3b}{4m}t$

$t=0,\ \omega=\omega_0 \Rightarrow c = -\frac{1}{\omega_0} \quad \therefore \omega = \left(\frac{1}{\omega_0} + \frac{3ka^3b}{4m}t\right)^{-1}$

ω 는 $\frac{\omega_0}{2}$ 가 되는 t 는 $\frac{2}{\omega_0}$ ∴ $\frac{3ka^4b}{4m}t = \frac{1}{\omega_0}$ ∴ $t = \frac{4m}{3ka^4b\omega_0}$ <7>

3.13



$\sigma = \frac{m}{2R\theta}$

$\therefore I_{zz} = \int \sigma \cdot (x^2+y^2)\, ds = \int_{-\theta}^{\theta} \sigma \cdot (2R\sin\frac{\varphi}{2})^2 \cdot R\, d\varphi$

$= 4\sigma R^3 \int_{-\theta}^{\theta} \sin^2\frac{\varphi}{2}\, d\varphi = 4\sigma R^3 \int_{-\theta}^{\theta} \frac{1}{2}(1-\cos\varphi)\, d\varphi$

$= 4\sigma R^3(\theta - \sin\theta) = 2mR^2\left(1-\frac{\sin\theta}{\theta}\right)$

$y_c = \frac{1}{m}\int \sigma y\, ds = \frac{1}{m}\cdot\frac{m}{2R\theta}\int_{-\theta}^{\theta}(R-R\cos\varphi)\cdot R\, d\varphi$

$= \frac{R}{2\theta}\int_{-\theta}^{\theta}(1-\cos\varphi)\, d\varphi = \frac{R}{2\theta}(2\theta - 2\sin\theta) = R\left(1-\frac{\sin\theta}{\theta}\right)$

$M_z = mgR\left(1-\frac{\sin\theta}{\theta}\right)\sin\varphi$

$I_{zz}\cdot\ddot{\varphi} = M_z$,

$2mR^2\left(1-\frac{\sin\theta}{\theta}\right)\ddot{\varphi} = -mgR\left(1-\frac{\sin\theta}{\theta}\right)\sin\varphi$

$\therefore \ddot{\varphi} = -\frac{g}{2R}\sin\varphi$

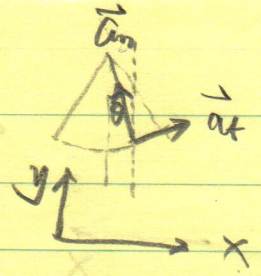
$\varphi \ll 1$ ∴ $\sin\varphi \sim \varphi$ ∴ $\ddot{\varphi} + \frac{g}{2R}\varphi \simeq 0$

$\therefore \varphi = \varphi_0 + \cos\left(\sqrt{\frac{g}{2R}}\,t + \varepsilon\right)$

$\therefore T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{2R}{g}}$

3.14

$I = mk^2$



机械能守恒 $\Rightarrow mgl(\cos\theta - \cos\theta_0) = \frac{1}{2} I \omega^2$

$= \frac{1}{2} m k^2 \omega^2$

$\omega < 0$

$\therefore \omega^2 = \frac{2gl}{k^2}(\cos\theta - \cos\theta_0)$ $\quad \therefore 2\omega\dot{\omega} = -\frac{2gl}{k^2}\sin\theta \cdot \omega$

$\therefore \dot{\omega} = -\frac{gl}{k^2}\sin\theta$

$\begin{cases} a_n = \omega^2 a \\ a_t = a\dot{\omega} \end{cases}$

水平方向: $R_x = m a_x = m(a_t \cos\theta - a_n \sin\theta)$

$= m \cdot a\left[-\frac{gl}{k^2}\sin\theta\cos\theta - \frac{2gl}{k^2}(\cos\theta - \cos\theta_0)\sin\theta\right]$

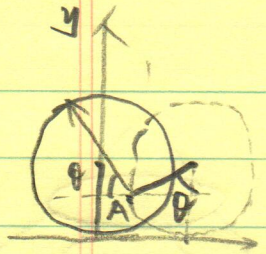
$= -\frac{ma^2}{k^2}(3\cos\theta - 2\cos\theta_0)\sin\theta$

$R_y - mg = m(a_n \cos\theta + a_t \sin\theta)$

$\therefore R_y = mg + m\left[\frac{2ga^2}{k^2}(\cos\theta - \cos\theta_0)\cos\theta - \frac{ga^2}{k^2}\sin^2\theta\right]$

$= mg\left\{1 + \frac{a^2}{k^2}\left[(3\cos\theta - 2\cos\theta_0)\cos\theta - 1\right]\right\}$

3.15



取轮心为运动系，则主动小轮的角速度为，

$$\omega = \frac{v_0}{r}, \quad \theta = \omega t = \frac{v_0 t}{r}$$

$$\therefore \vec{V}_A = \vec{V}_0 + \vec{V}_{A'} = v_0 \hat{e}_x + \omega r(-\cos\theta \hat{e}_x + \sin\theta \hat{e}_y)$$

$$= v_0\left(1 - \cos\frac{v_0 t}{r}\right)\hat{e}_x + v_0 \sin\frac{v_0 t}{r}\hat{e}_y$$

$$\therefore |\vec{V}_A|^2 = V_{A,x}^2 + V_{Ay}^2 = v_0^2\left(2 - 2\cos\frac{v_0 t}{r}\right) = 4v_0^2 \sin^2\frac{v_0 t}{2r}$$

$$= 4v_0^2 \sin^2\frac{\theta}{2}$$

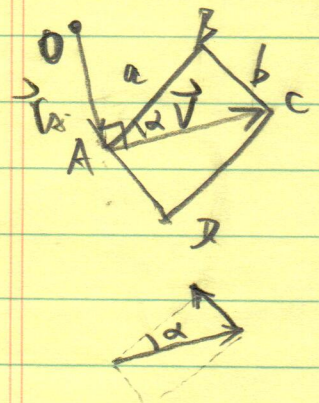
① 当 $t=0$，有 $|\vec{V}_A| = 0$

② $t = \frac{\pi r}{v_0}$，有 $|\vec{V}_A| = 2v_0$ ↑ $|\vec{V}_A| = 2v_0 \cdot \cos\frac{\theta'}{2}$

$\theta = \pi - \theta'$

轮边缘上为轮与地面接触点

3.16



$$\vec{V} = \vec{\omega} \times \vec{r}$$

$$\vec{V}_B = \vec{\omega} \times \vec{r}_B = \vec{\omega} \times (\vec{r}_A + \vec{r}_B')$$

$$\therefore = (V\cos\alpha)^2 + (V\sin\alpha - \omega a)^2$$

$$= V^2 + \omega^2 a^2 - 2V\omega a \cdot \frac{b}{\sqrt{a^2+b^2}}$$

十一题判别

$$\vec{r}_B' = -a\sin\alpha \hat{e}_r + a\cos\alpha \hat{e}_\theta, \quad \vec{\omega} = \omega \cdot \hat{e}_z$$

$$\vec{r}_A = r_A \cdot \hat{e}_r$$

$$\therefore \vec{\omega} \times \vec{r}_A = \omega r_A \hat{e}_\theta = V \hat{e}_\theta$$

$$\vec{\omega} \times \vec{r}_B = -\omega a \sin\alpha \hat{e}_\theta - \omega a \cos\alpha \hat{e}_r$$

$$\therefore \vec{V}_B = -\omega a\cos\alpha \hat{e}_r + (V - \omega a \sin\alpha)\hat{e}_\theta$$

$$\therefore V_B^2 = \omega^2 a^2 + V^2 - 2V\omega a \sin\alpha$$

$$= \omega^2 a^2 + V^2 - 2V\omega a \cdot \frac{b}{\sqrt{a^2+b^2}}$$

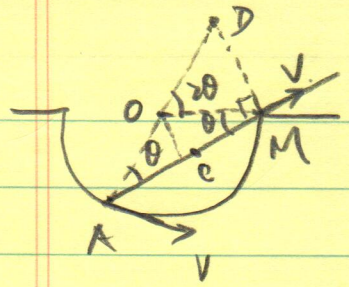
$$\vec{r}_{BC} = b\cos\alpha \hat{e}_r + b\sin\alpha \hat{e}_\theta$$

$$\vec{V} \cdot \vec{r}_{BC} = V\sin\alpha - \omega a$$

$$\sin = V_B^2 - (\vec{V}\hat{e}_{BC})^2 = V^2\cos^2\alpha$$

$$\tan = \frac{V\cos\alpha}{V\sin\alpha - \omega a}$$

3.17.



<07>

(1) 杆上任一点C，距A端为a，则以M为极，

∴ MO为极轴，则有

$$\rho = MC = 2r\cos\theta - a.$$

(2) 轨迹及速度：

A点轨迹为圆，速度沿圆切线，垂直于AO.

M点圆心，杆上与M接触点速度沿杆，垂直于杆

的线，即沿DV.

$$\left.\begin{aligned}\angle ODV &= \frac{\pi}{2} - \theta \\ \angle DVO &= \frac{\pi}{2} - \theta\end{aligned}\right\} \Rightarrow OD = OV = r$$

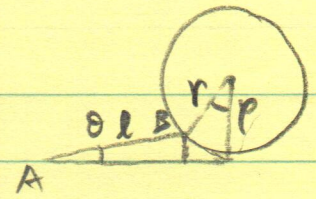
∴ D点轨迹（瞬时轨迹）为圆

以A为圆心方向，AB为x轴，易见 $AD = 2r =$ 半径

∴ A点在AB杆上下轨迹，本体轨迹为半径为r的圆.

3.18:

$$x = -l\cos\theta + r(\varphi - \sin\varphi)$$



$$l\sin\theta = r(1-\cos\varphi)$$

$$\therefore l\dot\theta\cos\theta = r\sin\varphi\,\dot\varphi \quad \therefore \dot\theta = \frac{r\sin\varphi}{l\cos\theta}\dot\varphi \qquad \dot\varphi = \frac{V_0}{r}$$

$$\sin\theta = \frac{r}{l}(1-\cos\varphi) \qquad \frac{1}{\cos\theta} = \frac{l}{\sqrt{l^2 - r^2(1-\cos\varphi)^2}} = \frac{l}{\sqrt{l^2 - 4r^2\sin^4\frac{\varphi}{2}}}$$

$$\therefore \dot x = l\dot\theta\sin\theta + r\dot\varphi(1-\cos\varphi)$$

$$= \dot\theta\, r(1-\cos\varphi) + r\dot\varphi(1-\cos\varphi) \qquad r\dot\varphi = V_0$$

$$= \frac{r\sin\varphi}{l}\cdot\frac{l}{\sqrt{l^2-4r^2\sin^2\frac{\varphi}{2}}}\cdot\dot\varphi\cdot r(1-\cos\varphi) + r\dot\varphi\cdot 2\sin^2\frac{\varphi}{2}$$

$$= \left[\frac{r\sin\varphi}{\sqrt{l^2-4r^2\sin^2\frac{\varphi}{2}}} + 1\right] 2V_0\sin^2\frac{\varphi}{2}$$

3.19. $I = \frac{4}{3}ma^2$ $\qquad mga = \frac{1}{2}I\omega^2$

$$\Rightarrow \omega^2 = \frac{2mga}{I} = \frac{2\times mga}{\frac{4}{3}ma^2} = \frac{3g}{2a}$$

$$V_{cl}(0) = \omega\cdot a = \sqrt{\frac{3ga}{2}}$$

$$h = \frac{1}{2}gt^2 \Rightarrow t = \sqrt{\frac{2h}{g}} \qquad \therefore n = \frac{\omega\cdot t}{2\pi} = \sqrt{\frac{3g}{2a}}\times\sqrt{\frac{2h}{g}}\times\frac{1}{2\pi}$$

$$= \frac{1}{2\pi}\sqrt{\frac{3h}{a}}$$

3.20. $I = \frac{1}{2}Ma^2 + Ma^2 = \frac{3}{2}Ma^2$ $\qquad \omega\cdot 2a = V \quad \therefore \omega a = \frac{V}{2}$

$$mgh = \frac{1}{2}mV^2 + \frac{1}{2}M\left(\frac{V}{2}\right)^2 + \frac{1}{2}\cdot\frac{1}{2}Ma^2\left(\frac{V}{2a}\right)^2 = \frac{V^2}{2}\left(m + \frac{3}{8}M\right) = \frac{V^2(3M+8m)}{16}$$

$$h = \frac{V^2}{2a_2} \quad \therefore a_2 = \frac{V^2}{2h} = \frac{8m}{3M+8m}g, \quad a_1 = \frac{a_2}{2} = \frac{4m}{3M+8m}g$$

$mg - T = ma_2$ $\quad \therefore T = mg - ma_2 = \frac{3Mm}{3M+8m} g$

3.21

① $mg - T = ma = mr\ddot{\varphi}$

② $Tr - G = I\ddot{\varphi}$

$\therefore mgr - mr^2\ddot{\varphi} - G = I\ddot{\varphi}$ $\quad \therefore \ddot{\varphi} = \frac{mgr - G}{I + mr^2} = \dot{\omega}$

$\varphi = \frac{1}{2}\alpha t^2 = \frac{\omega^2}{2\alpha}$

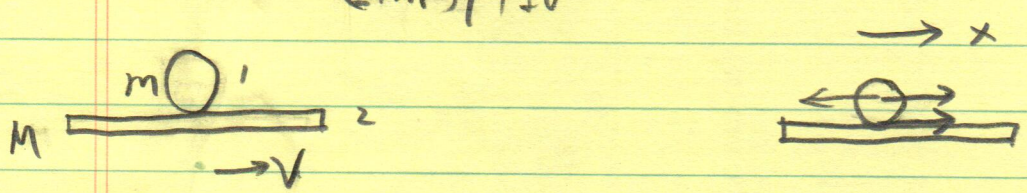
$\therefore \alpha = \frac{\omega^2}{2\varphi}$ $\quad \therefore \frac{\alpha_1}{\alpha_2} = \frac{\varphi_2}{\varphi_1}$

$\therefore \frac{\ddot{\varphi}}{G/I} = \frac{\varphi}{\theta}$ $\quad \therefore \frac{I(mgr - G)}{G(I + mr^2)} = \frac{\varphi}{\theta}$

$I\varphi G + \varphi G mr^2 = Imgr\theta - IG\theta$

$G = \frac{mgIr\theta}{(I+mr^2)\varphi + I\theta}$

3.22



$a_2 = -\frac{\mu m}{M} g$ $\quad V_2 = V - \frac{\mu mg}{M} t$

$\alpha = \frac{\mu mgr}{\frac{2}{5}mr^2} = \frac{5\mu g}{2r}$ $\quad V_1 = \mu gt + \frac{5}{2}\mu gt = \frac{7}{2}\mu gt$

$V_2 = V_1$, 가속

$\therefore V = \mu gt\left(\frac{m}{M} + \frac{7}{2}\right)$ $\quad \therefore t = \frac{V}{\mu g\left(\frac{m}{M} + \frac{7}{2}\right)}$

3.23:

$F-\mu(W_1+W_2)=f=m_1\cdot a_1$

$f=m_2a_2 \quad \therefore a_2=\frac{f}{m_2}$

$fr=\frac{1}{2}m_2r^2\cdot\alpha \Rightarrow \alpha=\frac{2f}{m_2r} \quad \therefore \omega=\frac{2ft}{m_2r}$

$\therefore V_2=V_{20}+\omega r$

$=\frac{ft}{m_2}+\frac{2ft}{m_2}=\frac{3ft}{m_2}$

$V_1=a_1t=\frac{F-\mu(W_1+W_2)}{m_1}t-\frac{f}{m_1}t$

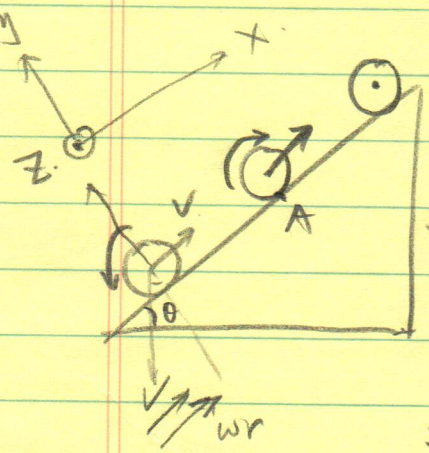
$V_1=V_2 \Rightarrow 3W_1f=W_2F-\mu W_2(W_1+W_2)-W_2f$

$f=\frac{W_2[F-\mu(W_1+W_2)W_2]}{3W_1+W_2}$

$a_1=\frac{F-\mu(W_1+W_2)-f}{M_1}=\frac{3[F-\mu(W_1+W_2)W_2]}{3W_1+W_2}\cdot g$

$=\frac{F-\mu(W_1+W_2)W_2}{W_1+\frac{W_2}{3}}\cdot g$

3.24



2阶段: ①滑动减速 ②滚动减速

①滑动阶段: 动量守恒 $\begin{cases} -[\mu mg\cos\theta+mg\sin\theta]=ma \\ \mu mg\cos\theta\cdot r=\frac{2}{5}mr^2\cdot\alpha \end{cases}$

$\Rightarrow \begin{cases} a=-g(\sin\theta+\mu\cos\theta) \\ \alpha=\frac{5\mu g\cos\theta}{2r} \end{cases}$

$\vec{V}_A=\vec{V}_C+\vec{\omega}\times\vec{r}$

$(\omega-\alpha t)\,\hat{e}_\theta\times(-r)\,\hat{e}_y=(\omega-\alpha t)\cdot r\,\hat{e}_x$

停止滑动时有: $V_A=0$ 即 $0=V+at-(\alpha t-\omega)r$

$$0 = v - g(\sin\theta + \mu\cos\theta)t_1 - \left(\frac{5\mu g\cos\theta}{2r}t_1 - \omega\right)r$$

$$\therefore t_1 = \frac{v + \omega r}{g(\sin\theta + \mu\cos\theta) + \frac{5}{2}\mu g\cos\theta} = \frac{v + \omega r}{g(\sin\theta + \frac{7}{2}\mu\cos\theta)}$$

② 无滑动时的角加速度, $I' = \frac{2}{5}mr^2 + mr^2 = \frac{7}{5}mr^2$

$$M = mg\sin\theta \cdot r$$

$$\therefore \alpha = \frac{M}{I'} = \frac{5g\sin\theta}{7r} \qquad \therefore \frac{\alpha'}{\alpha} = \frac{5\mu g\cos\theta}{2r} \cdot \frac{7r}{5g\sin\theta} = \frac{7\mu\cos\theta}{2\sin\theta}$$

$$\therefore t_2 = \frac{\alpha' t_1 - \omega}{\alpha}$$

$$\therefore t = t_1 + t_2 = \left(1 + \frac{\alpha'}{\alpha}\right)t_1 - \frac{\omega}{\alpha}$$

$$= \left(1 + \frac{7\mu\cos\theta}{2\sin\theta}\right)t_1 - \frac{7\omega r}{5g\sin\theta}$$

$$= \frac{2\sin\theta + 7\mu\cos\theta}{2\sin\theta} \cdot \frac{2v + 2\omega r}{g(2\sin\theta + 7\mu\cos\theta)} - \frac{7\omega r}{5g\sin\theta}$$

$$= \frac{v + \omega r}{g\sin\theta} - \frac{7\omega r}{5g\sin\theta}$$

$$= \frac{5v - 2\omega r}{5g\sin\theta}$$

3.25.

$$\frac{1}{2}\cdot\frac{2}{5}mr^2\cdot\omega^2+\frac{1}{2}m(\omega r)^2=mg(R+r)(1-\cos\varphi)$$

$$\therefore \boxed{\frac{7}{10}\omega^2r^2=g(R+r)(1-\cos\varphi)} \quad v^2=\frac{10g(R+r)(1-\cos\varphi)}{7}$$

两边求导，有 $\frac{7}{10}\cdot 2v\dot{v}r^2=g(R+r)\dot{\varphi}\sin\varphi$

$$\therefore \dot{\omega}=\frac{5}{7}\frac{g(R+r)}{r^2}\sin\varphi$$

对环：$\left.\begin{aligned} mg\sin\varphi-f&=m\dot{\omega}r\\ f\cdot r&=\frac{2}{5}mr^2\cdot\dot{\omega}\end{aligned}\right\}\Rightarrow mg\sin\varphi=\frac{7}{5}mr\dot{\omega}$

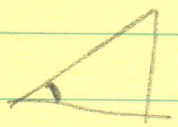
$$\therefore \dot{\omega}=\frac{5g\sin\varphi}{7r}$$

$$a=\dot{\omega}r=\frac{5g\sin\varphi}{7}$$

$$\therefore f=\frac{2}{5}mr\dot{\omega}=\frac{2}{7}mg\sin\varphi$$

$$m\frac{v^2}{R+r}=\frac{10}{7}mg(1-\cos\varphi)$$

$$\therefore mg\cos\varphi-N=m\frac{v^2}{R+r}=\frac{10}{7}mg(1-\cos\varphi)$$



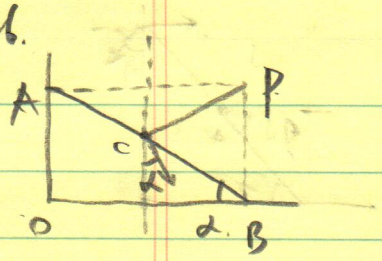
$$\therefore N=\frac{mg}{7}(17\cos\varphi-10)$$

$f=\mu N \Rightarrow \frac{\mu}{7}(17\cos\varphi-10)=\frac{2}{7}\sin\varphi$ $\quad \mu=\tan\lambda$

$$\therefore 17\sin\lambda\cos\varphi-10\sin\lambda=2\sin\varphi\cos\lambda$$

$$2\sin(\varphi-\lambda)=5\sin\lambda(3\cos\varphi-2)$$

3.26.



杆：$I=\frac{m}{2l}\int_{-l}^{l}x^2dx=\frac{1}{3}ml^2$

杆棒的动能：

$$mgl(\sin\alpha-\sin\beta)=\frac{1}{2}I\omega^2+\frac{1}{2}mv^2$$

$v=\omega r=\omega l$

$$=\frac{1}{2}\cdot\frac{1}{3}ml^2\omega^2+\frac{1}{2}m\cdot\omega^2l^2$$

$$=\frac{2}{3}ml^2\omega^2$$

$$\therefore \omega^2=\frac{3g}{2l}(\sin\alpha-\sin\beta)$$

$$2\omega\dot{\omega}=\frac{3g}{2l}(-\dot{\beta}\cos\beta)$$

$$\therefore \dot{\omega}=-\frac{3g}{4l}\cos\beta$$

$v_{Ax}=l\omega\sin\beta$

$$\therefore \dot{v}_{Ax}=l\dot{\omega}\sin\beta+l\omega^2\cos\beta$$

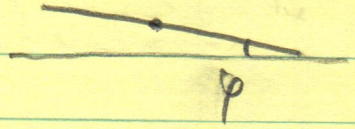
$$=-\frac{3g}{4}\sin\beta\cos\beta+\frac{3g}{2}(\sin\alpha\cos\beta-\sin\beta\cos\beta)$$

$$=\frac{3g}{4}[2\sin\alpha\cos\beta-3\sin\beta\cos\beta]$$

$\because N=m\dot{v}_{Ax}$ 刚起动时，有 $N=0\Rightarrow \dot{v}_{Ax}=0$

$$\Rightarrow \sin\beta=\frac{2}{3}\sin\alpha \quad \therefore \beta=\arcsin\frac{2}{3}\sin\alpha$$

3.27 分析：



$I=\frac{4}{3}ml^2$，$M=mg\cdot l\cos\rho$

$$\therefore \alpha=\frac{M}{I}=\frac{3g}{4l}\cos\rho$$

初始时刻，$\omega_0^2 = \frac{3g}{2l}\times\left(\sin\alpha - \frac{2}{3}\sin\alpha\right) = \frac{g\sin\alpha}{2l}$

$\therefore \varphi = \int \omega\,dt = \int(\omega_0 + \alpha t)\,dt = \omega_0 t + \frac{\alpha}{2}t^2 = \beta$

机械守恒：$\frac{1}{2}I\Omega^2 + \frac{1}{2}m(v_x^2 + v_y^2) = mgl\sin\alpha$

$v_x = \omega_2 l\sin\beta \quad \therefore v_x^2 = \frac{g\sin\alpha}{2l}\,l^2\cdot\frac{4}{9}\sin^2\alpha = \frac{2gl\sin^3\alpha}{9}$

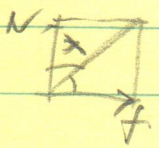
$v_y = \Omega l$

$\therefore \frac{1}{2}I\Omega^2 + \frac{1}{2}ml^2\Omega^2 + \frac{1}{2}mv_x^2 = mgl\sin\alpha$

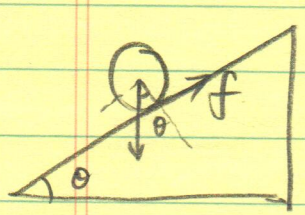
$\Rightarrow \left(\frac{1}{2}\cdot\frac{1}{3}ml^2 + \frac{1}{2}ml^2\right)\Omega^2 + \frac{1}{2}\cdot\frac{2}{9}mgl\sin^3\alpha = mgl\sin\alpha$

$\frac{2}{3}ml^2\cdot\Omega^2 = mgl\sin\alpha\left(1 - \frac{1}{9}\sin^2\alpha\right)$

$\therefore \Omega^2 = \frac{3g\sin\alpha}{2l}\left(1 - \frac{1}{9}\sin^2\alpha\right)$



3.28.



$mg\sin\theta - \mu\cdot mg\cos\theta = ma$

$\therefore a = g\sin\theta - \mu g\cos\theta \qquad \mu = \tan\lambda$

$= \left(g\sin\theta\cos\lambda - g\cdot\cos\theta\sin\lambda\right)\frac{1}{\cos\lambda}$

$= \frac{\sin(\theta-\lambda)}{\cos\lambda}g$

$fa = I\alpha \Rightarrow \mu mg\cos\theta\cdot a = \frac{1}{2}ma^2\alpha$

$\therefore \alpha = \frac{2\mu g}{a}\cos\theta$

3.29 球体: $I = \frac{2}{5}ma^2$.

圆环: $I = \frac{2}{3}m\cdot a^2$.

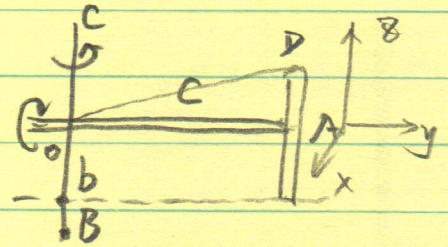
$$\frac{1}{2}m\cdot V_c^2 + \frac{1}{2}\cdot\frac{2}{5}ma^2\left(\frac{V_c}{a}\right)^2 = mgh$$

$$\frac{1}{2}mV_c'^2 + \frac{1}{2}\cdot\frac{2}{3}ma^2\left(\frac{V_c'}{a}\right)^2 = mgh$$

$\therefore \frac{7}{10}V_c^2 = gh = \frac{5}{6}V_c'^2$ 匀加速, $h \propto V\cdot t$

$$\frac{t}{t'} = \frac{V_c'}{V_c} = \sqrt{\frac{7}{10}\times\frac{6}{5}} = \frac{\sqrt{21}}{5}$$

3.30



$\omega\cdot c = \omega'\cdot b$

$\Rightarrow \omega' = \omega\cdot\frac{c}{b}$

$$\therefore \vec{\omega} = -\omega\frac{c}{b}\hat{e}_y + \omega\cdot\hat{e}_z$$

$$\vec{r}_D = c\cdot\hat{e}_y + b\,\hat{e}_z$$

$$\therefore \vec{V} = \vec{\omega}\times\vec{r}_D = -\omega c\,\hat{e}_x - \omega c\,\hat{e}_x = -2\omega c\,\hat{e}_x$$

$$\vec{a} = \dot{\vec{\omega}}\times\vec{r}_D + \vec{\omega}\times\vec{V}_D$$

$$= \omega^2\frac{c}{b}\hat{e}_x\times(c\hat{e}_y + b\hat{e}_z) + \left(-\omega\frac{c}{b}\hat{e}_y + \omega\hat{e}_z\right)\times(-2\omega c\hat{e}_x)$$

$$= \frac{\omega^2c^2}{b}\hat{e}_z - \omega^2c\,\hat{e}_y - \frac{2\omega^2c^2}{b}\hat{e}_z - 2\omega^2c\,\hat{e}_y$$

$$= -\frac{\omega^2c^2}{b}\hat{e}_z - 3\omega^2c\,\hat{e}_y$$

$$\therefore a = \omega^2c\sqrt{9 + \frac{c^2}{b^2}}$$

3.30

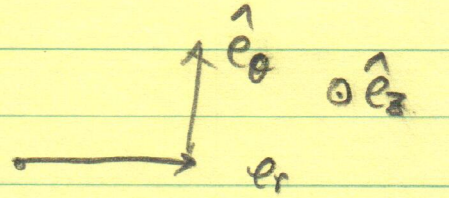
$OA\cdot\omega = OB\cdot\omega' \quad \therefore \omega' = \frac{OA}{OB}\omega = \frac{c}{b}\cdot\omega$

$\vec{V} = \vec{V}_A + \vec{\omega}'\times\vec{r}'$

$\vec{\omega}' = -\omega'\hat{e}_r$

$\dot{\vec{\omega}}' = -\omega\omega'\hat{e}_\theta$

$= \omega c\,\hat{e}_\theta + \omega' b\,\hat{e}_\theta = 2\omega c\,\hat{e}_\theta$



$\hat{e}_r\times\hat{e}_z = -\hat{e}_\theta$

$\vec{a} = \dot{\vec{V}} = \vec{\omega}\times\vec{V}_A + \dot{\vec{\omega}}'\times\vec{r}' + \vec{\omega}'\times\vec{V}'$

$I = \vec{\omega}\times\vec{V}_A = \omega\cdot\hat{e}_z\times(\omega c\,\hat{e}_\theta) = -\omega^2 c\,\hat{e}_r$

$II = \dot{\vec{\omega}}'\times\vec{r}' = -\omega\omega'\hat{e}_\theta\times b\hat{e}_z = -\omega^2 c\,\hat{e}_r$

$III = \vec{\omega}'\times\vec{V}' = -\omega'\hat{e}_r\times\omega' b\hat{e}_\theta = -\omega'^2 b\,\hat{e}_z = -\omega^2 c\cdot\frac{c}{b}\hat{e}_z$

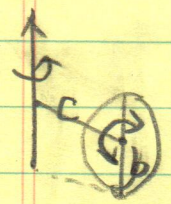
$\therefore \vec{a} = -2\omega^2 c\,\hat{e}_r - \omega^2 c\cdot\frac{c}{b}\hat{e}_z$

$\therefore a = \sqrt{a_r^2 + a_z^2} = \omega^2 c\sqrt{4+\frac{c^2}{b^2}}$ ×

錯誤原因：$\vec{V} = \vec{V}_A + \vec{\omega}\times\vec{r}'$

（$\vec{V}_A$：牽連動　$\vec{\omega}\times\vec{r}'$：所有部動）

取錯誤參考物打了 牽連+相對



$\vec{r}_A = c\,\hat{e}_r \qquad \vec{r}_m = c\hat{e}_r + b\hat{e}_z$

$\vec{\omega} = -\frac{c}{b}\omega\,\hat{e}_r + \omega\,\hat{e}_z$

$\hat{e}_r\times\hat{e}_z = -\hat{e}_\theta$

$\therefore \vec{V}_m = \vec{\omega}\times\vec{r}_m = -\omega c\,\hat{e}_r\times\hat{e}_z + \omega c\,\hat{e}_z\times\hat{e}_r$

$= 2\omega c\,\hat{e}_\theta$ 不管算 以為是用相對

$\vec{a}_m = \vec{\omega}\times\vec{r}_m + \dot{\vec{\omega}}\times\vec{r}_m$

$= -\frac{c}{b}\omega^2\hat{e}_\theta\times\vec{r}_m + \vec{\omega}\times 2\omega c\,\hat{e}_\theta$

$= \frac{c^2}{b}\omega^2\hat{e}_z - \omega^2 c\,\hat{e}_r - 2\omega^2\frac{c^2}{b}\hat{e}_z + 2\omega^2 c\,\hat{e}_r$

$= -\omega^2 c\cdot(3\hat{e}_r + \frac{c}{b}\hat{e}_z) \qquad \therefore |\vec{a}_m| = \omega^2 c\sqrt{9+\frac{c^2}{b^2}}$

3.31

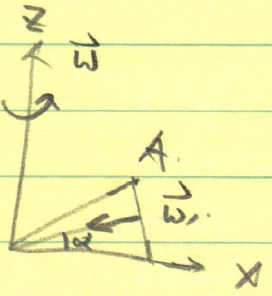
$$\vec{\omega} = (\omega_1 \cos\theta + \omega_2)\,\hat{e}_z + \omega_1 \sin\theta\,\hat{e}_y$$

$$\vec{r}_B = (b\sin\theta + a\cos\theta)\,\hat{e}_y + (b\cos\theta - a\sin\theta)\,\hat{e}_z$$

$$\therefore \vec{v} = \vec{\omega}\times\vec{r}_B = \omega_1 b\sin\theta\cos\theta - \omega_1 a\sin^2\theta - \omega_1 a\cos^2\theta - \omega_1 b\sin\theta\cos\theta$$

$$- \omega_2 b\sin\theta - \omega_2 a\cos\theta$$

$$= -\omega_1 a - \omega_2(b\sin\theta + a\cos\theta)$$



3.32 底面半径为 $r = h\tan\alpha$，侧边长度 $\frac{h}{\cos\alpha}$

由纯滚动，$\omega\cdot\frac{h}{\cos\alpha} = r\cdot\omega'$ $\therefore \omega' = \frac{h}{r\cos\alpha}\omega = \frac{\omega}{\sin\alpha}$

角速度为：$\vec{\omega} = \omega\,\hat{e}_z + \omega'(-\cos\alpha\,\hat{e}_x - \sin\alpha\,\hat{e}_z)$

$$= -\omega\cot\alpha\,\hat{e}_x$$

A点位矢为：$\vec{r}_A = \frac{h}{\cos\alpha}(\cos 2\alpha\,\hat{e}_x + \sin 2\alpha\,\hat{e}_z)$

$$\therefore \vec{v}_A = \vec{\omega}\times\vec{r}_A = -\omega\cot\alpha\cdot\frac{h}{\cos\alpha}\sin 2\alpha\cdot\hat{e}_x\times\hat{e}_z$$

$$= \omega\frac{\cos\alpha}{\sin\alpha}\frac{h}{\cos\alpha}\cdot 2\sin\alpha\cos\alpha\,\hat{e}_y$$

$$= 2\omega h\cos\alpha\,\hat{e}_y$$

$$\vec{a} = -\omega\cot\alpha\frac{d}{dt}\hat{e}_x = -\omega^2\cot\alpha\,\hat{e}_y$$

$\therefore$ 铁球加速度为

$$\vec{a}_1 = \vec{\omega}\times\vec{v}_A = \omega^2\cot\alpha\frac{h}{\cos\alpha}\cos 2\alpha\,\hat{e}_z - \omega^2\cot\alpha\frac{h}{\cos\alpha}\sin 2\alpha\,\hat{e}_x$$

$$= \frac{\omega^2 h}{\sin\alpha}(\cos 2\alpha\,\hat{e}_z - \sin 2\alpha\,\hat{e}_x) \qquad \therefore a_1 = \frac{\omega^2 h}{\sin\alpha}$$

科氏加速度为

$$\vec{a}_2 = \vec{\omega}\times\vec{v}_A = -\omega\cot\alpha\cdot 2\omega h\cos\alpha\,\hat{e}_z = -2\omega^2 h\frac{\cos^2\alpha}{\sin\alpha}\hat{e}_z = -2a_1\cos^2\alpha\,\hat{e}_z$$

7.33 刚体绕惯性主轴转动 → 欧拉-泊松解法；且 $I_1=I_2=2I_3$ <21>

$$\begin{cases} I_1\dot{\omega}_x-(I_2-I_3)\omega_y\omega_z=0 \\ I_2\dot{\omega}_y-(I_3-I_1)\omega_z\omega_x=0 \\ I_3\dot{\omega}_z=0 \end{cases}$$

① 欧拉运用法：$\theta=\theta_0,\ \varphi=\sec\theta_0(\Omega+n)t+\varphi_1$

$$\psi=\frac{\pi}{2}-(nt+\varepsilon)$$

$$n=\frac{I_3-I_1}{I_1}$$

对于本题，$n=-\frac{1}{2}$，$\Omega=\omega_1$，$\theta_0=\frac{\pi}{3}$，$n=\frac{I_3-I_1}{I_1}\Omega=-\frac{\Omega}{2}$

$$\therefore \omega_2=\dot{\varphi}=\sec\theta_0(\Omega+n)=2\left(\Omega-\frac{\Omega}{2}\right)=\Omega$$

$$\omega_1=\dot{\psi}=-n=\frac{\Omega}{2}$$

$$\therefore \omega_2=2\omega_1$$

② 泰勒法：

$$\begin{cases} 2\dot{\omega}_x-\omega_y\omega_z=0 \\ 2\dot{\omega}_y+\omega_y\omega_z=0 \\ \dot{\omega}_z=0 \end{cases} \Rightarrow \left\{\begin{array}{l} \omega_z=\Omega \\ \dot{\omega}_x-\frac{\Omega}{2}\omega_y=0 \\ \dot{\omega}_y+\frac{\Omega}{2}\omega_x=0 \end{array}\right\} \quad n\equiv\frac{\Omega}{2}$$

$$\begin{cases} \ddot{\omega}_x+n^2\omega_x=0 \\ \ddot{\omega}_y+n^2\omega_y=0 \end{cases} \Rightarrow \begin{cases} \omega_x=\omega_0\cos(nt+\varepsilon) \\ \omega_y=\omega_0\sin(nt+\varepsilon) \end{cases}$$

$$\left\{\begin{array}{l} J\sin\theta\sin\psi=I_1\omega_x=I_1\omega_0\cos(nt+\varepsilon) \\ J\sin\theta\cos\psi=I_2\omega_y=I_1\omega_0\sin(nt+\varepsilon) \\ J\cos\theta=I_3\omega_z=\frac{I_1}{2}\Omega \end{array}\right\} \Rightarrow \begin{array}{l} \cos\theta_0=\frac{I_1\Omega}{2J} \\ \sin\theta_0=\frac{I_1\omega_0}{J},\ \psi=\frac{\pi}{2}-(nt+\varepsilon) \\ \dot{\varphi}\cos\theta+\dot{\psi}=\omega_z \Rightarrow \varphi=\sec\theta_0(\Omega+n)t+\gamma \end{array}$$

同样的结果

3.34

$$\begin{cases} I_1\dot{\omega}_x - (I_2-I_3)\omega_y\omega_z = 0 & ① \\ I_2\dot{\omega}_y - (I_3-I_1)\omega_z\omega_x = 0 & ② \\ I_3\dot{\omega}_z - (I_1-I_2)\omega_x\omega_y = 0 & ③ \end{cases}$$

① $\times\omega_x$ + ② $\times\omega_y$ + ③ $\times\omega_z$，有

$$I_1\omega_x\dot{\omega}_x + I_2\omega_y\dot{\omega}_y + I_3\omega_z\dot{\omega}_z = 0$$

即 $\frac{d}{dt}\left(I_1\omega_x^2 + I_2\omega_y^2 + I_3\omega_z^2\right) = 0$

$\therefore T = \frac{1}{2}\left(I_1\omega_x^2 + I_2\omega_y^2 + I_3\omega_z^2\right) = \frac{1}{2}\vec{\omega}\cdot\hat{A}\cdot\vec{\omega} = C$ 动能守恒.

① $\hat{e}_x$ + ② $\hat{e}_y$ + ③ $\times\hat{e}_z$，有：

$$I_1\dot{\omega}_x\hat{e}_x + I_2\dot{\omega}_y\hat{e}_y + I_3\dot{\omega}_z\hat{e}_z + \vec{\omega}\times\vec{J} = 0$$

即 $\frac{d\vec{J}}{dt} = \text{LHS}$，故 $\frac{d\vec{J}}{dt} = 0$，$\vec{J} = \vec{C}$ 角动量守恒.

3.35. 本题属于刚体绕固定点转动问题

$$\begin{cases} I_1\dot{\omega}_x - (I_2 - I_3)\omega_y\omega_z = M_x \\ I_2\dot{\omega}_y - (I_3 - I_1)\omega_z\omega_x = M_y \\ I_3\dot{\omega}_z = 0 \end{cases}$$

对z轴上下运动轨迹有 (3.9.19, 3.9.21, 3.9.22)

$$\omega_z = \dot{\varphi}\cos\theta + \dot{\psi} = s \quad ①$$

$$\frac{1}{2}I_1(\dot{\theta}^2 + \dot{\varphi}^2\sin^2\theta) + \frac{1}{2}I_3 s^2 + mgl\cos\theta = E \quad ②$$

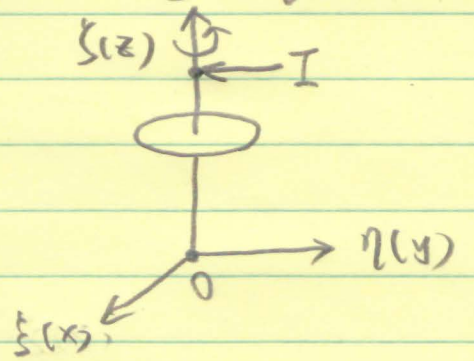
$$\hat{e}_3\cdot\vec{J} = I_1\dot{\varphi}\sin^2\theta + I_3 s\cos\theta = \alpha \quad ③$$

① 冲击之后, $\vec{\omega} = \omega_1\vec{k}$, 故 $s = \omega_1$

$$\theta = 0 \Rightarrow \alpha = I_3 s = I_3\omega_1$$

$$\dot{\theta} = \dot{\varphi} = 0,\ \theta = 0 \Rightarrow E = \frac{1}{2}I_3\omega_1^2 + mgl$$

② 冲击过程:



t=0时刻系统与xyz轴重合,

设冲击沿-y方向,

$\Delta\vec{J} = \vec{r}\times\vec{I} = Id\cdot\vec{i}$ 沿x方向

∴ 冲击后, $\vec{J} = \vec{J}_0 + \Delta\vec{J} = I_3\omega_1\vec{k} + Id\vec{i}$ ⇒ s不变仍为 ω_1

$\hat{e}_3\cdot\vec{J} = I_3\omega_1$ ⇒ α不变, 仍为 $I_3\omega_1$

由 $\vec{J} = I_1\omega_x\vec{i} + I_2\omega_y\vec{j} + I_3\omega_3\vec{k}$, 得 $\vec{\omega} = \frac{Id}{I_1}\vec{i} + \omega_1\vec{k}$

$$\therefore T = \frac{1}{2}\vec{\omega}\cdot\vec{J} = \frac{1}{2}I_3\omega_1^2 + \frac{I^2d^2}{2I_1}$$

∴ 冲击后 $E' = \frac{1}{2}I_3\omega_1^2 + \frac{I^2d^2}{2I_1} + mgl$, 多了一项 ω_x 导致的动能.

由②，知 $\dot{\varphi}\sin\theta = \frac{\alpha - I_3 s\cos\theta}{I_1\sin\theta} = \frac{I_3}{I_1}w_1\frac{1-\cos\theta}{\sin\theta} = \frac{I_3}{I_1}w_1\tan\frac{\theta}{2}$

将 E' 和 $\dot{\varphi}\sin\theta$ 代入 ③ 式，有

$$\frac{1}{2}I_1\left[\dot{\theta}^2 + \left(\frac{I_3}{I_1}w_1\tan\frac{\theta}{2}\right)^2\right] + \frac{1}{2}I_3 s^2 + mgl\cos\theta = \frac{1}{2}I_3 w_1^2 + \frac{I^2d^2}{2I_1} + mgl$$

当 θ 取极大值，有 $\dot{\theta}=0$，上式化为

$$\left(\frac{I_3 w_1}{I_1}\tan\frac{\theta}{2}\right)^2 = \frac{2mgl(1-\cos\theta)}{I_1} + \frac{I^2d^2}{I_1^2}$$

题目条件 w_1 很大，即 $T \gg V$，故右端第一项可忽略

$$\therefore \tan\frac{\theta}{2} = \frac{Id}{I_3 w_1} \Rightarrow \theta = 2\arctan\frac{Id}{I_3 w_1}$$

3.36 由题意，力偶矩平行于 $\vec{\omega}$ 轴，力偶矩 $\vec{M} = c\cdot\vec{\omega} = -I_3\lambda\vec{\omega}$.

代入欧拉动力学，有

$$\begin{cases} I_1\dot{w}_x - (I_2 - I_3)w_y w_z = -I_3\lambda w_x & ① \\ I_2\dot{w}_y - (I_3 - I_1)w_z w_x = -I_3\lambda w_y & ② \\ I_3\dot{w}_z = -I_3\lambda w_z & ③ \end{cases}$$

由③式，$\frac{dw_z}{w_z} = -\lambda dt \Rightarrow w_z = \Omega e^{-\lambda t}$.

①与②中，引入惯体 $\tilde{w}_x$ 与 w_x 联合考虑.

设 $\tilde{w}_x = w_x\cdot f$，则 $\dot{\tilde{w}}_x = \dot{w}_x f + w_x\dot{f} = \left(\dot{w}_x + \frac{\dot{f}}{f}w_x\right)\cdot f$

令其等于①中 $\dot{w}_x + \frac{I_3}{I_1}\lambda w_x$，即 $\frac{\dot{f}}{f} = \frac{I_3}{I_1}\lambda \Rightarrow f = e^{\frac{I_3}{I_1}\lambda t}$.

$\therefore$ ①式化为 $I_1\cdot\frac{\dot{\tilde{w}}_x}{f} - (I_1 - I_3)w_y\cdot\Omega e^{-\lambda t} = 0$，令 $n = \frac{I_3 - I_1}{I_1}\Omega$，则

$$\Rightarrow \dot{\tilde{w}}_x + ne^{-\lambda t}\tilde{w}_y = 0 \quad ④, \quad \text{同理} \quad \dot{\tilde{w}}_y - ne^{-\lambda t}\tilde{w}_x = 0 \quad ⑤$$

由④式, $\tilde{\omega}_y = -\frac{1}{n}\dot{\tilde{\omega}}_x e^{\lambda t} \Rightarrow \dot{\tilde{\omega}}_y = -\frac{1}{n}(\ddot{\tilde{\omega}}_x + \lambda\dot{\tilde{\omega}}_x)e^{\lambda t}$

代入⑤式, 整理, 有 $\ddot{\tilde{\omega}}_x + \lambda\dot{\tilde{\omega}}_x + n^2 e^{-\lambda t}\tilde{\omega}_x = 0$ ⑥

其为常系数, 设解为 $A\cos(\alpha t)$, 本系统为 ω_x

$$\tilde{\omega}_x = a\sin f$$

$$\Rightarrow \dot{\tilde{\omega}}_x = a\dot{f}\cos f, \quad \ddot{\tilde{\omega}}_x = a(\ddot{f}\cos f - \dot{f}^2\sin f)$$

代入⑥式, 有:

$$(\ddot{f} + \lambda\dot{f})\cos f + \underline{(-\dot{f}^2 + n^2 e^{-2\lambda t})}\sin f = 0$$

由 $\sin f$ 得到, 有 $\dot{f} = n e^{-\lambda t} \Rightarrow f = -\frac{n}{\lambda}e^{-\lambda t} + \varepsilon'$

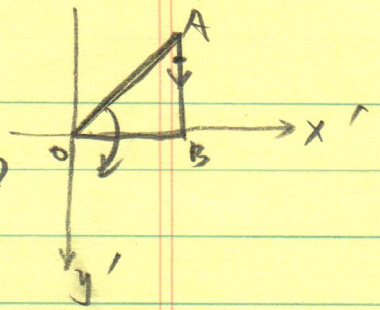
$$\therefore \tilde{\omega}_x = a\sin\left(-\frac{n}{\lambda}e^{-\lambda t} + \varepsilon'\right) = a'\sin\left(\frac{n}{\lambda}e^{-\lambda t} + \varepsilon\right)$$

$$\tilde{\omega}_y = a'\cos\left(\frac{n}{\lambda}e^{-\lambda t} + \varepsilon\right)$$

由 $\tilde{\omega}_x = \omega_x e^{\frac{I_3}{I_1}\lambda t}$, 有

$$\begin{cases} \omega_x = a' e^{-\frac{I_3}{I_1}\lambda t}\sin\left(\frac{n}{\lambda}e^{-\lambda t} + \varepsilon\right) \\ \omega_y = a' e^{-\frac{I_3}{I_1}\lambda t}\cos\left(\frac{n}{\lambda}e^{-\lambda t} + \varepsilon\right) \\ \omega_z = \Omega e^{-\lambda t} \end{cases}$$

4.1



$\vec{\omega} = \omega \hat{e}_z$, $OB = AB = b$

$v_A' = \frac{AB}{T} = \frac{\omega b}{2\pi}$, $\vec{v}_A' = \frac{\omega b}{2\pi} \hat{e}_{y'}$

P在初始时 $t=0$, $\theta_A = \frac{\pi}{4}$. $\vec{r}_A = b\hat{e}_x - b\hat{e}_y$, $\vec{v}_A' = \frac{\omega b}{2\pi}\hat{e}_y$

$\therefore \vec{v}_A = \vec{v}_A' + \vec{\omega} \times \vec{r}_A = \frac{\omega b}{2\pi}\hat{e}_y + \omega \hat{e}_z \times b(\hat{e}_x - \hat{e}_y)$

$= \omega b \hat{e}_x + \omega b(1 + \frac{1}{2\pi})\hat{e}_y$.

匀速运动，有科氏加速度，

$\vec{a}_A = \vec{a}_A' - \omega^2 \vec{r}_A + 2\vec{\omega} \times \vec{v}_A'$

$= 0 - \omega^2 b(\hat{e}_x - \hat{e}_y) + \underline{2\omega \hat{e}_z \times \frac{\omega b}{2\pi}\hat{e}_y} - \frac{\omega^2 b}{\pi}\hat{e}_x$

$= -\omega^2 b(1 + \frac{1}{\pi})\hat{e}_x + \omega^2 b \hat{e}_y$.

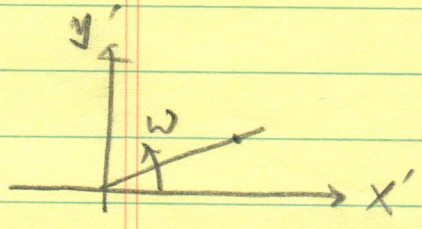
$\therefore v_A = \sqrt{(\omega b)^2 + (\omega b)^2(1+\frac{1}{2\pi})^2} = \frac{\omega b}{2\pi}\sqrt{4\pi^2 + (1+2\pi)^2} = \frac{\omega b}{2\pi}\sqrt{8\pi^2 + 4\pi + 1}$

$a_A = \omega^2 b \sqrt{(1+\frac{1}{\pi})^2 + 1} = \frac{\omega^2 b}{\pi}\sqrt{2\pi^2 + 2\pi + 1}$

$\vec{v}_A$ 与 $\vec{a}_A$ 与 OB 之间夹角分别为

$\theta_1 = \arctan\frac{v_{Ay}}{v_{Ax}} = \arctan\left(\frac{2\pi+1}{2\pi}\right)$ $\theta_2 = \arctan\frac{a_{Ay}}{a_{Ax}} = \arctan\left(\frac{\pi}{\pi+1}\right)$

4.2



P点在转动时相对O的速度为 u，则 $r = \int_0^t u\,dt$

$\vec{v} = u\hat{e}_r + \omega r \hat{e}_\theta$.

$v^2 = u^2 + \omega^2 r^2$ $\therefore r = \frac{1}{\omega}\sqrt{v^2 - u^2}$

$\therefore \dot{r} = \dot{u} = \frac{1}{\omega}\frac{-u\dot{u}}{\sqrt{v^2-u^2}}$ $\therefore \frac{\dot{u}}{\sqrt{v^2-u^2}} = -\omega$.

$\frac{dy}{\sqrt{v^2-u^2}} = -\omega dt$ $\quad \xi = \frac{u}{v}$

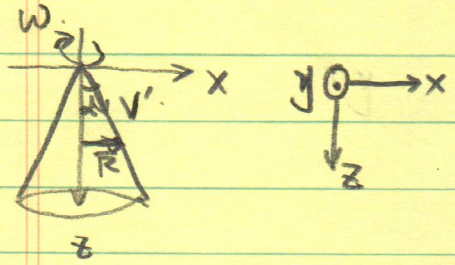
$\frac{d\xi}{\sqrt{1-\xi^2}} = -\omega dt$ $\quad C + \arcsin\xi = -\omega t$ $\quad t=0,\ \xi=1 \Rightarrow C = -\frac{\pi}{2}$

$\therefore \xi = \sin\left(\frac{\pi}{2} - \omega t\right) = \cos\omega t$

$\therefore u = v\xi = v\cos\omega t$

$r = \int_0^t u\,dt = \frac{v}{\omega}\sin\omega t$

4.3



角速度: $\vec{\omega} = -\omega\hat{e}_\theta$ $\quad \vec{\omega} = \omega\hat{e}_z$

$\vec{a} = \vec{a}' - \omega^2\vec{R} + 2\vec{\omega}\times\vec{v}'$

相对加速: $-\omega^2\vec{R} = -\omega^2 \cdot v't\sin\alpha\,\hat{e}_r$

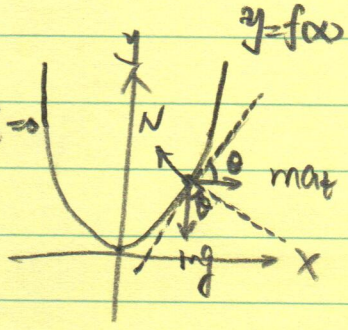
$2\vec{\omega}\times\vec{v}' = 2\omega\,\hat{e}_z \times (v'\sin\alpha\,\hat{e}_r + v'\cos\alpha\,\hat{e}_z)$

$= 2\omega v'\sin\alpha\,\hat{e}_\theta$

$\therefore a = \omega v'\sin\alpha\sqrt{\omega^2t^2+4}$

4.4

相对静止 $\Rightarrow \vec{F} = m\vec{a}_t$, $\vec{v}'=0$, $\vec{a}'=0$, $\vec{a}_c=0$



$\cos\theta = \frac{1}{\sqrt{1+\tan^2\theta}}$

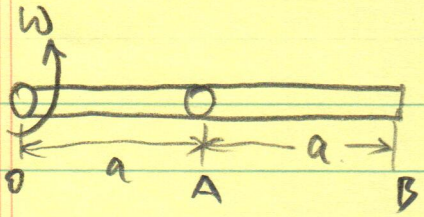
惯性力 $\Rightarrow m\vec{a}_t = +m\omega^2\vec{R}$

力平衡 $\Rightarrow m\omega^2 x\cos\theta = mg\sin\theta$

即 $m\omega^2 x = mg\cdot\tan\theta = mg\frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{\omega^2}{g}\cdot x$ $\quad \therefore y = \frac{\omega^2}{2g}x^2$

约束力: $R = m\omega^2 x\cdot\sin\theta + mg\cos\theta = mg\left(\frac{\sin^2\theta}{\cos\theta} + \cos\theta\right) = \frac{mg}{\cos\theta} = mg\sqrt{1+\tan^2\theta} = mg\sqrt{1+\frac{\omega^4x^2}{g^2}}$

4.5



$\vec{F}_{惯} = m\omega^2\vec{r} - 2m\vec{\omega}\times\vec{v}'$

惯性力　　科氏力

$\therefore \vec{a} = \frac{m\omega^2\vec{r}}{m} = \omega^2\vec{r} = \omega^2 r\,\hat{e}_r$

动能定理：$\Delta T = \frac{1}{2}mv'^2 = \int_a^{2a}\vec{F}\cdot d\vec{r} = \int_a^{2a} m\omega^2 r\,dr = \frac{3}{2}m\omega^2 a^2$

$\therefore v' = \sqrt{\frac{2T}{m}} = \sqrt{3}\omega a$　　$\therefore \vec{v} = \vec{v}' + \vec{\omega}\times\vec{r}_B = \sqrt{3}\omega a\,\hat{e}_r + 2\omega a\,\hat{e}_\theta$

$v = \sqrt{v'^2 + (\omega\cdot 2a)^2} = \sqrt{3\omega^2a^2 + 4\omega^2a^2} = \sqrt{7}\omega a$

$m\ddot{r} = m\omega^2 r$　　$\therefore r = C_1 e^{\omega t} + C_2 e^{-\omega t}$

$\begin{cases} t=0,\ r=a \Rightarrow C_1 + C_2 = a \\ t=\infty,\ \dot{r}=0 \Rightarrow C_1 - C_2 = 0 \end{cases} \Rightarrow C_1 = C_2 = \frac{a}{2}$

$\therefore r = \frac{a}{2}(e^{\omega t} + e^{-\omega t})$

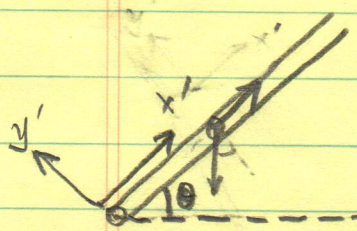
$r = 2a$，有　$2a = \frac{a}{2}(e^{\omega t} + e^{-\omega t})$

$\Rightarrow e^{2\omega t} - 4e^{\omega t} + 1 = 0$

$\therefore e^{\omega t} = \frac{1}{2}(4 \pm \sqrt{16-4}) = 2 \pm \sqrt{3}$

$t \to \infty$，$e^{\omega t} > 1$　　$2-\sqrt{3}$ 舍去　　$\therefore e^{\omega t} = 2+\sqrt{3}$　　$\therefore t = \frac{1}{\omega}\ln(2+\sqrt{3})$

4.6



匀速度，$\vec{F}_{惯} = m\omega^2\vec{r} - \vec{F}_C$；光滑管，$\vec{F}_C$ 无法与 m 同，仅能垂

$\vec{F}_t = m\omega^2\vec{r} = m\omega^2 r\,\hat{e}_{x'}$

$\vec{G} = mg(-\sin\theta\,\hat{e}_{x'} - \cos\theta\,\hat{e}_{y'})$

$\therefore$ x' 方向上运动方程为：

$m\omega^2 r - mg\sin\theta = m\ddot{r}$　　$\therefore \ddot{r} - \omega^2 r + g\sin\omega t = 0$

$\ddot{r} - \omega^2 r + g\sin\omega t$

上述方程为波动项 + 非齐次项，后者可视为强迫的小振动，故可

用试探解 $r = \tilde{r} + C\sin\omega t$，则有

$$(\ddot{\tilde{r}} - \omega^2\tilde{r}) - C\omega^2\sin\omega t - C\omega^2\sin\omega t + g\sin\omega t = 0$$

使非齐次项为0，有 $-2C\omega^2 + g = 0 \quad \therefore C = \frac{g}{2\omega^2}$，$\ddot{\tilde{r}} - \omega^2\tilde{r} = 0$

$$\therefore r = \frac{g}{2\omega^2}\sin\omega t + C_1 e^{\omega t} + C_2 e^{-\omega t}$$

初始条件：$t=0$ 时，$r=a \Rightarrow C_1 + C_2 = a$

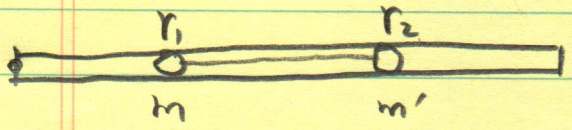
$t=0$ 时，$V=V_0 \Rightarrow \frac{g}{2\omega} + C_1\omega - C_2\omega = V_0$

解之，有：$C_1 = \frac{1}{2}\left(a + \frac{V_0}{\omega}\right) - \frac{g}{4\omega^2}$，$C_2 = \frac{1}{2}\left(a - \frac{V_0}{\omega}\right) + \frac{g}{4\omega^2}$

$$\therefore r = \left[\frac{1}{2}\left(a + \frac{V_0}{\omega}\right) - \frac{g}{4\omega^2}\right]e^{\omega t} + \left[\frac{1}{2}\left(a - \frac{V_0}{\omega}\right) + \frac{g}{4\omega^2}\right]e^{-\omega t} + \frac{g}{2\omega^2}\sin\omega t$$

4.7 匀角速，$\vec{F}_{惯} = m\omega^2\vec{r} + \vec{F}_c$

光滑管，$\vec{F}_c$ 垂直于管壁，对轨道方程（运动）无影响



$\vec{F}_H = m\omega^2 r_1 \hat{e}_{x'}$　　$\vec{F}_{H'} = m'\omega^2 r_2 \hat{e}_{x'}$

对 m 与 m' 分别列方程：

$$\begin{cases} m\ddot{r}_1 = m\omega^2 r_1 + k(r_2 - r_1 - a) & ① \\ m'\ddot{r}_2 = m'\omega^2 r_2 - k(r_2 - r_1 - a) & ② \end{cases}$$

②×m − ①×m'，上式为：

$$mm'(\ddot{r}_2-\ddot{r}_1)=mm'\omega^2(r_2-r_1)-(m+m')k(r_2-r_1-a)$$

令 $s=r_2-r_1$, $\mu=\frac{mm'}{m+m'}$, 上式化为

$$\mu(\ddot{s}-\omega^2 s)+k(s-a)=0 \qquad k=$$

$$\Downarrow k=2\mu\cdot\omega^2$$

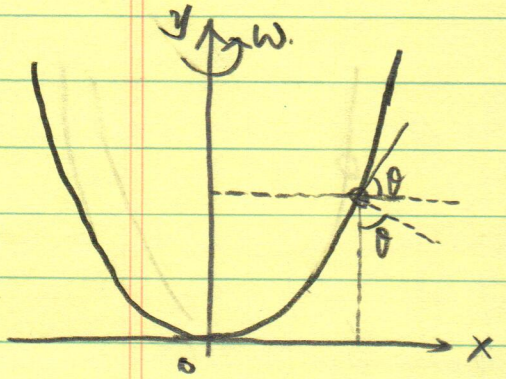
$$\ddot{s}+\omega^2 s-2\omega^2 a=0$$

$$\therefore s=2a+c_1\cdot\cos\omega t+c_2\sin\omega t$$

$$\left.\begin{array}{l} t=0,\ s=a \Rightarrow 2a+c_1=a \\ t=0,\ \dot{s}=0 \Rightarrow c_2=0 \end{array}\right\} \Rightarrow c_1=-a$$

$$\therefore s=a(2-\cos\omega t)$$

4.8



$$\vec{F}_{惯}=m\omega^2\vec{R}-2m\vec{\omega}\times\vec{v}'$$

$$\vec{R}=x\hat{e}_x$$

$$x^2=4ay \Rightarrow 2x\dot{x}=4a\dot{y} \quad \therefore \dot{y}=\frac{x}{2a}\dot{x}$$

$$\therefore \vec{v}'=\dot{x}\hat{e}_x+\dot{y}\hat{e}_y$$

$$=\dot{x}\hat{e}_x+\frac{x}{2a}\dot{x}\hat{e}_y$$

$$\therefore \vec{\omega}\times\vec{v}'=\omega\hat{e}_y\times\vec{v}'=-\omega\dot{x}\hat{e}_z$$

$\therefore -2m\vec{\omega}\times\vec{v}'$ 垂直于抛物线，在径向上无影响

$$\therefore \vec{F}_{惯}=m\omega^2 x\hat{e}_x$$

利用极坐标，由质点动力学得 $ma_\theta=F_\theta$

$$m\ddot{x}\cos\theta+m\ddot{y}\sin\theta=-mg\sin\theta+m\omega^2 x\cdot\cos\theta$$

$\therefore \ddot{x} + \ddot{y}\tan\theta - \omega^2 x + g\cdot\tan\theta = 0$

由题知, $\dot{y} = \frac{x}{2a}\dot{x}$ $\therefore$ $\ddot{y} = \frac{\dot{x}^2}{2a} + \frac{x\ddot{x}}{2a}$

$\therefore$ 运动微分方程 $\left(1 + \frac{x}{2a}\tan\theta\right)\ddot{x} + \frac{\dot{x}^2}{2a}\cdot\tan\theta - \omega^2 x + g\tan\theta = 0$

即 $\left(1 + \frac{x^2}{4a^2}\right)\ddot{x} + \frac{x}{4a^2}\dot{x}^2 - \omega^2 x + \frac{g}{2a}x = 0$

4.9 (1) 由方程可得, 有 $\dot{x} = \ddot{x} = 0$, $\therefore$ $-\omega^2 x + \frac{g}{2a}x = 0$ $\therefore$ $\omega^2 = \frac{g}{2a}$

(2) 由方程可得 $F_t = 0$

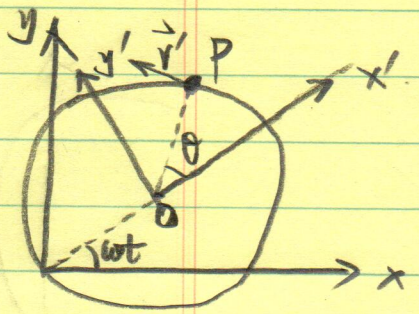
$\therefore$ $-mg\sin\theta + m\omega^2 x\cos\theta = 0$

$\therefore$ $g\tan\theta = \omega^2 x$ 即 $g\frac{x}{2a} = \omega^2 x$

$\therefore$ $\omega^2 = \frac{g}{2a}$

4.10 动点的绝对运动与牵连运动 $\Rightarrow$ $\begin{cases} \vec{v} = \vec{v}' + \vec{\omega}\times\vec{r} \\ \vec{a} = \vec{a}' + \vec{\omega}\times\vec{r} - \omega^2\vec{r} + 2\vec{\omega}\times\vec{v}' \end{cases}$

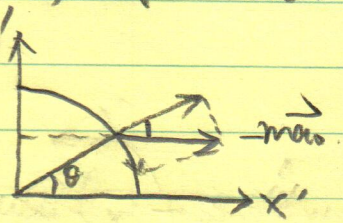
动点的绝对运动与牵连运动 $\Rightarrow$ $\begin{cases} \vec{v} = \vec{v}_0 + \vec{v}' + \vec{\omega}\times(\vec{r}-\vec{r}_0) \\ \vec{a} = \vec{a}_0 + \vec{a}' + \vec{\omega}\times(\vec{r}-\vec{r}_0) - \omega^2(\vec{r}-\vec{r}_0) + 2\vec{\omega}\times \end{cases}$



$\vec{F}_{惯} = -m\vec{a}_0 + \omega^2(\vec{r}-\vec{r}_0) - 2\vec{\omega}\times\vec{v}'$

牵连 径向, 科氏力, 垂直于 $\vec{v}'$, 无法确定

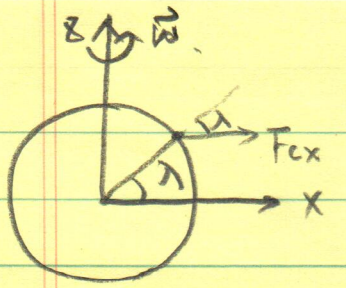
$-m\vec{a}_0 = -m\cdot\omega^2 a(-\hat{e}_x) = m\omega^2 a\hat{e}_x$



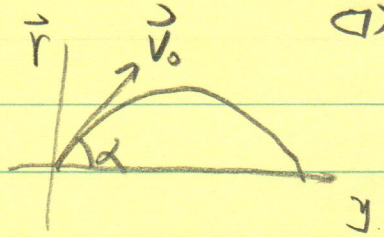
由动量矩定理 $-m\omega^2 a\cdot\sin\theta = ma\cdot\ddot{\theta}$

$\therefore$ $\ddot{\theta} + \omega^2\sin\theta = 0$

4.11



球体转动: $\vec{\omega} = \omega \hat{e}_z$



<7>

炮弹速度: $\vec{V} = V\cos\alpha\,\hat{e}_y + (V\sin\alpha - gt)\,\hat{e}_r$

$= V\cos\alpha\,\hat{e}_y + (V\sin\alpha - gt)(\cos\lambda\,\hat{e}_x + \sin\lambda\,\hat{e}_z)$

∴ 科氏力

$\vec{F}_c = -2m\vec{\omega}\times\vec{V}' = [-\omega V\cos\alpha\,\hat{e}_x + \omega(V\sin\alpha - gt)\cos\lambda\,\hat{e}_y] \times (-2m)$

引起炮弹横向偏离

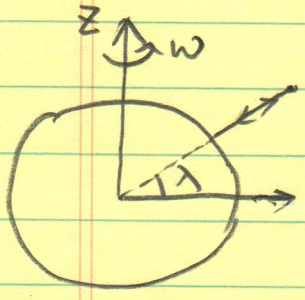
∴ 横向力为 $F_{cx}\cdot\sin\lambda = 2m\omega V\cos\alpha\cdot\sin\lambda$

$a_\perp = 2\omega V\cos\alpha\sin\lambda$ 为常数

∴ 炮弹偏离距离为:

$d = \frac{1}{2}a_\perp\cdot t^2 = \frac{1}{2}a_\perp\cdot\left(\frac{2V\sin\alpha}{g}\right)^2 = \frac{2a_\perp V^2\sin^2\alpha}{g^2} = \frac{4\omega V^3}{g^2}\sin^2\alpha\cos\alpha\sin\lambda$

4.12.



$\vec{V} = (V_0 - gt)\,\hat{e}_r = (V_0 - gt)(\cos\lambda\,\hat{e}_x + \sin\lambda\,\hat{e}_z)$

$\therefore \vec{F}_c = -2m\vec{\omega}\times\vec{V}'$

$= -2m\cdot\omega(V_0 - gt)\cos\lambda\,\hat{e}_y$

$\therefore \vec{a}_c = -2\omega\cos\lambda(V_0 - gt)\,\hat{e}_y$,

$\therefore \vec{V}_\perp = \int\vec{a}_c\,dt = -\omega\cos\lambda(2V_0 t - gt^2)\,\hat{e}_y$

$\therefore \vec{d} = \int_0^T \vec{v}_L \, dt = -\omega \cos\lambda \left(v_0 t^2 - \frac{g}{3} t^3 \right) \Big|_0^T \hat{e}_y$

$T = \frac{2v_0}{g}$，子弹上升再下降到开枪之前的时间，

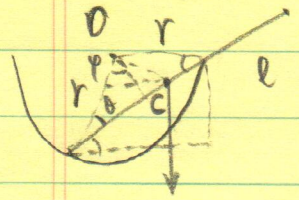
$\therefore d = -\omega \cos\lambda \left(\frac{4v_0^3}{g^2} - \frac{g}{3} \cdot \frac{8v_0^3}{g^3} \right) = -\omega \cos\lambda \frac{4v_0^3}{3g^2}$

竖直上抛，$v_0^2 = 2gh$

$\therefore d = -\omega \cdot \cos\lambda \frac{4 \cdot (\sqrt{2gh})^3}{3g^2} = -\frac{4}{3}\sqrt{\frac{8h^3}{g}} \omega \cos\lambda$

负表示位移与 y 方向，即位移方向向西.

5.1



$\delta W = 0$

$\therefore mg \cdot \delta y_c = 0$

$C = 2r\sin\varphi$

$y_c = (2r\sin\varphi - \frac{l}{2}) \cdot \sin(\frac{\pi}{2} - \varphi)$

$= r \cdot \sin 2\varphi - \frac{l}{2}\cdot\cos\varphi$

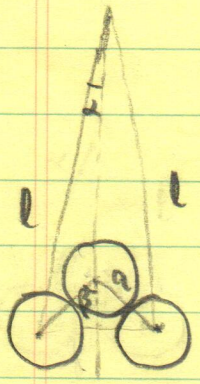
$\therefore \delta y = 2r\cos 2\varphi + \frac{l}{2}\sin\varphi = 0$

$2r - 4r\sin^2\varphi + \frac{l}{2}\sin\varphi = 0$

$2r - \frac{c^2}{r} + \frac{l}{4r}c = 0$

$\therefore l = \frac{4(c^2 - 2r^2)}{c}$

5.2



$y_c = \frac{1}{3m}(2lm\cos\alpha + m(l\cos\alpha - 2a\cos\beta))$

$= l\cos\alpha - \frac{2}{3}a\cdot\cos\beta$

$l\cdot\sin\alpha = 2a\sin\beta$

$\therefore l\cdot\cos\alpha\, d\alpha = 2a\cos\beta\, d\beta$

$\therefore \frac{d\alpha}{d\beta} = \frac{2a}{l}\cdot\frac{\cos\beta}{\cos\alpha}$

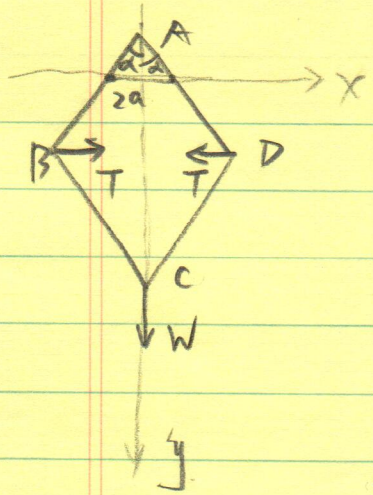
$\delta y_c = (-l\cdot\sin\alpha\cdot\frac{d\alpha}{d\beta} + \frac{2}{3}a\cdot\sin\beta)\,\delta\beta$

$= (-2a\cdot\frac{\cos\beta}{\cos\alpha}\cdot\sin\alpha + \frac{2}{3}a\sin\beta)\,\delta\beta$

$= -2a\cos\beta(\tan\alpha - \frac{1}{3}\tan\beta)\,\delta\beta = 0$

$\therefore 3\tan\alpha = \tan\beta$

5.3



$$\begin{cases} x_D = l\sin\alpha - a \\ y_D = l\cos\alpha - a\tan\alpha \end{cases}$$

$$\begin{cases} x_C = 0 \\ y_C = 2l\cos\alpha - a\cot\alpha \end{cases}$$

$$\frac{\cos\alpha}{\sin\alpha} = -\csc^2\alpha$$

$$\delta W = -2T\cdot\delta x_D + W\cdot\delta y_C = 0$$

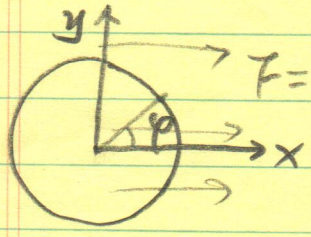
$$-2T\cdot l\cos\alpha\,\delta\alpha + W\cdot(-2l\sin\alpha + a\cdot\csc^2\alpha)\delta\alpha = 0$$

$$-T\cos\alpha - W\sin\alpha + W\cdot\frac{a}{2l}\cdot\csc^2\alpha = 0$$

$$T = -W\tan\alpha + W\cdot\frac{a}{2l}\cdot\frac{1}{\cos\alpha}\csc^2\alpha$$

$$= W\tan\alpha\left(\frac{a}{2l}\csc^3\alpha - 1\right)$$

5.4



$$x_A = r\cos\varphi \qquad \delta x_A = -r\sin\varphi\,\delta\varphi$$

$$y_A = r\sin\varphi \qquad \delta y_A = r\cos\varphi\,\delta\varphi$$

$$\left(F_\alpha + \lambda\frac{\partial f}{\partial x_\alpha}\right)\delta x_\alpha = 0 \quad \text{좌표계에 관계없이 성립.}$$

$$(k^2x + \lambda 2x)\cdot\delta x = 0$$

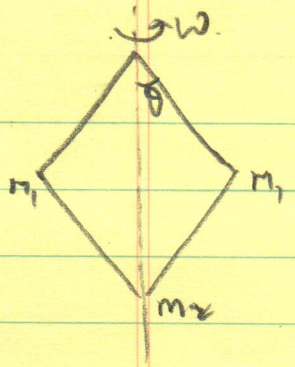
$$(W - \lambda 2y)\cdot\delta y = 0$$

$|\nabla f| = 2r$

① $x=0,\ y=\pm r,\ \lambda = \pm\frac{W}{2r}$, $R = |\lambda\nabla f| = W$

② $\lambda = -\frac{k^2}{2},\ y = -\frac{W}{k^2},\ x = \pm\sqrt{r^2 - \frac{W^2}{k^4}}$, $R = k^2 r$

5.5



$$\vec{V}_D = \vec{\Omega} \times \vec{r}_D + \vec{V}_D'$$

$$\therefore V_D^2 = (\Omega \cdot a\sin\theta)^2 + (a\dot{\theta})^2$$

$$= a^2(\Omega^2 \sin^2\theta + \dot{\theta}^2)$$

$$V_C^2 = [(2a\cos\theta)']^2 = 4a^2 \sin^2\theta\, \dot{\theta}^2$$

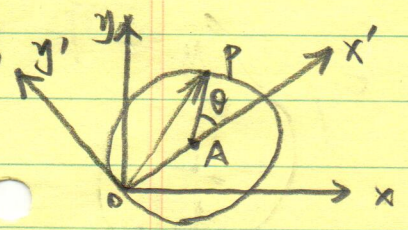
$$\therefore T = m_1 V_D^2 + \frac{1}{2} m_2 V_C^2$$

$$= m_1 a^2(\dot{\theta}^2 + \Omega^2 \sin^2\theta) + 2m_2 a^2 \sin^2\theta\, \dot{\theta}^2$$

$$V = -2m_1 g a\cos\theta - m_2 g \cdot 2a\cos\theta = -2(m_1+m_2) g a\cos\theta$$

$$L = T - V$$

5.6



数轴如 $ox'y'$ 选取如图，与静止系 Oxy 始终重合

$$\vec{V}_P = \vec{V}_P' + \vec{\omega} \times \vec{r}_P \qquad |\vec{r}_P| = 2a\cos\frac{\theta}{2}$$

$$= a\dot{\theta}(-\sin\theta\, \hat{e}_{x'} + \cos\theta\, \hat{e}_{y'}) + \omega \hat{e}_z \times 2a\cos\frac{\theta}{2}\left(\cos\frac{\theta}{2}\hat{e}_{x'} + \sin\frac{\theta}{2}\hat{e}_{y'}\right)$$

$$= -(\omega+\dot{\theta})\, a\sin\theta\, \hat{e}_x + [\omega a + (\omega+\dot{\theta})\, a\cos\theta]\, \hat{e}_y$$

$$V^2 = \omega^2 a^2 + (\dot{\theta}+\omega)^2 a^2 + 2a^2 \omega(\dot{\theta}+\omega)\cos\theta$$

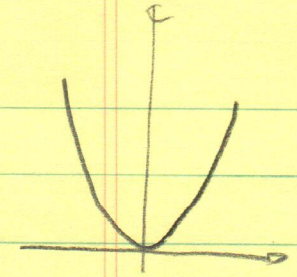
$$\frac{\partial T}{\partial \dot{\theta}} = Ma^2\{(\dot{\theta}+\omega) + \omega\cos\theta\}$$

$$\therefore \frac{d}{dt}\frac{\partial T}{\partial \dot{\theta}} = Ma^2(\ddot{\theta} - \omega\dot{\theta}\sin\theta)$$

$$\frac{\partial T}{\partial \theta} = -Ma^2 \underline{\omega(\dot{\theta}+\omega)}\sin\theta$$

代入得，$\ddot{\theta} + \omega^2 \sin\theta = 0$ 是一个非线性方程。

5.7.



$V^2 = \dot{x}^2 + \dot{y}^2 + \omega^2 x^2$

$x^2 = 4ay$

$\therefore 4a\dot{y} = 2x\dot{x}$

$\therefore \dot{y} = \frac{x}{2a}\dot{x}$

$\therefore V^2 = \dot{x}^2 + \frac{1}{4a^2}x^2\dot{x}^2 + \omega^2 x^2$

$\therefore L = T - V = \frac{1}{2}m\left(\dot{x}^2 + \frac{x^2\dot{x}^2}{4a^2} + \omega^2 x^2\right) - mg\frac{x^2}{4a}$

$\therefore \frac{\partial L}{\partial \dot{x}} = m\dot{x}\left(1 + \frac{x^2}{4a^2}\right)$

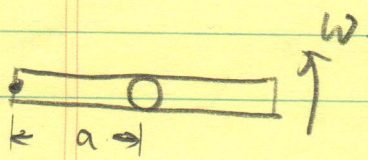
$\frac{d}{dt}\frac{\partial L}{\partial \dot{x}} = m\ddot{x}\left(1 + \frac{x^2}{4a^2}\right) + m\dot{x}\frac{2x}{4a^2}\dot{x}$

$= m\ddot{x}\left(1 + \frac{x^2}{4a^2}\right) + m\frac{x\dot{x}^2}{2a^2}$

$\frac{\partial L}{\partial x} = m\frac{x\dot{x}^2}{4a^2} + m\omega^2 x - mg\frac{x}{2a}$

$\therefore \ddot{x}\left(1 + \frac{x^2}{4a^2}\right) + \frac{x\dot{x}^2}{4a^2} - \left(\omega^2 - \frac{g}{2a}\right)x = 0$

5.8.



广义坐标：x.

$V = \omega x + \dot{x}$

$\therefore T = \frac{1}{2}mv^2 = \frac{1}{2}m(\omega x + \dot{x})^2$

$V = mg \cdot x \cdot \sin\omega t$

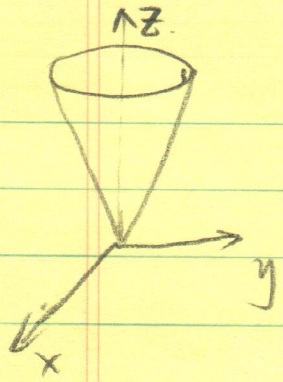
$L = T - V = \frac{1}{2}m(\omega x + \dot{x})^2 - mgx\sin\omega t$

$\frac{\partial L}{\partial \dot{x}} = m(\omega x + \dot{x})$ $\quad \frac{d}{dt}\frac{\partial L}{\partial \dot{x}} = m(\omega\dot{x} + \ddot{x})$

$\frac{\partial L}{\partial x} = m\omega(\omega x + \dot{x}) - mg\cdot\sin\omega t$ $\quad \therefore \ddot{x} - \omega^2 x + g\sin\omega t = 0$

$\rightarrow \dot{x} = C_1 e^{\omega t} + C_2 e^{-\omega t} + \frac{g}{2\omega}\sin\omega t$

5.9.



$$\begin{cases} x_A = r\cos\theta \\ y_A = r\sin\theta \\ z_A = r\cot\alpha \end{cases} \qquad \begin{cases} \dot{x}_A = \dot{r}\cos\theta - r\dot{\theta}\sin\theta \\ \dot{y}_A = \dot{r}\sin\theta + r\dot{\theta}\cos\theta \\ \dot{z}_A = \dot{r}\cot\alpha \end{cases}$$

$$\therefore v_A^2 = \dot{x}_A^2 + \dot{y}_A^2 + \dot{z}_A^2 = \dot{r}^2 + r^2\dot{\theta}^2 + \dot{r}^2\cot^2\alpha$$

$$\therefore L = \frac{1}{2}m\left[\frac{\dot{r}^2}{\sin^2\alpha} + r^2\dot{\theta}^2\right] - mgr\cot\alpha$$

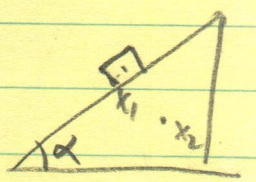
$$\frac{\partial L}{\partial \dot{r}} = m\dot{r}\frac{1}{\sin^2\alpha} \qquad \frac{d}{dt}\frac{\partial L}{\partial \dot{r}} = m\ddot{r}\frac{1}{\sin^2\alpha}$$

$$\frac{\partial L}{\partial r} = mr\dot{\theta}^2 - mg\cot\alpha$$

$$\frac{\ddot{r}}{\sin^2\alpha} - r\dot{\theta}^2 + g\frac{\cos\alpha}{\sin\alpha} = 0 \quad \text{즉} \quad \boxed{\ddot{r} - r\dot{\theta}^2\sin^2\alpha + gr\sin\alpha\cos\alpha = 0}$$

$$\frac{\partial L}{\partial \dot{\theta}} = mr^2\dot{\theta} \qquad \frac{\partial L}{\partial \theta} = 0 \qquad \therefore \frac{d}{dt}(mr^2\dot{\theta}) = 0 \quad \therefore r^2\dot{\theta} = h \text{ (일정)}$$

5.10 위치좌표는, 질량 x_1, x_2로 취한다.



$$v_2 = \dot{x}_2$$

$$v_1^2 = \dot{x}_1^2 + \dot{y}_1^2 = \dot{x}_1^2 + (\dot{x}_1 - \dot{x}_2)^2\tan^2\alpha$$

$$y_1 = -(x_1 - x_2)\tan\alpha$$

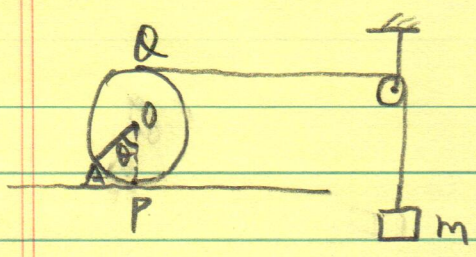
$$L = \frac{1}{2}m_1\left[\dot{x}_1^2 + (\dot{x}_1 - \dot{x}_2)^2\tan^2\alpha\right] + \frac{1}{2}m_2\dot{x}_2^2 + mg(x_1 - x_2)\tan\alpha$$

$$\frac{\partial L}{\partial x_1} = m_1\left[\dot{x}_1 + (\dot{x}_1 - \dot{x}_2)\tan^2\alpha\right] \qquad \frac{d}{dt}\frac{\partial L}{\partial \dot{x}_1} = m_1\left[(1+\tan^2\alpha)\ddot{x}_1 - \tan^2\alpha\,\ddot{x}_2\right]$$

$$\frac{\partial L}{\partial x_1} = m_1 g\tan\alpha$$

$$\Rightarrow \quad \ddot{x}_1 - \ddot{x}_2\sin^2\alpha - g\sin\alpha\cos\alpha = 0$$

5.11



初始时刻圆柱与地面接触点为A，任意时刻OA转动角度θ为坐标

由无滑滚动条件，$V_P=0 \Rightarrow V_O-r\dot\theta=0 \Rightarrow V_O=r\dot\theta$

∴圆柱顶端速度 $V_Q=V_O+r\dot\theta=2r\dot\theta$

∴圆柱动能 $T_1=\frac{1}{2}I_P\omega^2=\frac{1}{2}\left(\frac{1}{2}Mr^2+Mr^2\right)\cdot\dot\theta^2=\frac{3}{4}Mr^2\dot\theta^2$

重物动能 $T_2=\frac{1}{2}mV^2=\frac{1}{2}mV_Q^2=2mr^2\dot\theta^2$

势能零点，取重物初始时刻为势能零点，

则零点到Q点，任意时刻有 $V_Q=2V_O$

∴绳子下降距离为 $\int_0^t V_Q\,dt=\int_0^t 2r\dot\theta\,dt=2r\theta$

∴ $V=-mgh=-2mgr\theta$

∴系统拉格朗日量为

$$L=\left(\frac{3}{4}M+2m\right)r^2\dot\theta^2+2mgr\theta$$

$$\frac{d}{dt}\frac{\partial L}{\partial\dot\theta}=\frac{d}{dt}\left[\left(\frac{3}{4}M+2m\right)\cdot 2r^2\dot\theta\right]=\left(\frac{3}{2}M+4m\right)r^2\ddot\theta$$

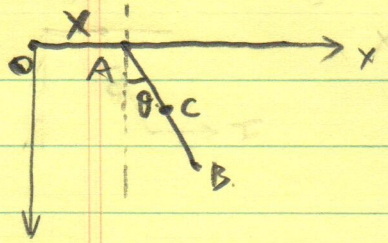
$$\frac{\partial L}{\partial\theta}=2mgr$$

由拉格朗日方程，$\left(\frac{3}{2}M+4m\right)r^2\ddot\theta=2mgr$

$$a_1=\frac{d}{dt}(r\dot\theta)=r\ddot\theta=\frac{2mg}{\frac{3}{2}M+4m}=\frac{4m}{3M+8m}\cdot g$$

$$a_2=\frac{d}{dt}(2r\dot\theta)=2r\ddot\theta=2a_1$$

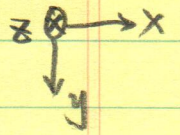
5.12



易见自由度为 $s=2$，取 x 与 θ 为广义坐标.

$$T=\frac{1}{2}mV_c^2+\frac{1}{2}I\omega^2$$

$$\vec{V}_c=\vec{V}_A+\vec{\omega}\times\vec{r}_C$$



$$=\dot{x}\,\hat{e}_x+(-\dot{\theta}\,\hat{e}_z)\times(a\sin\theta\,\hat{e}_x+a\cos\theta\,\hat{e}_y)$$

$$=(\dot{x}+a\dot{\theta}\cos\theta)\,\hat{e}_x-a\dot{\theta}\sin\theta\,\hat{e}_y$$

$$\therefore V_c^2=\dot{x}^2+a^2\dot{\theta}^2+2a\dot{x}\dot{\theta}\cos\theta$$

$$I=\frac{1}{12}ml^2=\frac{1}{12}m\cdot(2a)^2=\frac{1}{3}ma^2$$

$$\therefore T=\frac{1}{2}m(\dot{x}^2+a^2\dot{\theta}^2+2a\dot{x}\dot{\theta}\cos\theta)+\frac{1}{6}ma^2\dot{\theta}^2$$

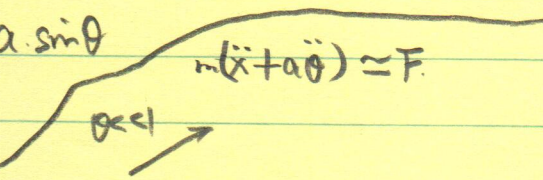
$$\begin{cases}\vec{F}_B=F\cdot\hat{e}_x\\ \vec{F}_C=mg\cdot\hat{e}_y\end{cases}\quad\because\quad\begin{aligned}\vec{r}_B&=(x+2a\sin\theta)\,\hat{e}_x+2a\cos\theta\,\hat{e}_y\\ \vec{r}_C&=(x+a\sin\theta)\,\hat{e}_x+a\cos\theta\,\hat{e}_y\end{aligned}$$

$$Q_x=\vec{F}_B\cdot\frac{\partial\vec{r}_B}{\partial x}+\vec{F}_C\cdot\frac{\partial\vec{r}_C}{\partial x}=F_{Bx}+F_{Cx}=F$$

$$Q_\theta=\vec{F}_B\frac{\partial\vec{r}_B}{\partial\theta}+\vec{F}_C\cdot\frac{\partial\vec{r}_C}{\partial\theta}=F\frac{\partial r_{Bx}}{\partial\theta}+mg\cdot\frac{\partial r_{Cy}}{\partial\theta}$$

$$=F\cdot2a\cos\theta-mg\,a\sin\theta$$

$\therefore$ 运动方程 $\frac{d}{dt}\cdot\frac{\partial T}{\partial\dot{q}_\alpha}-\frac{\partial T}{\partial q_\alpha}=Q_\alpha$，有.



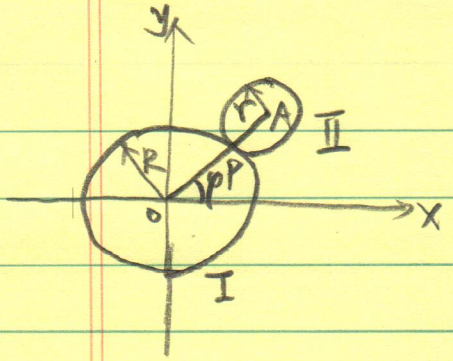
$q_\alpha=x$，有 $\frac{d}{dt}(m\dot{x}+ma\dot{\theta}\cos\theta)=F$ 即 $m(\ddot{x}+a\ddot{\theta}\cos\theta-a\dot{\theta}^2\sin\theta)=F$

$q_\alpha=\theta$，有 $\frac{d}{dt}\left(ma^2\dot{\theta}+ma\dot{x}\cos\theta+\frac{1}{3}ma^2\dot{\theta}\right)-\left(-ma\dot{x}\dot{\theta}\sin\theta\right)=F\cdot2a\cos\theta-mg\cdot a\sin\theta$

即 $ma\ddot{\theta}+m\ddot{x}\cos\theta-m\dot{x}\dot{\theta}\sin\theta+\frac{1}{3}ma\ddot{\theta}+m\dot{x}\dot{\theta}\sin\theta=2F\cos\theta-mg\sin\theta$

$$m\left(\ddot{x}\cos\theta+\frac{4}{3}a\ddot{\theta}\right)=2F\cos\theta-g\sin\theta\xrightarrow{\theta\ll1}m\left(\ddot{x}+\frac{4}{3}a\ddot{\theta}\right)=2F-mg\theta$$

5.13.



由无滑滑动条件，Ⅰ轮与Ⅱ轮接触点速度相等.

$\therefore \quad V_A - rw = 0$

即 $(R+r)\dot{\varphi} - rw = 0 \quad \therefore w = (\frac{R}{r}+1)\dot{\varphi}$

Ⅱ齿轮Ⅱ动能：$T_1 = \frac{1}{2} I_P w^2 = \frac{1}{2}\left(\frac{1}{2}m_2 r^2 + m_2 r^2\right)\left(\frac{R}{r}+1\right)^2 \dot{\varphi}^2 = \frac{3}{4} m_2 (R+r)^2 \dot{\varphi}^2$

杆OA动能 $T_2 = \frac{1}{2} I_O \dot{\varphi}^2 = \frac{1}{2}\left[\frac{1}{12} m_1 (R+r)^2 + \frac{1}{4} m_1 (R+r)^2\right]\dot{\varphi}^2 = \frac{1}{6} m_1 (R+r)^2 \dot{\varphi}^2$

$\therefore$ 总动能 $T = T_1 + T_2 = \left(\frac{3}{4} m_2 + \frac{1}{6} m_1\right)(R+r)^2 \dot{\varphi}^2$

拉氏方程 $\frac{d}{dt}\frac{\partial T}{\partial \dot{\varphi}} - \frac{\partial T}{\partial \varphi} = Q \qquad ①$

本题中广义坐标为角度φ，对应广义力为力矩M

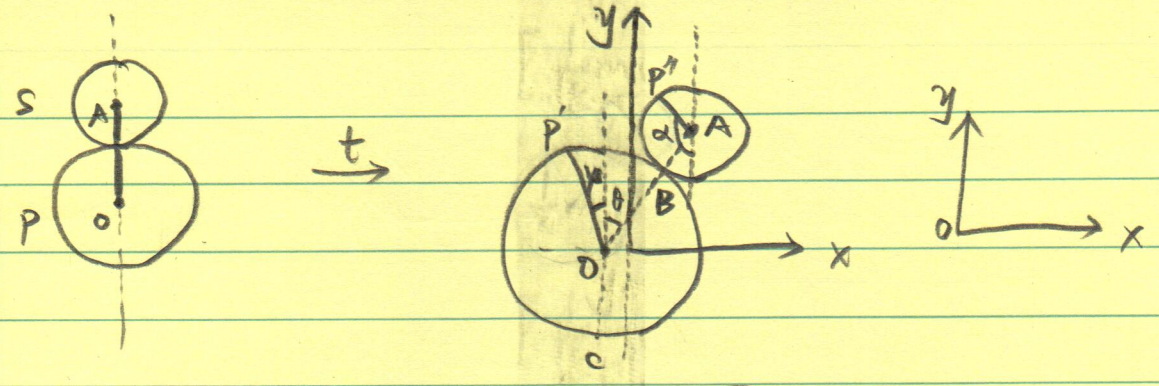
$\frac{\partial T}{\partial \dot{\varphi}} = \left(\frac{3}{2} m_2 + \frac{1}{3} m_1\right)(R+r)^2 \dot{\varphi}$

$\frac{d}{dt}\frac{\partial T}{\partial \dot{\varphi}} = \left(\frac{3}{2} m_2 + \frac{1}{3} m_1\right)(R+r)^2 \ddot{\varphi} = \frac{3}{2} m_2 (R+r)^2 \ddot{\varphi}\left(1 + \frac{2}{9}\frac{m_1}{m_2}\right)$

代入方程①，并考虑到 $Q = M$，得到

$$\ddot{\varphi} = \frac{2M}{3m_2(R+r)^2\left(1+\frac{2}{9}\frac{m_1}{m_2}\right)}.$$

5.14



无滑动 $\therefore$ 弧长 $\widehat{BP'} = \widehat{BP''}$

$\therefore R(\theta+\varphi) = r(\alpha-\theta)$

$\therefore (R+r)\theta = r\alpha - R\varphi \quad \therefore c\dot\theta = r\dot\alpha - R\dot\varphi$

圆柱体P点动能 $T_1 = \frac{1}{2} I_C \dot\varphi^2 = \frac{1}{2}\left(\frac{1}{2}MR^2 + MR^2\right)\dot\varphi^2 = \frac{3}{4}MR^2\dot\varphi^2$

圆球体S点动能 $T_2 = \frac{1}{2}mV_A^2 + \frac{1}{2}I_A\dot\alpha^2$

$$\vec V_A = \vec V_O + \vec\omega_A \times \vec r_{OA} = -R\dot\varphi\,\hat e_x + (\dot\theta\,\hat e_z)\times c(\sin\theta\,\hat e_x + \cos\theta\,\hat e_y)$$

$$= (-R\dot\varphi + c\dot\theta\cos\theta)\,\hat e_x - c\dot\theta\sin\theta\,\hat e_y$$

$\therefore V_A^2 = R^2\dot\varphi^2 + c^2\dot\theta^2 - 2Rc\dot\varphi\dot\theta\cos\theta$

$\therefore T_2 = \frac{1}{2}m\cdot(R^2\dot\varphi^2 + c^2\dot\theta^2 - 2Rc\dot\varphi\dot\theta\cos\theta) + \underline{\frac{1}{4}mr^2\dot\alpha^2}$

$\frac{1}{4}m(c^2\dot\theta^2 + R^2\dot\varphi^2 + 2Rc\dot\theta\dot\varphi)$

$$= \frac{3}{4}mR^2\dot\varphi^2 + \frac{3}{4}mc^2\dot\theta^2 + \frac{1}{2}mRc\dot\theta\dot\varphi\,(1-2\cos\theta)$$

$V = mgc\cdot\cos\theta$

$\therefore L = T_1 + T_2 - V = \frac{3}{4}(M+m)R^2\dot\varphi^2 + \frac{3}{4}mc^2\dot\theta^2 + \frac{1}{2}mRc\dot\theta\dot\varphi(1-2\cos\theta) - mgc\cos$

易见 $\frac{\partial L}{\partial\varphi} = 0$，由拉格朗日方程，$\frac{d}{dt}\frac{\partial L}{\partial\dot\varphi} = \dot P_\varphi = 0 \quad \therefore P_\varphi =$ 常数

$\therefore \frac{3}{2}(M+m)R^2\dot{\varphi} + \frac{1}{2}mRC\dot{\theta}(1-2\cos\theta) = C_1$

$t=0$ 时，圆柱的初始，有 $\dot{\varphi}=\dot{\theta}=0$ $\therefore C_1=0$

两边同乘以 dt，有

$\frac{3}{2}(M+m)R^2 d\varphi = -\frac{1}{2}mRC(1-2\cos\theta)d\theta$

积分后有 $\frac{3}{2}(M+m)R^2\varphi = -\frac{1}{2}mRC(\theta-2\sin\theta) + C_2$

$t=0$ 时，有 $\theta=\varphi=0$， $\therefore C_2=0$

$\therefore 3(M+m)R\varphi + mC(\theta-2\sin\theta)=0$

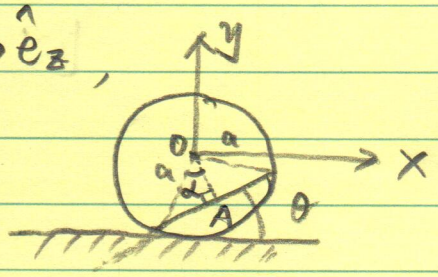
$\therefore R\varphi = -\frac{mC(\theta-2\sin\theta)}{3(M+m)}$

$\therefore$ 杆A点坐标为：

$$\begin{cases} x_A = -R\varphi + C\sin\theta = \frac{mC(\theta-2\sin\theta)}{3(M+m)} + C\sin\theta = C\frac{m\theta+(3M+m)\sin\theta}{3(M+m)} \\ y_A = C\cos\theta \end{cases}$$

E.15. 杆与球壳组成一个系统，球壳可以作运动，杆在球壳内也可以绕球心O运动。设球壳角速度为 $\omega\hat{e}_z$，

则 $\vec{V}_O = -\omega a\hat{e}_x$，球壳 $I_O = \frac{2}{5}Ma^2$



$\therefore$ 球壳动能 $T_1 = \frac{1}{2}(I_O + Ma^2)\cdot\omega^2 = \frac{7}{10}Ma^2\omega^2$

对杆，其角速度 $\vec{\omega}_1 = \dot{\theta}\hat{e}_z$，杆质心A点的速度为

$\vec{V}_A = \vec{V}_O + \vec{\omega}_1\times\vec{r}_{OA} = -\omega a\hat{e}_x + \dot{\theta}\hat{e}_z \times a\cdot\cos\alpha(\sin\theta\hat{e}_x - \cos\theta\hat{e}_y)$

$$= (a\dot\theta\cos\theta\cos\alpha - \omega a)\hat e_x + a\dot\theta\sin\theta\cos\alpha\,\hat e_y$$

∴ 棒 2 的动能为:

$$T_2 = \frac{1}{2}I_A\dot\theta^2 + \frac{1}{2}m\cdot\vec V_A^2$$

$$= \frac{1}{2}\cdot\frac{1}{12}m\cdot(2a\sin\alpha)^2\cdot\dot\theta^2 + \frac{m}{2}\left(\omega^2a^2 + a^2\dot\theta^2\cos^2\alpha - 2\omega a^2\cos\alpha\cdot\dot\theta\cos\theta\right)$$

$\frac{1}{6}ma^2\dot\theta^2\sin^2\alpha$

$$= \frac{1}{2}m\omega^2a^2 + ma^2\left(\frac{1}{2}\cos^2\alpha + \frac{1}{6}\sin^2\alpha\right)\dot\theta^2 - m\omega a^2\cos\alpha\,\dot\theta\cos\theta$$

棒 1 的势能:

$$V = -mg\cdot a\cos\alpha\cdot\cos\theta$$

取过点 O 的角度 φ 和棒 1 的角度 θ 为广义坐标, $\omega = \dot\varphi$

$$\therefore L = T_1 + T_2 - V = \frac{1}{6}(5M+3m)a^2\dot\varphi^2 + \frac{1}{6}ma^2(3\cos^2\alpha + \sin^2\alpha)\dot\theta^2$$

$$- ma^2\cos\alpha\,\dot\varphi\dot\theta\cos\theta + mga\cos\alpha\cos\theta$$

因为 $\frac{\partial L}{\partial\varphi} = 0$, 故 $P_\varphi = c$ 常数

$$\therefore c = P_\varphi = \frac{\partial L}{\partial\dot\varphi} = \frac{1}{3}(5M+3m)a^2\dot\varphi - ma^2\cos\alpha\,\dot\theta\cos\theta$$

$t=0$ 时, 有 $\dot\theta = \dot\varphi = 0$, 故 $c = 0$ $\therefore \dot\varphi = \frac{3m}{5M+3m}\cos\alpha\cos\theta\cdot\dot\theta$

因为 $\frac{\partial H}{\partial t} = 0$ $\therefore$ 机械能守恒. 将 $\dot\varphi$ 表达式代回 T, 得到

$$\frac{1}{6}a^2\dot\theta^2\left[\frac{9m^2}{5M+3m}\cos^2\alpha\cos^2\theta + m(3\cos^2\alpha+\sin^2\alpha) - \frac{18m^2}{5M+3m}\cos^2\alpha\cos^2\theta\right]$$

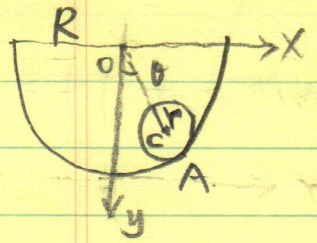
$$+ mga\cos\alpha\cos\theta = -mga\cos\alpha\cos\beta \quad \text{(上式左边为系统时刻 } t \text{ 的机械能)}$$

整理, 得到

$$6g(5M+3m)(\cos\theta - \cos\beta)\cos\alpha$$

$$= \left[(5M+3m)(3\cos^2\alpha+\sin^2\alpha) - 9m\cos^2\alpha\cos^2\theta\right]a\dot\theta^2$$

5.16



取θ为广义坐标，则对应的y坐标为：

$$y_c = (R-r)\cos\theta$$

$$\therefore V = -mgy_c = -mg(R-r)\cos\theta$$

由A点处速度为0，有：$v_c - \omega r = 0$

即 $v_c = \dot{\theta}(R-r)$，故小球的角速度为

$$\omega = \frac{v_c}{r} = \left(\frac{R}{r}-1\right)\dot{\theta}$$

$$\therefore T = \frac{1}{2}I_A\omega^2 = \frac{1}{2}\left(\frac{2}{5}mr^2 + mr^2\right)\left(\frac{R}{r}-1\right)^2\dot{\theta}^2$$

$$= \frac{7}{10}m(R-r)^2\dot{\theta}^2$$

$$\therefore L = T - V = \frac{7}{10}m(R-r)^2\dot{\theta}^2 + mg(R-r)\cos\theta$$

拉氏方程：$\frac{d}{dt}\frac{\partial L}{\partial \dot{\theta}} - \frac{\partial L}{\partial \theta} = 0$

$$\frac{\partial L}{\partial \dot{\theta}} = \frac{7}{5}m(R-r)^2\dot{\theta}, \quad \frac{d}{dt}\frac{\partial L}{\partial \dot{\theta}} = \frac{7}{5}m(R-r)^2\ddot{\theta}, \quad \frac{\partial L}{\partial \theta} = -mg(R-r)\sin\theta$$

代入方程，

$$\frac{7}{5}m(R-r)^2\ddot{\theta} + mg(R-r)\sin\theta = 0$$

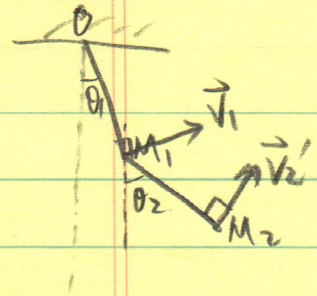
$$\ddot{\theta} + \frac{5g}{7(R-r)}\sin\theta = 0$$

微振动，$\theta \ll \pi$，$\sin\theta \simeq \theta$，上式变为 $\ddot{\theta} + \frac{5g}{7(R-r)}\theta = 0$

$$\therefore \theta = C_1\cos\omega t + C_2\sin\omega t \qquad \omega = \sqrt{\frac{5g}{7(R-r)}}$$

$$\therefore \text{周期 } T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{5}{7}\frac{R-r}{g}}$$

5.17



$$V = -m_1 g l\cos\theta_1 - m_2 g(l\cos\theta_1 + l_2\cos\theta_2)$$

$$= -(m_1+m_2) g l_1 \cos\theta_1 - m_2 g l_2 \cos\theta_2 .$$

$V_1 = l\dot{\theta}_1$, $\quad \vec{V}_2 = \vec{V}_1 + \vec{V}_2'$

因为 l_1 与 l_2 夹角为 $\theta_2-\theta_1$ ∴ $\vec{V}_1$ 与 $\vec{V}_2'$ 夹角为 $\theta_2-\theta_1$

$$\therefore V_2^2 = V_1^2 + V_2'^2 + 2V_1V_2'\cos(\theta_2-\theta_1)$$

$$= l_1^2\dot{\theta}_1^2 + l_2^2\dot{\theta}_2^2 + 2l_1l_2\dot{\theta}_1\dot{\theta}_2\cos(\theta_2-\theta_1)$$

$$\therefore L = T - V = \frac{1}{2}(m_1+m_2)l_1^2\dot{\theta}_1^2 + \frac{1}{2}m_2\left[l_2^2\dot{\theta}_2^2 + 2l_1l_2\dot{\theta}_1\dot{\theta}_2\cos(\theta_2-\theta_1)\right]$$

$$+ (m_1+m_2) g l_1\cos\theta_1 + m_2 g l_2\cos\theta_2$$

$$\frac{\partial L}{\partial\theta_1} = \underline{m_2 l_1 l_2\dot{\theta}_1\dot{\theta}_2\sin(\theta_2-\theta_1)} - (m_1+m_2) g l_1\sin\theta_1$$

$$\frac{d}{dt}\frac{\partial L}{\partial\dot{\theta}_1} = \frac{d}{dt}\left[(m_1+m_2)l_1^2\dot{\theta}_1 + m_2 l_1 l_2\dot{\theta}_2\cos(\theta_2-\theta_1)\right]$$

$$= (m_1+m_2)l_1^2\ddot{\theta}_1 + m_2 l_1 l_2\ddot{\theta}_2\cos(\theta_2-\theta_1) - \underline{m_2 l_1 l_2\dot{\theta}_2(\dot{\theta}_2-\dot{\theta}_1)\sin(\theta_2-\theta}$$

微振动, $\sin\theta\simeq\theta$, $\cos\theta\simeq 1$, $\theta^2\ll\theta$, 故有

$$(m_1+m_2)l_1^2\ddot{\theta}_1 + m_2 l_1 l_2\ddot{\theta}_2 - (m_1+m_2) g l_1\theta_1 = 0$$

即

$$\boxed{\ddot{\theta}_1 + \frac{m_2}{m_1+m_2}\frac{l_2}{l_1}\ddot{\theta}_2 - \frac{g}{l_1}\theta_1 = 0}$$

$$\frac{\partial L}{\partial\theta_2} = \underline{-m_2 l_1 l_2\dot{\theta}_1\dot{\theta}_2\sin(\theta_2-\theta_1)} - m_2 g l_2\sin\theta_2$$

$$\frac{d}{dt}\frac{\partial L}{\partial\dot{\theta}_2} = \frac{d}{dt}\left[m_2 l_2^2\dot{\theta}_2 + m_2 l_1 l_2\dot{\theta}_1\cos(\theta_2-\theta_1)\right]$$

$$= m_2 l_2^2\ddot{\theta}_2 + m_2 l_1 l_2\ddot{\theta}_1\cos(\theta_2-\theta_1) - \underline{m_2 l_1 l_2\dot{\theta}_1(\dot{\theta}_2-\dot{\theta}_1)\sin(\theta_2-\theta_1}$$

纵振动，下列运动方程成立，有：

$$\ddot{\theta}_2 + \frac{l_1}{l_2}\ddot{\theta}_1 + \frac{g}{l_2}\cdot\theta_2 = 0$$

$m_1 = m_2 = m$，$l_1 = l_2 = l$，则有

$$\ddot{\theta}_1 + \frac{1}{2}\ddot{\theta}_2 + \frac{g}{l}\theta_1 = 0 \qquad ①$$

$$\ddot{\theta}_2 + \ddot{\theta}_1 + \frac{g}{l}\theta_2 = 0 \qquad ②$$

②−①，$\frac{\ddot{\theta}_2}{2} + \frac{g}{l}(\theta_2 - \theta_1) = 0$ ③

①×2−②，$\ddot{\theta}_1 + \frac{g}{l}(2\theta_1 - \theta_2) = 0$ ④

$$\ddot{\theta}_1 + \lambda\ddot{\theta}_2 + \frac{g}{l}\left[(2-2\lambda)\theta_1 + (2\lambda-1)\theta_2\right] = 0$$

为使上式组合 $\theta_1 + \lambda\theta_2$ 形式，则有 $\frac{1}{\lambda} = \frac{2-2\lambda}{2\lambda-1}$（此式中表述有误，应为 $\theta_1+\lambda\theta_2$）

$\therefore 2\lambda^2 - 1 = 0$ $\therefore \lambda = \pm\frac{1}{\sqrt{2}}$

$\therefore 2 - 2\lambda = 2 \mp \sqrt{2}$

取 $\xi = \theta_1 + \lambda\theta_2$，则上式化为 $\ddot{\xi} + (2 \mp \sqrt{2})\frac{g}{l}\xi = 0$

$\therefore T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{l}{(2\mp\sqrt{2})g}}$

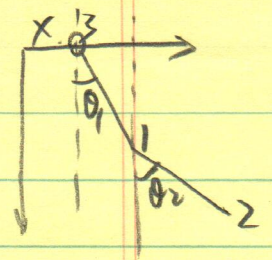
本题利用的方法与参考书答案上不同，基本思路为：

对于 $\ddot{\theta}_1 + a\theta_2 + b\theta_1 = 0$，$\ddot{\theta}_2 + c\theta_2 + d\theta_1 = 0$

寻找 θ_1 与 θ_2 的线性组合 $\xi = \theta_1 + \theta_2$，

使上述方程化为 $\begin{cases}\ddot{\xi}_1 + e\xi_1 = 0\\ \ddot{\xi}_2 + e'\xi_2 = 0\end{cases}$

5.18



$$T_3 = \frac{1}{2}\cdot 2m\cdot \dot{x}^2 = m\dot{x}^2$$

$$T_1 = \frac{1}{2}\cdot m\cdot(\vec{V}_3+\vec{\omega}_1\times\vec{r}_{13})^2 = \frac{1}{2}m(\dot{x}^2+l\dot{\theta}_1^2+2\dot{x}l\dot{\theta}_1\cos\theta_1)$$

$$T_2 = \frac{1}{2}m(\vec{V}_3+\vec{\omega}_1\times\vec{r}_{13}+\vec{\omega}_2\times\vec{r}_{24})^2$$

$$= \frac{1}{2}m[\dot{x}^2+l^2\dot{\theta}_1^2+l^2\dot{\theta}_2^2+2\dot{x}l(\dot{\theta}_1\cos\theta_1+\dot{\theta}_2\cos\theta_2)$$
$$+2l^2\dot{\theta}_1\dot{\theta}_2\cos(\theta_2-\theta_1)]$$

$$\therefore L=T-V = 2m\dot{x}^2+ml^2\left(\dot{\theta}_1^2+\frac{\dot{\theta}_2^2}{2}\right)+m\dot{x}l(2\dot{\theta}_1\cos\theta_1+\dot{\theta}_2\cos\theta_2)$$
$$+ml^2\dot{\theta}_1\dot{\theta}_2\cos(\theta_1-\theta_2)$$
$$+mgl(2\cos\theta_1+\cos\theta_2)$$

$$\frac{d}{dt}\frac{\partial L}{\partial\dot{x}} = 4m\ddot{x}+ml(2\dot{\theta}_1\cos\theta_1+\dot{\theta}_2\cos\theta_2)' = 0$$

∴ 化简后，有：$\boxed{4\ddot{x}+l(2\ddot{\theta}_1+\ddot{\theta}_2)=0}$ ∴ $x\sim l\theta$ 大量级.

$$\frac{d}{dt}\frac{\partial L}{\partial\dot{\theta}_1}\Big|_{\text{大量级}} = 2ml^2\ddot{\theta}_1+2m\ddot{x}l+ml^2\ddot{\theta}_2 \qquad \frac{\ddot{x}}{l}=-\frac{\ddot{\theta}_1}{2}-\frac{\ddot{\theta}_2}{4}$$

$$\frac{\partial L}{\partial\theta_1}\Big|_{\text{大量级}} = -2mgl\theta_1$$

$$\therefore \ddot{\theta}_1+\frac{\ddot{x}}{l}+\frac{\ddot{\theta}_2}{2}+\frac{g}{l}\theta_1=0 \Rightarrow \boxed{\ddot{\theta}_1+\frac{\ddot{\theta}_2}{2}+\frac{2g}{l}\theta_1=0} \quad ①$$

$$\frac{d}{dt}\frac{\partial L}{\partial\dot{\theta}_2}\Big|_{\text{大量级}} = ml^2\ddot{\theta}_2+m\ddot{x}l+ml^2\ddot{\theta}_1, \quad \frac{\partial L}{\partial\theta_2}\Big|_{\text{大量级}} = -mgl\theta_2$$

$$\therefore \ddot{\theta}_2+\frac{\ddot{x}}{l}+\ddot{\theta}_1+\frac{g}{l}\theta_2=0 \qquad \boxed{\ddot{\theta}_1+\frac{3\ddot{\theta}_2}{2}+\frac{2g}{l}\theta_2=0} \quad ②$$

②−①，有：$\ddot{\theta}_2 + \frac{2g}{l}(\theta_2 - \theta_1) = 0$ ③

①×3−②，有：$\ddot{\theta}_1 + \frac{g}{l}(3\theta_1 - \theta_2) = 0$ ④

取 $\xi = \theta_1 + \lambda\theta_2$，则 ④+③×λ，有：

$$(\ddot{\theta}_1 + \lambda\ddot{\theta}_2) + \frac{g}{l}[(3-2\lambda)\theta_1 + (2\lambda-1)\theta_2] = 0$$

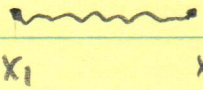
$\therefore \frac{1}{\lambda} = \frac{3-2\lambda}{2\lambda-1}$ $\quad \lambda = 1$ 或 $\lambda = -\frac{1}{2}$

$\therefore 3-2\lambda = 1$ 或 4

$\therefore \ddot{\xi} + (3-2\lambda)\frac{g}{l}\xi = 0$

$\therefore T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{l}{g}}$ 或 $\pi\sqrt{\frac{l}{g}}$

5.19



$$L = T - V = \frac{1}{2}m_1\dot{x}_1^2 + \frac{1}{2}m_2\dot{x}_2^2 - \frac{1}{2}k(x_2 - x_1 - a)^2$$

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{x}_1} = m_1\ddot{x}_1, \quad \frac{\partial L}{\partial x_1} = k(x_2 - x_1 - a) \Rightarrow m_1\ddot{x}_1 - k(x_2 - x_1 - a) = 0 \quad ①$$

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{x}_2} = m_2\ddot{x}_2, \quad \frac{\partial L}{\partial x_2} = -k(x_2 - x_1 - a) \Rightarrow m_2\ddot{x}_2 + k(x_2 - x_1 - a) = 0 \quad ②$$

①+②, 有 $m_1\ddot{x}_1 + m_2\ddot{x}_2 = 0$ 即 $m\ddot{x}_c = 0$

$$\therefore x_c = \frac{m_1x_1 + m_2x_2}{m} = A + Bt \quad ③$$

$m_1 \times ② - m_2 \times ①$, 有

$$m_1m_2(\ddot{x}_2 - \ddot{x}_1) + (m_1 + m_2)k(x_2 - x_1 - a) = 0$$

即 $\mu\ddot{\xi} + k(\xi - a) = 0$

$$\nu = \sqrt{\frac{k}{\mu}} = \sqrt{\frac{(m_2 + m_1)k}{m_1 \cdot m_2}} \quad ④$$

$$\therefore \xi = x_2 - x_1 = a + C\sin(\nu t + \varepsilon)$$

$③ \times \frac{m}{m_1} + ④$, 有 $\left(\frac{m_2}{m_1} + 1\right)x_2 = A\frac{m}{m_1} + B\frac{m}{m_1}t + a + C\sin(\nu t + \varepsilon)$

$$x_2 = \left(A + a\frac{m_1}{m}\right) + Bt + C\frac{m_1}{m}\sin(\nu t + \varepsilon)$$

$③ \times \frac{m}{m_2} - ④$, 有 $\left(\frac{m_1}{m_2} + 1\right)x_2 = A\frac{m}{m_2} + B\frac{m}{m_2}t - a - C\sin(\nu t + \varepsilon)$

$$x_1 = \left(A - a\frac{m_2}{m}\right) + Bt - C\frac{m_2}{m}\sin(\nu t + \varepsilon)$$

令 $\tilde{A} = A - a\frac{m_2}{m}$, $\tilde{C} = -C\frac{m_2}{m}$, 则

$$\begin{cases} x_1 = \tilde{A} + Bt + \tilde{C}\sin(\nu t + \varepsilon) \\ x_2 = A + a + Bt - \tilde{C}\frac{m_1}{m_2}\sin(\nu t + \varepsilon) \end{cases}$$

5.20 解：直角坐标 $\vec{r}$ 为广义坐标，$s=3$，$v^2=\dot{x}^2+\dot{y}^2+\dot{z}^2$

广义动量由定义 $p_x=\frac{\partial L}{\partial \dot{x}}=m\dot{x}+qA_x$

$p_y=\frac{\partial L}{\partial \dot{y}}=m\dot{y}+qA_y$

$p_z=\frac{\partial L}{\partial \dot{z}}=m\dot{z}+qA_z$

$\Rightarrow \vec{p}=m\vec{v}+q\vec{A}$

$\vec{v}=\frac{1}{m}(\vec{p}-q\vec{A})$

$H=\sum_{\alpha=1}^{3}p_\alpha\dot{r}_\alpha-L=\dot{x}p_x+\dot{y}p_y+\dot{z}p_z-L$

$=m\dot{x}^2+m\dot{y}^2+m\dot{z}^2+q(A_x\dot{x}+A_y\dot{y}+A_z\dot{z})-L$

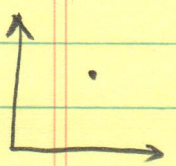
$=mv^2-q\vec{A}\cdot\vec{v}-L$

$=mv^2-q\vec{A}\cdot\vec{v}-(\frac{1}{2}mv^2-q\varphi+q\vec{A}\cdot\vec{v})$

$=\frac{1}{2}mv^2+q\varphi$

$=\frac{1}{2m}(\vec{p}-q\vec{A})^2+q\varphi$

5.21



匀速转动参考系，$\vec{v}=\vec{v}'+\vec{\Omega}\times\vec{r}$

$L=T-V=\frac{1}{2}mv^2=\frac{1}{2}m(\vec{v}'+\vec{\Omega}\times\vec{r})^2$

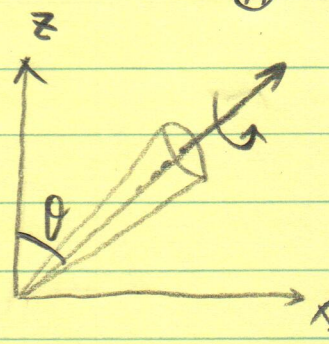
$\vec{p}=\frac{\partial L}{\partial \vec{v}'}=m(\vec{v}'+\vec{\Omega}\times\vec{r})$

$\therefore H=\vec{p}\cdot\vec{v}'-L=\vec{p}\cdot\vec{v}-\vec{p}\cdot(\vec{\Omega}\times\vec{r})-\frac{1}{2}mv^2=\frac{1}{2}mv^2-\vec{\Omega}\cdot(\vec{r}\times\vec{p})$

$=\frac{p^2}{2m}-\vec{\Omega}\cdot(\vec{r}\times\vec{p})$

有势场中 $H=\frac{p^2}{2m}-\vec{\Omega}\cdot(\vec{r}\times\vec{p})+V$

5.22 质量为m长为l的均匀细棒的一端固定的欧拉角和欧拉动力学方程



$$\begin{cases} \omega_x = \dot{\varphi}\sin\theta\sin\psi + \dot{\theta}\cos\psi \\ \omega_y = \dot{\varphi}\sin\theta\cos\psi - \dot{\theta}\sin\psi \\ \omega_z = \dot{\varphi}\cos\theta + \dot{\psi} \end{cases}$$

$$\therefore T = \frac{1}{2}I_1\omega_x^2 + \frac{1}{2}I_2\omega_y^2 + \frac{1}{2}I_3\omega_z^2$$

$$= \frac{1}{2}I(\dot{\varphi}^2\sin^2\theta + \dot{\theta}^2) + \frac{1}{2}I_3(\dot{\varphi}\cos\theta + \dot{\psi})^2$$

$$V = mgh = mgl\cdot\cos\theta$$

$$\therefore L = \frac{1}{2}I(\dot{\varphi}^2\sin^2\theta + \dot{\theta}^2) + \frac{1}{2}I_3(\dot{\varphi}\cos\theta + \dot{\psi})^2 - mgl\cos\theta$$

$$P_\theta = \frac{\partial L}{\partial\dot{\theta}} = I\dot{\theta}, \quad P_\psi = \frac{\partial L}{\partial\dot{\psi}} = I_3(\dot{\varphi}\cos\theta + \dot{\psi})$$

$$P_\varphi = \frac{\partial L}{\partial\dot{\varphi}} = I\dot{\varphi}\sin^2\theta + I_3\cos\theta(\dot{\varphi}\cos\theta + \dot{\psi}) = I\dot{\varphi}\sin^2\theta + P_\psi\cos\theta$$

$$H = \dot{\theta}P_\theta + \dot{\varphi}P_\varphi + \dot{\psi}P_\psi - L$$

$$= I\dot{\theta}^2 + I\dot{\varphi}^2\sin^2\theta + I_3\dot{\varphi}\cos\theta(\dot{\varphi}\cos\theta + \dot{\psi}) + I_3\dot{\psi}(\dot{\varphi}\cos\theta + \dot{\psi}) - L$$

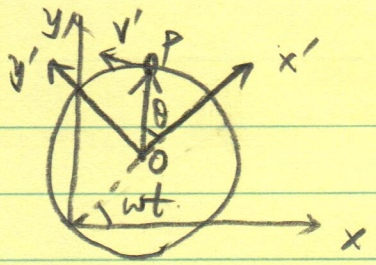
$$= I(\dot{\varphi}^2\sin^2\theta + \dot{\theta}^2) + I_3(\dot{\varphi}\cos\theta + \dot{\psi})^2 - L$$

$$\therefore = \frac{1}{2}I(\dot{\varphi}^2\sin^2\theta + \dot{\theta}^2) + \frac{1}{2}I_3(\dot{\varphi}\cos\theta + \dot{\psi})^2 + mgl\cos\theta$$

$$= \frac{I}{2}\left\{\left[\frac{1}{I\sin^2\theta}(P_\varphi - P_\psi\cos\theta)\right]^2\sin^2\theta + \left(\frac{P_\theta}{I}\right)^2\right\} + \frac{1}{2}I_3\cdot\left(\frac{P_\psi}{I_3}\right)^2 + mgl\cdot\cos\theta$$

$$= \frac{P_\theta^2}{2I} + \frac{(P_\varphi - P_\psi\cos\theta)^2}{2I\cdot\sin^2\theta} + \frac{P_\psi^2}{2I_3} + mgl\cos\theta$$

5.23 利用正则方程解 4.10 题



(20)

前几章的结论：拉氏量 $L=T-V$.

运动学如图所示

(这里利用：坐标基矢与 S 系不同，请注意)

$$\vec{V}_P=\vec{V}_0+\vec{V}'+\vec{\omega}\times(\vec{r}_P-\vec{r}_0)$$

$$=\omega a\,\hat{e}_{y'}+a\dot{\theta}(-\sin\theta\,\hat{e}_{x'}+\cos\theta\,\hat{e}_{y'})+\omega a(-\sin\theta\,\hat{e}_{x'}+\cos\theta\,\hat{e}_{y'})$$

$$=-(\omega+\dot{\theta})a\sin\theta\,\hat{e}_{x'}+[\omega a+(\omega+\dot{\theta})a\cos\theta]\,\hat{e}_{y'}$$

$$L=T-V=T=\frac{1}{2}m\vec{V}_P^2=\frac{1}{2}ma^2[\omega^2+(\omega+\dot{\theta})^2+2\omega(\omega+\dot{\theta})\cos\theta]$$

$$\therefore P_\theta=\frac{\partial L}{\partial\dot{\theta}}=ma^2(\omega+\dot{\theta})+ma^2\omega\cos\theta \qquad (2)$$

$$H=\dot{\theta}P_\theta-L=\frac{1}{2}ma^2[2\dot{\theta}(\omega+\dot{\theta})+2\dot{\theta}\omega\cos\theta-\omega^2-(\omega+\dot{\theta})^2-2\omega(\omega+\dot{\theta})\cos\theta]$$

$$=\frac{1}{2}ma^2[\dot{\theta}^2-2\omega^2(\cos\theta+1)]$$

$$=\frac{1}{2}ma^2\left\{\left[\frac{P_\theta}{ma^2}-\omega(1+\cos\theta)\right]^2-2\omega^2(\cos\theta+1)\right\}$$

$$=\frac{P_\theta^2}{2ma^2}-P_\theta\omega(1+\cos\theta)-\frac{1}{2}ma^2\omega^2\sin^2\theta$$

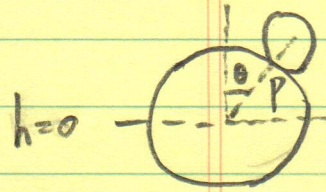
哈密顿方程：(注意正负号)

$$\begin{cases}\dot{P}_\theta=-\dfrac{\partial H}{\partial\theta}=-P_\theta\omega\sin\theta+ma^2\omega^2\sin\theta\cos\theta \qquad (1)\\ \dot{\theta}=\dfrac{\partial H}{\partial P_\theta}=\dfrac{P_\theta}{ma^2}-\omega(1+\cos\theta)\end{cases}$$

由(1)，$ma^2\ddot{\theta}-ma^2\omega\dot{\theta}\sin\theta=-ma^2(\omega+\dot{\theta})\omega\sin\theta-ma^2\omega^2\cos\theta\sin\theta+ma^2\omega^2\sin\theta\cos\theta$

$$\therefore\ \ddot{\theta}=-\omega^2\sin\theta \quad 即\quad \ddot{\theta}+\omega^2\sin\theta=0$$

5.24



以坐标 θ 为广义坐标，质心加速度 $a_t = \dot{v}_c = (b+c)\ddot{\theta}$

无滑滚动，$v_p = 0$

$$\therefore \dot{\theta}(b+c) = \omega c \qquad \therefore \omega = \left(1+\frac{b}{c}\right)\dot{\theta}$$

$$\therefore T = \frac{1}{2}I_p\omega^2 = \frac{1}{2}\left(\frac{2}{5}mc^2 + mc^2\right)\omega^2 = \frac{7}{10}m(b+c)^2\dot{\theta}^2$$

$$V = mgh = mg(b+c)\cos\theta$$

$$\therefore L = T - V = \frac{7}{10}m(b+c)^2\dot{\theta}^2 - mg(b+c)\cos\theta$$

$$p_\theta = \frac{\partial L}{\partial \dot{\theta}} = \frac{7}{5}m(b+c)^2\dot{\theta} \qquad \therefore \dot{\theta} = \frac{5}{7}\frac{p_\theta}{m(b+c)^2}$$

$$\therefore H = \dot{\theta}p_\theta - L = \frac{7}{10}m(b+c)^2\dot{\theta}^2 + mg(b+c)\cos\theta$$

$$= \frac{5}{14}\frac{p_\theta^2}{m(b+c)^2} + mg(b+c)\cos\theta$$

由 $\dot{p}_\theta = -\frac{\partial H}{\partial \theta}$，得

$$\frac{7}{5}m(b+c)^2\ddot{\theta} = -\frac{\partial H}{\partial \theta} = mg(b+c)\sin\theta$$

$$\therefore a_t = (b+c)\ddot{\theta} = \frac{5}{7}g\sin\theta$$

5.25 角动量分量：泊松括号 $[J_x, J_y]$，$[J_x, J_z]$，$[J_y, J_z]$

泊松括号定义：$[A,B] = \sum_{\alpha=1}^{s}\left(\frac{\partial A}{\partial q_\alpha}\frac{\partial B}{\partial p_\alpha} - \frac{\partial A}{\partial p_\alpha}\frac{\partial B}{\partial q_\alpha}\right)$

性质：$[C, A] = 0$

$[A,B] + [B,A] = 0 \Rightarrow [A,A] = 0$

$\left[\sum_{i=1}^{n} c_i A_i, B\right] = \sum_{i=1}^{n} c_i [A_i, B]$

$\frac{\partial}{\partial t}[A,B] = \left[\frac{\partial A}{\partial t}, B\right] + \left[A, \frac{\partial B}{\partial t}\right]$

$[A,[B,\psi]] + [B,[\psi,A]] + [\psi,[A,B]] = 0$

$[q_\alpha, p_\beta] = \begin{cases} \delta_{\alpha\beta} & (\alpha=\beta) \\ 0 & (\alpha\neq\beta) \end{cases}$

根据角动量定义: $J_x = yp_z - zp_y$, $J_y = zp_x - xp_z$, $J_z = xp_y - yp_x$

$$[J_x, J_y] = \sum_{\alpha=1}^{s}\left(\frac{\partial J_x}{\partial q_\alpha}\frac{\partial J_y}{\partial p_\alpha} - \frac{\partial J_x}{\partial p_\alpha}\frac{\partial J_y}{\partial q_\alpha}\right)$$

$$= \left(\frac{\partial J_x}{\partial x}\frac{\partial J_y}{\partial p_x} - \frac{\partial J_x}{\partial p_x}\frac{\partial J_y}{\partial x}\right) + \left(\frac{\partial J_x}{\partial y}\frac{\partial J_y}{\partial p_y} - \frac{\partial J_x}{\partial p_y}\frac{\partial J_y}{\partial y}\right) + \left(\frac{\partial J_x}{\partial z}\frac{\partial J_y}{\partial p_z} - \frac{\partial J_x}{\partial p_z}\frac{\partial J_y}{\partial z}\right)$$

$$= 0 + 0 + (-p_y)\cdot(-x) - y\cdot p_x = xp_y - yp_x = J_z$$

$\therefore [J_x, J_y] = J_z$

同理, 可证 $[J_y, J_z] = J_x$, $[J_z, J_x] = J_y$

引入记号 $\epsilon_{ijk} = \begin{cases} 1, & (ijk\text{是}123\text{的偶排列}) \\ -1, & (ijk\text{是}123\text{的奇排列}) \\ 0, & (ijk\text{中有相同的指标}) \end{cases}$

则三个关系可以统一写为 $[J_\alpha, J_\beta] = \sum_{\gamma=1}^{3}\epsilon_{\alpha\beta\gamma}J_\gamma$

例如, $\alpha=1, \beta=2$, 有

$$[J_x, J_y] = \underbrace{\epsilon_{121}}_{0}J_1 + \underbrace{\epsilon_{122}}_{0}J_2 + \epsilon_{123}J_3 = J_3.$$

5.26 关于哪个分量的泊松括号:

注意到 $\frac{\partial J_x}{\partial x} = \frac{\partial J_x}{\partial p_x} = 0$, $\frac{\partial p_\alpha}{\partial p_\beta} = \delta_{\alpha\beta} = \begin{cases} 1, & (\alpha=\beta) \\ 0, & (\alpha\neq\beta) \end{cases}$, $\frac{\partial p_\alpha}{\partial q_\beta} = 0$.

$$[J_x, p_y] = \sum_{\alpha=1}^{s}\left(\frac{\partial J_x}{\partial q_\alpha}\frac{\partial p_y}{\partial p_\alpha} - \frac{\partial J_x}{\partial p_\alpha}\frac{\partial p_y}{\partial q_\alpha}\right) = \sum_{\alpha=1}^{s}\frac{\partial J_x}{\partial q_\alpha}\frac{\partial p_y}{\partial p_\alpha} = \frac{\partial J_x}{\partial y}\frac{\partial p_y}{\partial p_y} = p_z.$$

$$[J_y, p_z] = \sum_{\alpha=1}^{s}\left(\frac{\partial J_y}{\partial q_\alpha}\frac{\partial p_z}{\partial p_\alpha} - \frac{\partial J_y}{\partial p_\alpha}\frac{\partial p_z}{\partial q_\alpha}\right) = \sum_{\alpha=1}^{s}\frac{\partial J_y}{\partial q_\alpha}\frac{\partial p_z}{\partial p_\alpha} = \frac{\partial J_y}{\partial z}\frac{\partial p_z}{\partial p_z} = p_x$$

$$[J_z, p_x] = \sum_{\alpha=1}^{s}\left(\frac{\partial J_z}{\partial q_\alpha}\frac{\partial p_x}{\partial p_\alpha} - \frac{\partial J_z}{\partial p_\alpha}\frac{\partial p_x}{\partial q_\alpha}\right) = \frac{\partial J_z}{\partial x} = p_y$$

$$[J_i, p_j] = \sum_{\alpha=1}^{s}\left(\frac{\partial J_i}{\partial q_\alpha}\frac{\partial p_j}{\partial p_\alpha} - \frac{\partial J_i}{\partial p_\alpha}\frac{\partial p_j}{\partial q_\alpha}\right) = \sum_{\alpha=1}^{s}\frac{\partial J_i}{\partial q_\alpha}\delta_{j\alpha} = \frac{\partial J_i}{\partial q_j}$$

由此可知：

$$[J_x, P_z] = \frac{\partial J_x}{\partial z} = -P_y, \quad [J_y, P_x] = \frac{\partial J_y}{\partial x} = -P_z, \quad [J_z, P_y] = \frac{\partial J_z}{\partial y} = -P_x.$$

$$[J_x, P_x] = \frac{\partial J_x}{\partial x} = 0, \quad [J_y, P_y] = \frac{\partial J_y}{\partial y} = 0, \quad [J_z, P_z] = \frac{\partial J_z}{\partial z} = 0.$$

例7 $\varphi = a\vec{r}^2 + b\vec{r}\cdot\vec{p} + c\vec{p}^2$, $J_z = xP_y - yP_x$.

$$\frac{\partial \varphi}{\partial x} = 2ax + bP_x, \quad \frac{\partial \varphi}{\partial P_x} = bx + 2cP_x.$$

$$\frac{\partial \varphi}{\partial y} = 2ay + bP_y, \quad \frac{\partial \varphi}{\partial P_y} = by + 2cP_y.$$

$$\therefore [\varphi, J_z] = \sum_{\alpha=1}^{3} \frac{\partial \varphi}{\partial q_\alpha}\frac{\partial J_z}{\partial P_\alpha} - \frac{\partial \varphi}{\partial P_\alpha}\frac{\partial J_z}{\partial q_\alpha}$$

$$\boxed{\frac{\partial J_z}{\partial z} = \frac{\partial J_z}{\partial P_z} = 0}$$

$$= \left(\frac{\partial \varphi}{\partial x}\frac{\partial J_z}{\partial P_x} - \frac{\partial \varphi}{\partial P_x}\frac{\partial J_z}{\partial x}\right) + \left(\frac{\partial \varphi}{\partial y}\frac{\partial J_z}{\partial P_y} + \frac{\partial \varphi}{\partial P_y}\frac{\partial J_z}{\partial y}\right)$$

$$= \underline{(2ax + bP_x)\cdot(-y)} - \underline{(bx + 2cP_x)P_y} + \underline{(2ay + bP_y)\cdot x} - \underline{(by + 2cP_y)(-P_x)}$$

$$= 0$$

若引用矢量形式容可以计算：

$$J_z = \sum_{i,j=1}^{3} \epsilon_{zij} q_i P_j \quad \therefore \frac{\partial J_z}{\partial q_\alpha} = \sum_{j=1}^{3} \epsilon_{z\alpha j} P_j, \quad \frac{\partial J_z}{\partial P_\alpha} = \sum_{i=1}^{3} \epsilon_{zi\alpha} q_i$$

$$\frac{\partial \varphi}{\partial q_\alpha} = 2aq_\alpha + bP_\alpha, \quad \frac{\partial \varphi}{\partial P_\alpha} = bq_\alpha + 2cP_\alpha$$

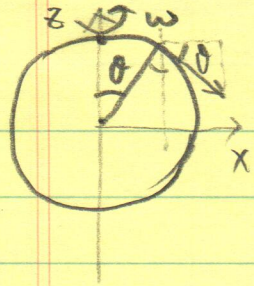
$$\therefore [\varphi, J_z] = \sum_{\alpha=1}^{3} \left[\underbrace{(2aq_\alpha + bP_\alpha)}\sum_{i=1}^{3}\underbrace{\epsilon_{zi\alpha} q_i}_{0} - \underbrace{(bq_\alpha + 2cP_\alpha)}_{0}\sum_{j=1}^{3}\underbrace{\epsilon_{z\alpha j} P_j}\right]$$

$$= \sum_{i,\alpha=1}^{3} \left(b\,\epsilon_{zi\alpha} q_i P_\alpha - b\,\epsilon_{z\alpha i} q_\alpha P_i\right)$$

$$= \sum_{i,\alpha=1}^{3} \left(b\,\epsilon_{zi\alpha} q_i P_\alpha - b\,\epsilon_{zi\alpha} q_i P_\alpha\right) = 0$$

最后一步交换了 $i \leftrightarrow \alpha$ 后仍表示求和小哑标

5.28.



$$\vec{V} = \omega a \sin\theta\, \hat{e}_y + a\dot{\theta}\cos\theta\, \hat{e}_x + a\dot{\theta}\sin\theta\, \hat{e}_z$$

$$\therefore \vec{V}^2 = a^2(\omega^2\sin^2\theta + \dot{\theta}^2)$$

$$\therefore T = \frac{1}{2}m\vec{V}^2 = \frac{1}{2}ma^2(\omega^2\sin^2\theta + \dot{\theta}^2)$$

$$V = mg \cdot a\cos\theta$$

$$\therefore L = T - V = \frac{1}{2}ma^2(\omega^2\sin^2\theta + \dot{\theta}^2) - mga\cdot\cos\theta$$

$$S = \int_0^t L\,dt \qquad \delta S = 0$$

$$\therefore \delta S = \int_0^t \delta L\,dt$$

$$= \int_0^t \left[\frac{1}{2}ma^2(\omega^2 \cdot 2\sin\theta\cdot\cos\theta\,\delta\theta + 2\dot{\theta}\,\delta\dot{\theta}) + mga\cdot\sin\theta\,\delta\theta\right]$$

$$\dot{\theta}\,\delta\dot{\theta} = \dot{\theta}\left(\frac{d}{dt}\delta\theta\right) = \frac{d}{dt}(\dot{\theta}\delta\theta) - \ddot{\theta}\delta\theta$$

$$\therefore \delta S = \int_0^t \left[ma^2(\omega^2\sin\theta\cos\theta - \ddot{\theta}) + mga\cdot\sin\theta\right]\cdot\delta\theta\,dt + \dot{\theta}\,\delta\theta\Big|_0^t$$

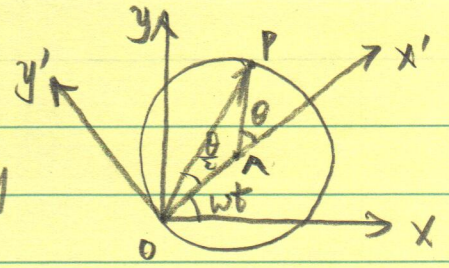
$\dot{\theta}\,\delta\theta\big|_0^t \to 0$

∴ 运动微分方程为

$$ma\ddot{\theta} = ma\omega^2\sin\theta\cos\theta + mg\sin\theta$$

注意：$\frac{d}{dt}$ 和 δ 可以交换，也即是说：$\frac{d}{dt}\delta\theta = \delta\cdot\frac{d\theta}{dt} = \delta\dot{\theta}$

5.29 用拉格朗日方程 解 4.10 题.



转动参考系, 刚体质点与刚体平动坐标原点重合, 则

$$\vec{v}_p = \vec{v}_{p'} + \vec{\omega} \times \vec{r}_p$$

相对速度: $\vec{v}_{p'} = \vec{\omega}_0 \times \vec{r}_{PA} = a\dot{\theta}(-\sin\theta\,\hat{e}_{x'} + \cos\theta\,\hat{e}_{y'})$

$$\vec{\omega} \times \vec{r}_p = \omega \hat{e}_z \times \underbrace{2a\cos\frac{\theta}{2}}_{|\vec{r}_p|} \cdot \left(\cos\frac{\theta}{2}\hat{e}_{x'} + \sin\frac{\theta}{2}\hat{e}_{y'}\right)$$

$$= \omega a\left[-\sin\theta\,\hat{e}_{x'} + (\cos\theta + 1)\hat{e}_{y'}\right]$$

$$\therefore \vec{v}_p = -(\omega+\dot{\theta})a\sin\theta\,\hat{e}_x + \left[\omega a + (\omega+\dot{\theta})a\cos\theta\right]\hat{e}_{y'}$$

$$\therefore \vec{v}_p^{\,2} = \omega^2 a^2 + (\omega+\dot{\theta})^2 a^2 + 2\omega(\omega+\dot{\theta})a^2\cos\theta$$

$$\therefore L = T = \frac{1}{2}m\left[\omega^2 a^2 + (\omega+\dot{\theta})^2 a^2 + 2\omega(\omega+\dot{\theta})a^2\cos\theta\right]$$

$$0 = \delta S = \delta\int_{t_1}^{t_2} L\,dt = \int_{t_1}^{t_2}\delta L\cdot dt$$

$$\boxed{\delta f(\theta) = \frac{\partial f}{\partial\theta}\delta\theta}$$

$$= \int_{t_1}^{t_2}\Big[\underbrace{m(\omega+\dot{\theta})a^2\cdot\delta\dot{\theta}}_{-ma^2\ddot{\theta}\delta\theta} + \underbrace{ma^2\cos\theta\cdot\delta\dot{\theta}}_{-ma^2\dot{\theta}\sin\theta\delta\theta} - m\omega(\omega+\dot{\theta})a^2\sin\theta\cdot\delta\theta\Big]dt$$

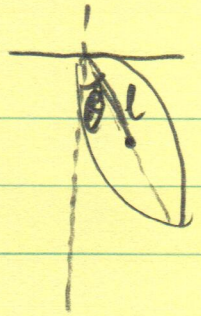
$$= \left[m(\omega+\dot{\theta})a^2\delta\theta + 2\omega a^2\cos\theta\,\delta\theta\right]\Big|_{t_1}^{t_2}$$

$$+ \int_{t_1}^{t_2} ma^2\left(-\ddot{\theta} - \omega\dot{\theta}\sin\theta - \omega(\omega+\dot{\theta})\sin\theta\right)\delta\theta\,dt$$

$$= -\int_{t_1}^{t_2} ma^2\left[\ddot{\theta} + \omega^2\sin\theta\right]\delta\theta\,dt = 0$$

$\therefore \delta\theta$ 任意 $\therefore \ddot{\theta} + \omega^2\sin\theta = 0.$

5.30 哈密顿原理求解单摆周期



单摆是转动惯量 $I=ml^2$，θ 为广义坐标

$$L=T-V=\frac{1}{2}I\dot{\theta}^2+mgl\cos\theta$$

$$\therefore \delta L=\frac{1}{2}I\,\delta(\dot{\theta}^2)+mgl\,\delta(\cos\theta)$$

$$\boxed{\delta(f(x))=\frac{\partial f}{\partial x}\delta x}$$

$$=I\dot{\theta}\,\delta\dot{\theta}-mgl\sin\theta\,\delta\theta$$

$$\therefore \delta I=\delta\int_{t_1}^{t_2}L\,dt=\int_{t_1}^{t_2}\delta L\,dt=\int_{t_1}^{t_2}(I\dot{\theta}\,\delta\dot{\theta}-mgl\sin\theta\,\delta\theta)\,dt$$

$$I\dot{\theta}\,\delta\dot{\theta}=I\dot{\theta}\,\delta\frac{d\theta}{dt}=I\dot{\theta}\frac{d}{dt}\delta\theta=\frac{d}{dt}(I\dot{\theta}\,\delta\theta)-I\ddot{\theta}\,\delta\theta$$

$$\boxed{\delta\frac{d}{dt}\theta=\frac{d}{dt}\delta\theta}\quad \delta\text{与}\frac{d}{dt}\text{可交换}$$

$$\delta I=\int_{t_1}^{t_2}\left[\frac{d}{dt}(I\dot{\theta}\,\delta\theta)-I\ddot{\theta}\,\delta\theta-mgl\sin\theta\,\delta\theta\right]dt$$

$$=I\dot{\theta}\,\delta\theta\Big|_{t_1}^{t_2}-\int_{t_1}^{t_2}(I\ddot{\theta}+mgl\sin\theta)\,\delta\theta\,dt$$

作变分的前后端点不变

$\Rightarrow \delta\theta|_{t=t_1}=\delta\theta|_{t=t_2}=0$

$$=-\int_{t_1}^{t_2}(I\ddot{\theta}+mgl\sin\theta)\,\delta\theta\,dt$$

哈密顿原理 $\Rightarrow \delta I=0$

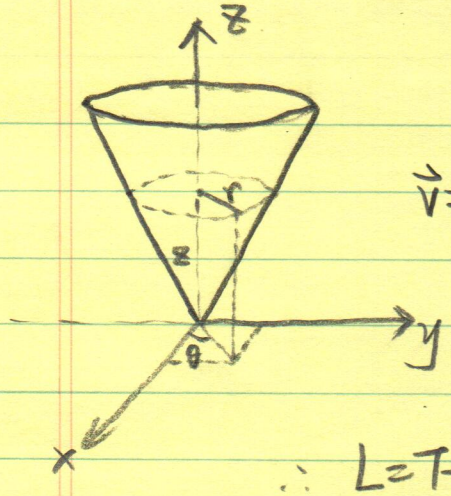
对所有 $\delta\theta$ 的成立

$$\therefore I\ddot{\theta}+mgl\sin\theta=0$$

微振动时，$\sin\theta\sim\theta$，$\therefore I\ddot{\theta}+mgl\theta=0$，$\omega=\sqrt{\frac{mgl}{I}}$

$\therefore T=\frac{2\pi}{\omega}=2\pi\sqrt{\frac{I}{mgl}}$

5.31



(27)

$$\vec{r} = r\cos\theta\,\hat{e}_x + r\sin\theta\,\hat{e}_y + z\hat{e}_z$$

$$\vec{v} = (\dot{r}\cos\theta - r\dot{\theta}\sin\theta)\hat{e}_x + (\dot{r}\sin\theta + r\dot{\theta}\cos\theta)\hat{e}_y + \dot{z}\hat{e}_z$$

$$\therefore v^2 = \dot{r}^2 + r^2\dot{\theta}^2 + \dot{z}^2 \qquad z = r\cot\alpha$$

$$= \dot{r}^2\csc^2\alpha + r^2\dot{\theta}^2$$

$$\therefore L = T - V = \frac{1}{2}m(\dot{r}^2\csc^2\alpha + r^2\dot{\theta}^2) - mgr\cot\alpha$$

$$\delta L = \frac{m\dot{r}}{\sin^2\alpha}\delta\dot{r} + mr\dot{\theta}^2\delta r + mr^2\dot{\theta}\delta\dot{\theta} - mg\delta r\cdot\cot\alpha$$

依{对}为{变} $\Rightarrow \delta S = 0$

即 $0 = \delta\int_{t_1}^{t_2} L\,dt = \int_{t_1}^{t_2}\delta L\,dt$

$$= \int_{t_1}^{t_2}\left(\frac{m\dot{r}}{\sin^2\alpha}\delta\dot{r} + mr\dot{\theta}^2\delta r + mr^2\dot{\theta}\delta\dot{\theta} - mg\delta r\cot\alpha\right)dt$$

之用分步积分

$$= \left(\frac{m\dot{r}}{\sin^2\alpha}\delta r + mr^2\dot{\theta}\cdot\delta\theta\right)\Bigg|_{t_1}^{t_2}$$

$$+ \int_{t_1}^{t_2}\left[\left(-\frac{m\ddot{r}}{\sin^2\alpha} + mr\dot{\theta}^2 - mg\cdot\cot\alpha\right)\delta r - \frac{d}{dt}(mr^2\dot{\theta})\delta\theta\right]dt$$

上式右-项为0，考虑到 δr 与 $\delta\theta$ 相互独立，且为任意变化量，故有

$$\begin{cases} \ddot{r}\csc^2\alpha - r\dot{\theta}^2 + g\cot\alpha = 0 & \text{即 } \ddot{r} - r\dot{\theta}^2\sin^2\alpha + g\sin\alpha\cos\alpha = 0 \\ mr^2\dot{\theta} = h \ (\text{常数}) \end{cases}$$

5.32 证明 $Q=\ln(\frac{1}{q}\sin p)$, $P=q\cot p$ 为正则变换

正则变换条件为下列四式之一：

$$p_\alpha dq_\alpha - P_\alpha dQ_\alpha + (H^*-H)dt = dU, \quad U=U(q,Q)$$

$$-q_\alpha dp_\alpha - P_\alpha dQ_\alpha + (H^*-H)dt = dU_1, \quad U_1=U_1(p,Q)$$

$$p_\alpha dq_\alpha + Q_\alpha dP_\alpha + (H^*-H)dt = dU_2, \quad U_2=U_2(q,P)$$

$$-q_\alpha dp_\alpha + Q_\alpha dP_\alpha + (H^*-H)dt = dU_3, \quad U_3=U_3(p,P)$$

如果 H 和 Q_α P_α 不显含 t，则 H^* 和 U 也可以显含 t，$H^*-H=0$

因此，证明以下等式，表明其为一全微分即可.

$$p\,dq - P\,dQ = p\,dq - (q\cot p)\left[\frac{q}{\sin p}\cdot\frac{1}{q^2}\sin p\cdot dq + \frac{q}{\sin p}\cdot\frac{1}{q}\cos p\,dp\right]$$

$$= p\,dq + \cot p\,dq - q\cdot\cot^2 p\,dp$$

$$= (p+\cot p)\,dq - q\cdot\cot^2 p\,dp$$

判定全微分式 $\mathrm{I} = \frac{d}{dp}(p+\cot p) = 1+\frac{-1}{\sin^2 p} = -\cot^2 p$

$$\mathrm{II} = \frac{d}{dq}(-q\cot^2 p) = -\cot^2 p$$

$\mathrm{I}=\mathrm{II}$ $\therefore$ $(p+\cot p)\,dq - q\cot^2 p\,dp = dU$

用 p 和 q 作为独立相空间，则可取 $U = q\cdot(p+\cot p)$. 证毕.

5.33 等用 = 外积式 $-Q\,dP - p\,dq = dU$ 来证明.

$$LHS = -Q\,dP - \sqrt{2Qk}\sin P\left[-\sqrt{\frac{Q}{k}}\sin P\,dP + \frac{1}{\sqrt{2Qk}}\cos P\,dQ\right]$$

$$= Q(2\sin^2 P - 1)\,dP - \sin P\cos P\,dQ$$

$$= -Q\cos 2P\,dP - \frac{1}{2}\sin 2P\,dQ$$

$$= d\left(-\frac{Q}{2}\sin 2P\right)$$

$\therefore q = \sqrt{\frac{2Q}{k}}\cos P$, $p = \sqrt{2Qk}\sin P$ 是正则变换.

$$H^* = H = \frac{1}{2}(p^2 + k^2 q^2) = \frac{1}{2}\left(2Qk\sin^2 P + k^2\cdot\frac{2Q}{k}\cos^2 P\right) = Qk.$$

$\because$ 正则变换 $\therefore$ 正则方程形式不变,

$$\begin{cases} \dot{P} = -\dfrac{\partial H^*}{\partial Q} \\ \dot{Q} = \dfrac{\partial H^*}{\partial P} \end{cases}$$

5.34 根据正则变换判断方式不变性

$(p,q)\to(P,Q)$ 为正则变换 等价于
$(P,Q)\to(p,q)$ 为正则变换

只须判断 $PQ\to pq$ 是否为正则变换:

$$P\,dQ - p\,dq = P\,dQ - p_2\left(\frac{\partial p_1}{\partial P}dP + \frac{\partial p_1}{\partial Q}dQ\right)$$

$$= -p_2\frac{\partial p_1}{\partial P}\,dP + \left(P - p_2\frac{\partial p_1}{\partial Q}\right)dQ$$

上式为全微分, 有

$$\frac{\partial}{\partial Q}\left(-p_2\frac{\partial p_1}{\partial P}\right) = \frac{\partial}{\partial P}\left(P - p_2\frac{\partial p_1}{\partial Q}\right)$$

$$\text{即 } -\frac{\partial p_2}{\partial Q}\frac{\partial p_1}{\partial P} - p_2\frac{\partial^2 p_1}{\partial P\partial Q} = 1 - \frac{\partial p_2}{\partial P}\frac{\partial p_1}{\partial Q} - p_2\frac{\partial^2 p_1}{\partial P\partial Q}$$

$$\therefore \frac{\partial p_1}{\partial Q}\frac{\partial p_2}{\partial P} - \frac{\partial p_1}{\partial P}\frac{\partial p_2}{\partial Q} = \begin{vmatrix} \partial p_1/\partial Q & \partial p_1/\partial P \\ \partial p_2/\partial Q & \partial p_2/\partial P \end{vmatrix} = \frac{\partial(p,q)}{\partial(P,Q)} = 1.$$

5.35 $H=T+V=\frac{P^2}{2m}+mgq$, $U=mg\left(\frac{1}{2}gQ^2+qQ\right)$

$U=U(q,Q)$, 由全微分一阶不变性, 全微分:

$$p\,dq-P\,dQ=dU$$

$$\therefore P=\frac{\partial U}{\partial Q}=mg\left(\frac{1}{2}gQ^2+q\right) \Rightarrow q=\frac{P}{mg}-\frac{1}{2}gQ^2$$

$$p=-\frac{\partial U}{\partial q}=-mgQ \Rightarrow Q=\frac{-p}{mg}$$

$$\therefore H=\frac{p^2}{2m}+mgq=\frac{1}{2m}(-mgQ)^2+mg\left(\frac{P}{mg}-\frac{1}{2}gQ^2\right)$$

$$=P$$

正则方程: $\dot{Q}=\frac{\partial H}{\partial P}=1 \Rightarrow \frac{-\dot{p}}{mg}=1$ 即 $\frac{-m\ddot{q}}{mg}=1$ $\therefore \ddot{q}=-g$ 加速度

$\dot{P}=-\frac{\partial H}{\partial Q}=0$ $\therefore P=C$ 即 $H=\frac{p^2}{2m}+mgq=C$. 能量守恒

5.36 $H=T+V=\frac{1}{2m}\left(p_r^2+\frac{p_\theta^2}{r^2}+\frac{p_\varphi^2}{r^2\sin^2\theta}\right)+\left(\frac{\alpha}{r^2}-\frac{F}{r^2}\cos\theta\right)$

哈密顿一那可比方程:

(H不显含t, 即 $\partial H/\partial t=0$)

$$S=-Et+W+C$$

$$H\left(r,\theta,\varphi,\frac{\partial W}{\partial r},\frac{\partial W}{\partial\theta},\frac{\partial W}{\partial\varphi}\right)=E$$

即 $\frac{1}{2m}\left[\left(\frac{\partial W}{\partial r}\right)^2+\frac{1}{r^2}\left(\frac{\partial W}{\partial\theta}\right)^2+\frac{1}{r^2\sin^2\theta}\left(\frac{\partial W}{\partial\varphi}\right)^2\right]+\left(\frac{\alpha}{r^2}-\frac{F}{r^2}\cos\theta\right)=E$

采用分离变量法, 首先方程对 φ 不显含仅在 $\left(\frac{\partial W}{\partial\varphi}\right)^2$ 中, 故可取 $\frac{\partial W}{\partial\varphi}=\alpha_3$

上式可化为 $\left[\frac{r^2}{2m}\left(\frac{\partial W}{\partial r}\right)^2-Er^2\right]+\left[\frac{1}{2m}\left(\frac{\partial W}{\partial\theta}\right)^2+\frac{1}{2m}\frac{\alpha_3^2}{\sin^2\theta}+\alpha-F\cos\theta\right]=0$

由分离变量方法, 可取 $W=f(r)+g(\theta)+\alpha_3\varphi$ $\therefore$ $\begin{cases}\frac{r^2}{2m}\left(\frac{df}{dr}\right)^2-Er^2=-\alpha_2\\ \frac{1}{2m}\left(\frac{dg}{d\theta}\right)^2+\frac{1}{2m}\frac{\alpha_3^2}{\sin^2\theta}+\alpha-F\cos\theta=\alpha_2\end{cases}$

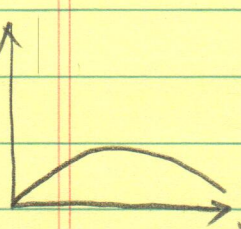
$$\therefore \frac{df}{dr} = \sqrt{2mE - \frac{\alpha_2}{r^2}}$$

$$\frac{dg}{d\theta} = \sqrt{\alpha_2 + 2mF\cos\theta - \frac{\alpha_3^2}{\sin^2\theta} - 2m\alpha}$$

$$\therefore W = \int \frac{df}{dr}dr + \int \frac{dg}{d\theta}d\theta$$

$$\therefore S = -Et + W + C = -Et + \int \sqrt{2mE - \frac{\alpha_2}{r^2}}\,dr + \int \sqrt{\alpha_2 + 2mF\cos\theta - \frac{\alpha_3}{\sin^2\theta} - 2m\alpha}\,d\theta + \alpha_3\varphi + C$$

5.37. 利用Hamilton-Jacobi方程求抛射体在真空中运动的轨迹方程.



方便起见取 x 和 y.

$$L = T - V = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2) - mgy$$

$$P_x = \frac{\partial L}{\partial \dot{x}} = m\dot{x}, \quad P_y = \frac{\partial L}{\partial \dot{y}} = m\dot{y}$$

$$\therefore H = \dot{x}P_x + \dot{y}P_y - L = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2) + mgy = \frac{1}{2m}(P_x^2 + P_y^2) + mgy$$

$$\frac{\partial H}{\partial t} = 0, \quad \frac{\partial H}{\partial x} = 0 \quad \therefore S = -Et + \alpha_2 x + W(y) + C$$

$$H\left(x, y; \alpha_2, \frac{dW}{dy}\right) = E$$

即 $$\frac{1}{2m}\left[\alpha_2^2 + \left(\frac{dW}{dy}\right)^2\right] + mgy = E$$

$$\frac{dW}{dy} = \sqrt{2m(E - mgy) - \alpha_2^2}$$

$$\therefore W' = \int \frac{dW}{dy}dy = \frac{2}{3}\cdot\frac{1}{2m^2g}\left[2m(E - mgy) - \alpha_2^2\right]^{\frac{3}{2}}$$

$$\therefore W = \alpha_2 x + W' = \alpha_2 x - \frac{1}{3m^2g}\left[2m(E - mgy) - \alpha_2^2\right]^{\frac{3}{2}}$$

$P_\alpha = \frac{\partial S}{\partial q_\alpha}$, $Q_\alpha = \frac{\partial S}{\partial P_\alpha}$.

$\beta_2 = \frac{\partial S}{\partial \alpha_2} = \frac{\partial W}{\partial \alpha_2} = x + \frac{1}{2m^2g} \cdot \frac{1}{2} \cdot 2\alpha_2 \cdot [2m(E-mgy) - \alpha_2^2]^{\frac{1}{2}}$

$= x + \frac{\alpha_2}{m^2 g} [2m(E-mgy) - \alpha_2^2]^{\frac{1}{2}}$

$\therefore 2m(E-mgy) - \alpha_2^2 = (\beta - x)^2 \cdot \frac{m^4 g^2}{\alpha_2^2}$

$\therefore y = \frac{E}{mg} - \frac{\alpha_2^2}{2m^2 g} - \frac{m^2 g}{2\alpha_2^2}(\beta_2 - x)^2$

$t=0$ 时, $x=y=0$, $\frac{E}{mg} - \frac{\alpha_2^2}{2m^2 g} - \frac{m^2 g}{2\alpha_2^2}\beta_2^2 = 0$

$\dot{x} = V_0 \cos\alpha$, $\dot{y} = V_0 \sin\alpha$,

$V_0 \sin\alpha = + \frac{m^2 g}{2\alpha_2^2} 2(\beta_2 - x) \cdot \dot{x}$ ($\beta_2 - x = 0$, $\dot{x} = V_0\cos\alpha$) $\Rightarrow \tan\alpha = \frac{m^2 g \beta_2}{\alpha_2^2}$

$\therefore \beta_2 = \frac{\alpha_2^2}{m^2 g} \tan\alpha$

$\frac{E}{mg} - \frac{\alpha_2^2}{2m^2 g}(1 + \tan^2\alpha) = 0$ $\quad \therefore \alpha_2^2 = 2mE \cdot \cos^2\alpha$,

$E = \frac{1}{2} m V_0^2$ $\quad \therefore \alpha_2 = m V_0 \cos\alpha$

$\beta_2 = \frac{V_0^2}{g} \cdot \sin\alpha\cos\alpha$

$\frac{\alpha_2^2}{m^2 g} = \frac{2m \cdot \frac{1}{2} m V_0^2 \cdot \cos^2\alpha}{m^2 g} = \frac{V_0^2}{g} \cdot \cos^2\alpha$

$\therefore y = \frac{V_0^2}{2g} - \frac{V_0^2}{2g}\cos^2\alpha - \frac{g}{2V_0^2\cos^2\alpha}\left(\frac{V_0^2}{g}\sin\alpha\cos\alpha - x\right)^2$

$= \frac{V_0^2}{2g}\sin^2\alpha + \frac{g}{2V_0^2\cos^2\alpha}\left(\frac{V_0^2}{g}\sin\alpha\cos\alpha - x\right)^2$.

5.38

$$L = T - V = \frac{1}{2}(A_1+A_2+\cdots+A_s)(B_1\dot{q}_1^2+B_2\dot{q}_2^2+\cdots+B_s\dot{q}_s^2) - \frac{V_1+V_2+\cdots+V_s}{A_1+A_2+\cdots+A_s}$$

因为 T 是 $\dot{q}^2$ 的齐次，故 $\dot{q}\cdot\frac{\partial T}{\partial \dot{q}} = 2T$，$H = \dot{q}p - L = T+V$

$$\therefore H = \frac{1}{2}(A_1+A_2+\cdots+A_s)(B_1\dot{q}_1^2+B_2\dot{q}_2^2+\cdots+B_s\dot{q}_s^2) + \frac{V_1+V_2+\cdots+V_s}{A_1+A_2+\cdots+A_s}$$

$$p_i = \frac{\partial L}{\partial \dot{q}_i} = (A_1+A_2+\cdots+A_s)B_i\dot{q}_i \qquad \therefore \dot{q}_i = \frac{p_i}{(A_1+A_2+\cdots+A_s)B_i}$$

$$\therefore H = \sum_{i=1}^{s}\frac{p_i^2}{2(A_1+A_2+\cdots+A_s)B_i} + \frac{V_1+V_2+\cdots+V_s}{A_1+A_2+\cdots+A_s}$$

$\frac{\partial H}{\partial t} = 0$ ∴ 设取 $S = -Et + W$

$$H\left(q_1, q_2, \cdots q_s; \frac{\partial W}{\partial q_1}, \frac{\partial W}{\partial q_2}, \cdots \frac{\partial W}{\partial q_s}\right) = E$$

即
$$\sum_{i=1}^{s}\frac{(\partial W/\partial q_i)^2}{2(A_1+A_2+\cdots+A_s)B_i} + \frac{V_1+V_2+\cdots+V_s}{A_1+A_2+\cdots+A_s} = E$$

$$\therefore \sum_{i=1}^{s}\left[\frac{1}{2B_i}\left(\frac{\partial W}{\partial q_i}\right)^2 + V_i - EA_i\right] = 0$$

可用分离变量法，$W = W_1(q_1)+W_2(q_2)+\cdots+W_s(q_s)$

$$\frac{1}{2B_i}\left(\frac{dW_i}{dq_i}\right)^2 + V_i - EA_i = \alpha_i, \quad i=1,2,\cdots,s, \quad \sum_{i=1}^{s}\alpha_i = 0$$

$$\therefore \frac{dW_i}{dq_i} = \sqrt{2B_i(\alpha_i+EA_i-V_i)} \qquad \therefore W_i = \int\sqrt{2B_i(\alpha_i+EA_i-V_i)}\,dq_i$$

$$\therefore S = -Et + \sum_{i=1}^{s}\int\sqrt{2B_i(\alpha_i+EA_i-V_i)}\,dq_i, \quad 其中 \sum_{i=1}^{s}\alpha_i = 0$$

5.31 行星绕日运动

有心力，平面运动，$\vec{v}^2 = v_r^2 + v_\theta^2 = \dot{r}^2 + r^2\dot{\theta}^2$

$\therefore T = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2)$，$V = -\frac{k^2 m}{r}$

$\therefore p_r = \frac{\partial L}{\partial \dot{r}} = m\dot{r}$，$p_\theta = \frac{\partial L}{\partial \dot{\theta}} = mr^2\dot{\theta}$

$\because T = T_2$，对 T 是以速度 $\dot{r}$ 和 $\dot{\theta}$ 齐二次 $\therefore \dot{r}\frac{\partial T}{\partial \dot{r}} + \dot{\theta}\frac{\partial T}{\partial \dot{\theta}} = 2T$

$\therefore H = \dot{r}p_r + \dot{\theta}p_\theta - L = T + V = \frac{p_r^2}{2m} + \frac{p_\theta^2}{2mr^2} - \frac{mk^2}{r}$

$\because \frac{\partial H}{\partial \theta} = 0$ $\therefore \dot{p}_\theta = -\frac{\partial H}{\partial \theta} = 0$ $\therefore p_\theta = mr^2\dot{\theta} = h$（常数）

$\because \frac{\partial H}{\partial t} = 0$ $\therefore S = -Et + h\theta + W(r) + C$

$$\frac{1}{2m}\left(\frac{dW}{dr}\right)^2 + \frac{h^2}{2mr^2} - \frac{mk^2}{r} = E$$

$$\therefore \frac{dW}{dr} = \sqrt{2mE + \frac{2m^2k^2}{r} - \frac{h^2}{r^2}}$$

$$\therefore S = -Et + h\theta + \int \sqrt{2mE + \frac{2m^2k^2}{r} - \frac{h^2}{r^2}}\,dr + C$$

$$\therefore \beta = \frac{\partial S}{\partial h} = \theta - \int \frac{1}{2}\frac{2h}{r^2}\cdot \frac{dr}{\sqrt{2mE + \frac{2m^2k^2}{r} - \frac{h^2}{r^2}}} + C \quad 令\ u = \frac{1}{r}$$

$$= \theta + h\int \frac{du}{\sqrt{2mE + 2m^2k^2u - h^2u^2}}$$

令 $= h^2\left(\frac{2mE}{h^2} + \frac{2m^2k^2}{h^2}u - u^2\right) = h^2\left[\underbrace{\frac{2mE}{h^2} + \frac{m^4k^4}{h^4}}_{C_1} - \underbrace{\left(\frac{m^2k^2}{h^2} - u\right)^2}_{\tilde{u}}\right]$

$$\beta = \theta + \int \frac{d\tilde{u}}{\sqrt{C_1 - \tilde{u}^2}} = \theta + \int \frac{dx}{\sqrt{1-x^2}} = \theta + \arccos x + C_2$$

$x = \frac{\tilde{u}}{\sqrt{C_1}}$

$\therefore x = \cos(\theta - \beta - C_2) = \cos(\theta - \theta_0)$

3.7 $\left(\frac{1}{r}-\frac{m^2k^2}{h^2}\right)\Big/\sqrt{\frac{2mE}{h^2}+\frac{m^4k^4}{h^4}} = \cos(\theta-\theta_0)$

$\therefore \frac{1}{r} = \frac{m^2k^2}{h^2} + \sqrt{\frac{m^4k^4}{h^4}+\frac{2mE}{h^2}}\cos(\theta-\theta_0)$

$= \underbrace{\frac{m^2k^2}{h^2}}_{1/p}\left[1+\underbrace{\sqrt{1+\frac{2Eh^2}{m^2k^4}}}_{e}\cdot\cos(\theta-\theta_0)\right]$

$p=\frac{h^2}{m^2k^2}, \quad e=\sqrt{1+\frac{2Eh^2}{m^2k^4}}$

$\therefore r = \frac{p}{1+e\cos(\theta-\theta_0)}$

当 $\theta=0$ 时, $r=r_{min}$, 故 $\cos(\theta-\theta_0)$ 取极大值, $\theta_0=0$

$\therefore r = \frac{p}{1+e\cos\theta}$

5.40 $L=\frac{1}{2}m(\dot r^2+r^2\dot\theta^2+\dot z^2)-V(r,\theta,z)$

椭圆坐标变换: $r=\sigma\sqrt{(\xi^2-1)(1-\eta^2)}$, $z=\sigma\xi\eta$

$\xi\in(1,\infty)$
$\eta\in(-1,1)$

$V=\frac{a(\xi)+b(\eta)}{\xi^2-\eta^2}$

写出用 (ξ,η,θ) 表示的哈密顿函数 (5.31), 并求主动量积分 (5.32).

解: 首先根据 r 和 z 求导.

$\dot r = \frac{\sigma}{2}\frac{2\xi\dot\xi(1-\eta^2)-(\xi^2-1)2\eta\dot\eta}{\sqrt{(\xi^2-1)(1-\eta^2)}} \Rightarrow \dot r^2 = \frac{\sigma^2[\xi(1-\eta^2)\dot\xi-(\xi^2-1)\eta\dot\eta]^2}{(\xi^2-1)(1-\eta^2)}$

$\therefore r^2\dot\theta^2 = \sigma^2(\xi^2-1)(1-\eta^2)\dot\theta^2$

$\dot z = \sigma(\xi\dot\eta+\eta\dot\xi) \Rightarrow \dot z^2 = \sigma^2(\xi\dot\eta+\eta\dot\xi)^2$

$\therefore L = \frac{m\sigma^2[\xi(1-\eta^2)\dot\xi-(\xi^2-1)\eta\dot\eta]^2}{2(\xi^2-1)(1-\eta^2)} + \frac{m\sigma^2}{2}(\xi^2-1)(1-\eta^2)\dot\theta^2 + \frac{m\sigma^2}{2}(\xi\dot\eta+\eta\dot\xi)^2 - \frac{a(\xi)+b(\eta)}{\xi^2-\eta^2}$

$$P_\xi = \frac{\partial L}{\partial \dot\xi} = \frac{m\sigma^2\xi}{\xi^2-1}\left[\xi(1-\eta^2)\dot\xi - (\xi^2-1)\eta\dot\eta\right] + m\sigma^2\eta(\xi\dot\eta+\eta\dot\xi) \quad ①$$

$$P_\eta = \frac{\partial L}{\partial \dot\eta} = -\frac{m\sigma^2\eta}{1-\eta^2}\left[\xi(1-\eta^2)\dot\xi - (\xi^2-1)\eta\dot\eta\right] + m\sigma^2\xi(\xi\dot\eta+\eta\dot\xi) \quad ②$$

$$\frac{\xi^2-1}{\xi}P_\xi + \frac{1-\eta^2}{\eta}P_\eta = m\sigma^2(\xi\dot\eta+\eta\dot\xi)\left(\frac{\xi^2-1}{\xi}\eta + \frac{1-\eta^2}{\eta}\xi\right)$$

$$\left(-\frac{\eta}{\xi}+\frac{\xi}{\eta}\right) = \frac{\xi^2-\eta^2}{\xi\eta}$$

$$\therefore \boxed{\xi\dot\eta+\eta\dot\xi = \frac{1}{m\sigma^2}\frac{1}{\xi^2-\eta^2}\left[\eta(\xi^2-1)P_\xi + \xi(1-\eta^2)P_\eta\right]} \quad ③$$

$$\frac{1}{\eta}P_\xi - \frac{1}{\xi}P_\eta = \left(\frac{m\sigma^2}{\xi^2-1}\frac{\xi}{\eta} + \frac{m\sigma^2}{1-\eta^2}\frac{\eta}{\xi}\right)\left[\xi(1-\eta^2)\dot\xi - (\xi^2-1)\eta\dot\eta\right]$$

$$\frac{m\sigma^2}{(\xi^2-1)(1-\eta^2)}\frac{1}{\xi\eta}\left[\xi^2(1-\eta^2) + (\xi^2-1)\eta^2\right] = \frac{m\sigma^2}{(\xi^2-1)(1-\eta^2)}\frac{\xi^2-\eta^2}{\xi\eta}$$

$$\therefore \boxed{\xi(1-\eta^2)\dot\xi - (\xi^2-1)\eta\dot\eta = \frac{(\xi^2-1)(1-\eta^2)}{m\sigma^2}\frac{1}{\xi^2-\eta^2}(\xi P_\xi - \eta P_\eta)} \quad ④$$

$$P_\theta = \frac{\partial L}{\partial \dot\theta} = m\sigma^2(\xi^2-1)(1-\eta^2)\dot\theta \Rightarrow \boxed{\dot\theta = \frac{1}{m\sigma^2}\frac{P_\theta}{(\xi^2-1)(1-\eta^2)}} \quad ⑤$$

∴ 将②③④代回T表达式中，有（因为T是$\dot\xi,\dot\eta,\dot\theta$二次型，故 $H=T+V$）

$$H = T+V = \frac{m\sigma^2}{2(\xi^2-1)(1-\eta^2)}\left[\frac{(\xi^2-1)(1-\eta^2)}{m\sigma^2}\frac{1}{\xi^2-\eta^2}(\xi P_\xi - \eta P_\eta)\right]^2 + \frac{m\sigma^2}{2}(\xi^2-1)(1-\eta^2)\left[\frac{1}{m\sigma^2}\frac{P_\theta}{(\xi^2-1)(1-\eta^2)}\right]^2$$

$$+\frac{m\sigma^2}{2}\left\{\frac{1}{m\sigma^2}\frac{1}{\xi^2-\eta^2}\left[\eta(\xi^2-1)P_\xi + \xi(1-\eta^2)P_\eta\right]\right\}^2 + \frac{a(\xi)+b(\eta)}{\xi^2-\eta^2}$$

$$= \frac{1}{2m\sigma^2(\xi^2-\eta^2)^2}\left\{(\xi^2-1)(1-\eta^2)(\xi P_\xi - \eta P_\eta)^2 + \left[\eta(\xi^2-1)P_\xi + \xi(1-\eta^2)P_\eta\right]^2\right\}$$

$$+\frac{P_\theta^2}{2m\sigma^2(\xi^2-1)(1-\eta^2)} + \frac{a(\xi)+b(\eta)}{\xi^2-\eta^2}$$

大括号中公项为

$$\{\} = P_\xi^2\cdot\frac{[(\xi^2-1)(1-\eta^2)\xi^2+\eta^2(\xi^2-1)^2]}{(\xi^2-1)(\xi^2-\eta^2)} + P_\eta^2\frac{[(\xi^2-1)(1-\eta^2)\eta^2+\xi^2(1-\eta^2)^2]}{(1-\eta^2)(\xi^2-\eta^2)}$$

$$+2P_\xi P_\eta\underbrace{[(\xi^2-1)(1-\eta^2)\xi\eta+(\xi^2-1)(1-\eta^2)\eta\xi]}_{0}$$

$$=(\xi^2-\eta^2)[(\xi^2-1)P_\xi^2+(1-\eta^2)P_\eta^2]$$

$$\boxed{\frac{1}{(\xi^2-1)(1-\eta^2)}=\frac{1}{\xi^2-\eta^2}\left(\frac{1}{\xi^2-1}+\frac{1}{1-\eta^2}\right)}$$

$$\therefore H=\frac{1}{2m\sigma^2(\xi^2-\eta^2)}[(\xi^2-1)P_\xi^2+(1-\eta^2)P_\eta^2]+\frac{1}{2m\sigma^2(\xi^2-\eta^2)}\left(\frac{1}{\xi^2-1}+\frac{1}{1-\eta^2}\right)P_\theta^2+V$$

$$=\frac{1}{2m\sigma^2(\xi^2-\eta^2)}\left[(\xi^2-1)P_\xi^2+(1-\eta^2)P_\eta^2+\left(\frac{1}{\xi^2-1}+\frac{1}{1-\eta^2}\right)P_\theta^2\right]+\frac{a(\xi)+b(\eta)}{\xi^2-\eta^2}$$

此即为(5.9.31)式.

因为 $\frac{\partial H}{\partial t}=0$, 故主函数 $S=-Et+W$.

因为 $\frac{\partial H}{\partial\theta}=0$, 故 θ 为循环变量, $\dot{P}_\theta=0$, $P_\theta=$常数 $\therefore S=-Et+P_\theta\theta+W_2$,

$W_2=W_2(\xi,\eta)$

哈密顿-雅可比方程 $H=E$ 变为

$$\frac{1}{2m\sigma^2(\xi^2-\eta^2)}\left[(\xi^2-1)\left(\frac{\partial W_2}{\partial\xi}\right)^2+(1-\eta^2)\left(\frac{\partial W_2}{\partial\eta}\right)^2+\left(\frac{1}{\xi^2-1}+\frac{1}{1-\eta^2}\right)P_\theta^2\right]+\frac{a(\xi)+b(\eta)}{\xi^2-\eta^2}=E$$

整理后变为:

$$\left[(\xi^2-1)\left(\frac{\partial W_2}{\partial\xi}\right)^2+\frac{P_\theta^2}{\xi^2-1}+2m\sigma^2a(\xi)-2m\sigma^2E\xi^2\right]$$

$$+\left[(1-\eta^2)\left(\frac{\partial W_2}{\partial\eta}\right)^2+\frac{P_\theta^2}{1-\eta^2}+2m\sigma^2b(\eta)+2m\sigma^2E\eta^2\right]=0$$

上式中 P_θ 为常数. 从上式, 第一括号中仅与 ξ 有关, 第二括号中仅与 η 有关, 可分离变量 $W_2=f(\xi)+b(\eta)$, 则 $\frac{\partial W_2}{\partial\xi}=\frac{df}{d\xi}$, $\frac{\partial W_2}{\partial\eta}=\frac{dg}{d\eta}$

代回上式, 可得.

$$(\xi^2-1)\left(\frac{df}{d\xi}\right)^2+\frac{P_\theta^2}{\xi^2-1}+2m\sigma^2 a(\xi)-2m\sigma^2 E\xi^2=\alpha$$

$$(1-\eta^2)\left(\frac{dg}{d\eta}\right)^2+\frac{P_\theta^2}{1-\eta^2}+2m\sigma^2 b(\eta)+2m\sigma^2 E\eta^2=-\alpha.$$

$\xi^2=(\xi^2-1)+1$

$$\therefore \frac{df}{d\xi}=\sqrt{2m\sigma^2 E+\frac{\alpha+2m\sigma^2 E-2m\sigma^2 a(\xi)}{\xi^2-1}+\frac{P_\theta^2}{(\xi^2-1)^2}}$$

$$\frac{dg}{d\eta}=\sqrt{2m\sigma^2 E-\frac{\alpha+2m\sigma^2 E+2m\sigma^2 b(\eta)}{1-\eta^2}+\frac{P_\theta^2}{(1-\eta^2)^2}}$$

令 $\boxed{\alpha+2m\sigma^2 E\equiv\alpha_2}$

$$\therefore S=-Et+P_\theta\theta+\int\frac{df}{d\xi}d\xi+\int\frac{dg}{d\eta}d\eta+C$$

$$=-Et+P_\theta\theta+\int\sqrt{2m\sigma^2 E+\frac{\alpha_2-2m\sigma^2 a(\xi)}{\xi^2-1}+\frac{P_\theta^2}{(\xi^2-1)^2}}\cdot d\xi$$

$$+\int\sqrt{2m\sigma^2 E-\frac{\alpha_2+2m\sigma^2 b(\eta)}{1-\eta^2}+\frac{P_\theta^2}{(1-\eta^2)^2}}\,d\eta+C$$

由 (5.9.24), 抛物线坐标下的动能为

$$H=T+V=\frac{2}{m}\frac{\xi P_\xi^2+\eta P_\eta^2}{\xi+\eta}+\frac{P_\theta^2}{2m\xi\eta}+V(\xi,\eta,\theta)$$

抛物线坐标满足: $z=\frac{1}{2}(\xi-\eta)$, $r=\sqrt{\xi\eta}$.

$\boxed{\xi\in(0,\infty)\quad\eta\in(0,\infty)}$

$$\therefore R=\sqrt{r^2+z^2}=\sqrt{\frac{1}{4}(\xi-\eta)^2+\xi\eta}=\frac{1}{2}(\xi+\eta)$$

$$\therefore V=\frac{\alpha}{R}-Fz=\frac{2\alpha}{\xi+\eta}-\frac{F}{2}(\xi-\eta)$$

$$\therefore H=\frac{2}{m}\frac{\xi P_\xi^2+\eta P_\eta^2}{\xi+\eta}+\frac{P_\theta^2}{2m\xi\eta}+\frac{2\alpha}{\xi+\eta}-\frac{F}{2}(\xi-\eta)$$

$\because \frac{\partial H}{\partial t}=0$ $\therefore S=-Et+W$. $\because \frac{\partial H}{\partial\theta}=0$ $\therefore P_\theta=$常数.

$\therefore S = -Et + P_\theta \cdot \theta + W_2 \quad , \quad W_2 = W_2(\xi, \eta)$

W_2 满足哈密顿-雅可比方程

$H = E$

$$\frac{2}{m(\xi+\eta)}\left[\xi\left(\frac{\partial W_2}{\partial \xi}\right)^2 + \eta\left(\frac{\partial W_2}{\partial \eta}\right)^2\right] + \frac{P_\theta^2}{2m\xi\eta} + \frac{2\alpha}{\xi+\eta} - \frac{F}{2}(\xi-\eta) = E$$

方程两边乘以 $\frac{m}{2}(\xi+\eta)$，上式变为

$$\xi\left(\frac{\partial W_2}{\partial \xi}\right)^2 + \eta\left(\frac{\partial W_2}{\partial \eta}\right)^2 + \frac{P_\theta^2}{4}\left(\frac{1}{\xi}+\frac{1}{\eta}\right) + m\alpha - \frac{mF}{4}(\xi^2-\eta^2) = \frac{mE}{2}(\xi+\eta)$$

$$\therefore \left[\xi\left(\frac{\partial W_2}{\partial \xi}\right)^2 + \frac{P_\theta^2}{4\xi} - \frac{mF}{4}\xi^2 - \frac{mE}{2}\xi + \frac{m\alpha}{2}\right]$$

$$+ \left[\eta\left(\frac{\partial W_2}{\partial \eta}\right)^2 + \frac{P_\theta^2}{4\eta} + \frac{mF}{4}\eta^2 - \frac{mE}{2}\eta + \frac{m\alpha}{2}\right] = 0$$

上式可用分离变量法，$W_2(\xi,\eta) = f(\xi) + g(\eta)$，$\frac{\partial W_2}{\partial \xi} = \frac{df}{d\xi}$，$\frac{\partial W_2}{\partial \eta} = \frac{dg}{d\eta}$

$$\therefore \xi\left(\frac{df}{d\xi}\right)^2 + \frac{P_\theta^2}{4\xi} - \frac{mF}{4}\xi^2 - \frac{mE}{2}\xi + \frac{m\alpha}{2} = \frac{\alpha_2}{2}$$

$$\eta\left(\frac{dg}{d\eta}\right)^2 + \frac{P_\theta^2}{4\eta} + \frac{mF}{4}\eta^2 - \frac{mE}{2}\eta + \frac{m\alpha}{2} = -\frac{\alpha_2}{2}$$

$$\therefore \frac{df}{d\xi} = \sqrt{\frac{mE}{2} - \frac{m\alpha-\alpha_2}{2\xi} - \frac{P_\theta^2}{4\xi^2} + \frac{mF\xi}{4}}$$

$$\frac{dg}{d\eta} = \sqrt{\frac{mE}{2} - \frac{m\alpha+\alpha_2}{2\eta} - \frac{P_\theta^2}{4\eta^2} - \frac{mF\eta}{4}}$$

$$\therefore S = -Et + P_\theta \cdot \theta + \int \frac{df}{d\xi}\,d\xi + \int \frac{dg}{d\eta}\,d\eta$$

$$= -Et + P_\theta \cdot \theta + \int \sqrt{\frac{mE}{2} - \frac{m\alpha-\alpha_2}{2\xi} - \frac{P_\theta^2}{4\xi^2} + \frac{mF\xi}{4}}\,d\xi$$

$$+ \int \sqrt{\frac{mE}{2} - \frac{m\alpha+\alpha_2}{2\eta} - \frac{P_\eta^2}{4\eta^2} - \frac{mF\eta}{4}}\,d\eta$$